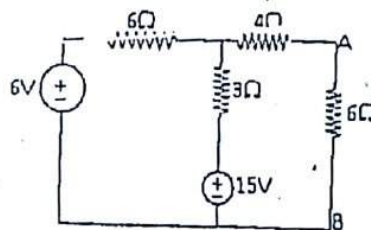


OLD QUESTION SOLUTION

1. Determine the current passing through the 6Ω resistor connected across A-B terminals in the circuit using Thevenin's theorem. [PU 2002(BEE)]



Solution:

Here $R_L = 6\Omega$

Calculation of V_{Th} :

From fig (i),

Since V_b is reference node,

$$\therefore V_b = 0$$

At node 'a'

$$\frac{V_a - 6}{6} + \frac{V_a - 15 - V_b}{3} = 0$$

$$V_a - 6 + 2V_a - 30 - 0 = 0$$

$$3V_a = 36$$

$$V_a = 12V$$

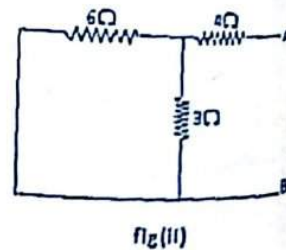
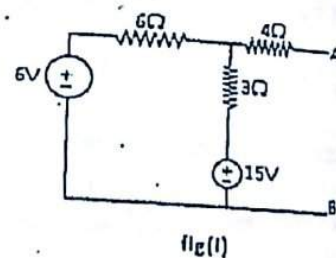
$$\therefore V_{Th} = V_{ab} = V_a - V_b = 12 - 0 = 12V$$

Calculation of R_{Th}

From fig (ii),

$$R_{Th} = (6 \parallel 3) + 4 = 2 + 4 = 6\Omega$$

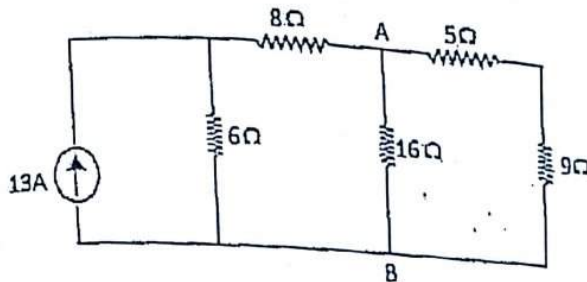
Now, current passing through the load resistance R_L is:



D.C. Circuit Analysis

$$I_L = \frac{V_{Th}}{R_{eq} + R_L} = \frac{12}{6+4} = 1.2 \text{ A}$$

2. Using Norton's theorem to calculate the current in 16Ω resistor in the given network.
[PU 2008 BE]



Solution:

Here $R_L = 16\Omega$

Calculation of I_N

From fig (i),

In mesh 1,

$$I_1 = 13\text{A}$$

In mesh 2,

$$-6I_2 + 6I_1 - 8I_2 = 0$$

$$6I_1 = 14I_2$$

$$I_2 = 5.57\text{A}$$

In mesh 3,

$$-5I_3 - 9I_3 = 0$$

$$-14I_3 = 0$$

$$I_3 = 0$$

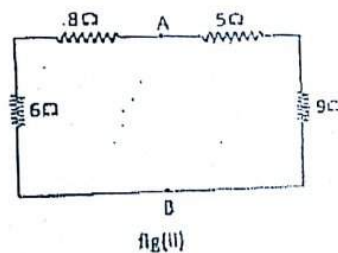
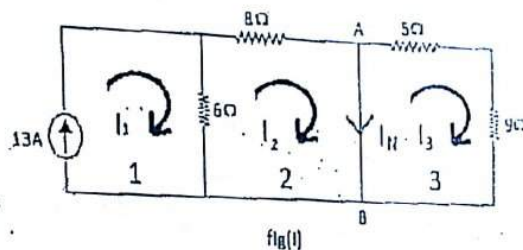
$$\therefore I_N = I_2 - I_3 = 5.57 \text{ A}$$

Calculation of R_{eq}

From fig (ii),

$$R_{eq} = (6+8) \parallel (5+9) = 14 \parallel 14 = 7\Omega$$

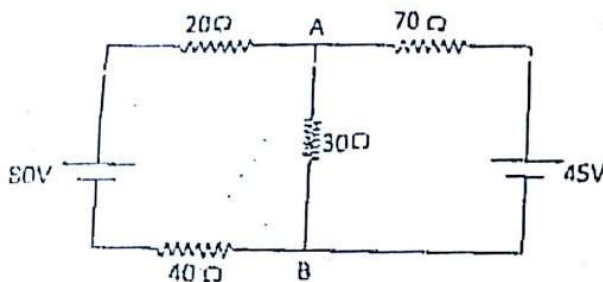
Hence, current through branch AB is,



D.C. Circuit Analysis

$$I_{AB} = I_1 \cdot \frac{R_{eq}}{R_{eq} + R_L} = 5.75 \cdot \frac{7}{7+16} = 1.75 \text{ A}$$

3. Use superposition theorem to find the current through 30Ω resistance in the following circuit. [PU 2008 S]



Solution:

- c) Take 80V source only, (as shown in fig.(i))

$$R_{eq} = (70 \parallel 30) + 20 + 40$$

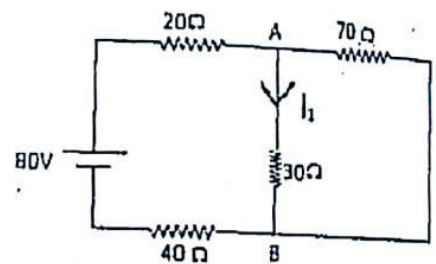
$$= 21 + 20 + 40$$

$$= 81\Omega$$

$$I = \frac{V}{R_{eq}} = \frac{80}{81} = 0.987 \text{ A}$$

Current flowing through AB branch,

$$I_1 = I \cdot \frac{70}{30 + 70} = 0.987 \cdot \frac{70}{100} = 0.691 \text{ A (A to B)}$$



fig(i)

- d) Take 45V source only, (as shown in fig.(ii))

$$R'_{eq} = (60 \parallel 30) + 70$$

$$= 90\Omega$$

$$I' = \frac{V'}{R'_{eq}} = \frac{45}{90} = 0.5 \text{ A}$$

Current flowing through AB branch,

$$I_2 = I' \cdot \frac{60}{60+30} = 0.5 \cdot \frac{60}{90} = 0.334 \text{ A (A to B)}$$

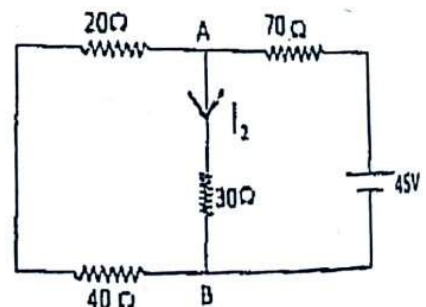
Now,

Total current flowing through 30Ω branch is,

$$I_{30\Omega} = I_1 + I_2$$

$$= 0.691 + 0.334$$

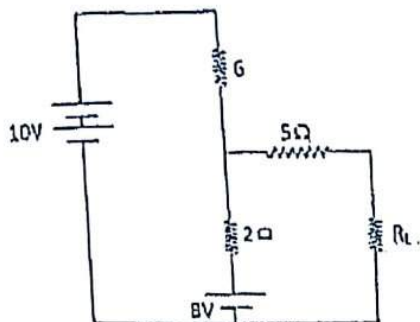
$$= 1.024 \text{ A (A to B)}$$



fig(ii)

4. Find the Thevenin's equivalent circuit for the following figure, and find the maximum power transferred to the load resistance. [PU 2008F]

D.C. Circuit Analysis



Solution:

Calculation of V_{Th} :

From fig (i), using Nodal analysis,

At node 'a'

$$\frac{v_a - 8}{2} + \frac{v_a - 10}{6} = 0$$

$$\Rightarrow 3v_a - 24 + v_a - 10 = 0$$

$$\Rightarrow 4v_a = 34$$

$$\Rightarrow v_a = 34/4$$

$$\Rightarrow v_a = 8.5 \text{ V}$$

Node 'b' is taken as reference node,

$$\therefore v_b = 0$$

$$\text{Now, } V_{Th} = v_a - v_b = 8.5 \text{ V}$$

Calculation of R_{eq}

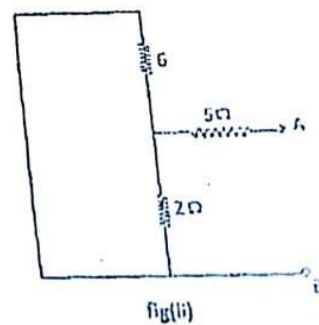
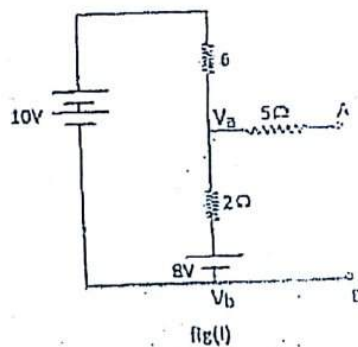
From fig (ii),

$$R_{eq} = R_{Th} = 6 \parallel 2 + 5 = 6.5 \Omega$$

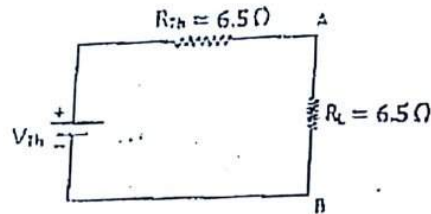
For maximum power,

$$R_L = R_{Th} = 6.5 \Omega$$

Now, Thevenin's equivalent circuit is:



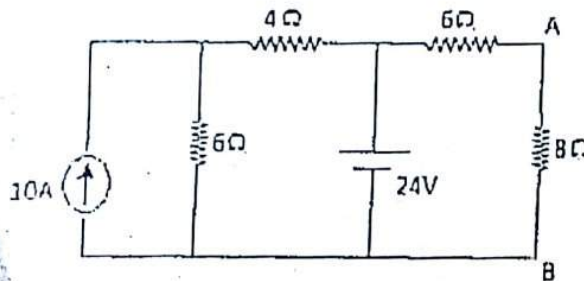
D.C. Circuit Analysis



And, maximum power transferred to the load is:

$$P_{L(max)} = \frac{V_{Th}^2}{4R_{Th}} = \frac{0.5^2}{4 \times 6.5} = 2.77 \text{ W}$$

5. Calculate the power consumed by 8Ω resistor using superposition theorem. [PU2008F]



Solution:

- a) Take 24V source only, (as shown in fig.(i))

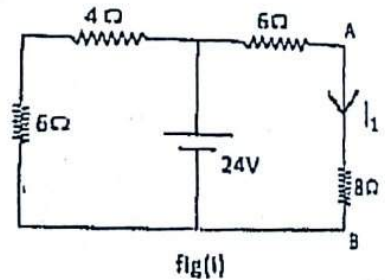
$$R_{eq} = (4 + 6) \parallel (6 + 8)$$

$$= 5.83 \Omega$$

$$I = \frac{V}{R_{eq}} = \frac{24}{5.83} = 4.116 \text{ A}$$

Current flowing through AB branch,

$$I_1 = I \times \frac{10}{10 + 14} = 4.116 \times \frac{10}{24} = 1.715 \text{ A (A to B)}$$



fig(i)

- b) Take 10A source only, (as shown in fig.(ii))

Due to short circuit,

Current flowing through AB branch,

$$I_2 = 0$$

Now,

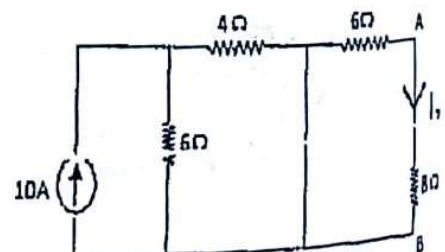
Total current flowing through 8Ω branch is,

$$I_{8\Omega} = I_1 + I_2$$

$$= 1.715 + 0$$

$$= 1.715 \text{ A (A to B)}$$

Now, power consumed by 8Ω resistors is:

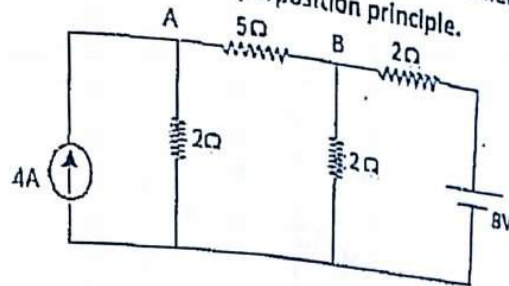


fig(II)

D.C. Circuit Analysis

$$I^2 \times R = 1.715^2 \times 8 = 23.5298 \text{ W}$$

6. Find out the current through 5Ω resistors connected across the terminals AB in the given network using superposition principle. [PU 2004 S]



Solution:

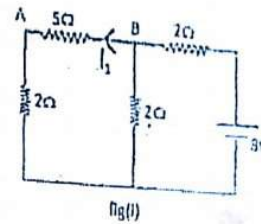
- a) Take 8V source only, (as shown in fig.(i))

$$R_{eq} = (5+2) \parallel (2+2) = 3.55 \Omega$$

$$I = \frac{V}{R_{eq}} = \frac{8}{3.55} = 2.253 \text{ A}$$

Current flowing through AB branch,

$$I_1 = I \times \frac{2}{7+2} = 2.253 \times \frac{2}{9} = 0.5 \text{ A (B to A)}$$



- b) Take 4A Source only, (as shown in fig.(ii))

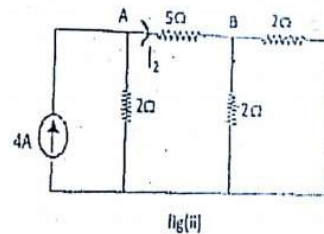
Current flowing through AB branch,

$$I_2 = 4 \times \frac{2}{2+(5+(2 \parallel 2))} = 4 \times \frac{2}{2+5+1} = 1 \text{ A (A to B)}$$

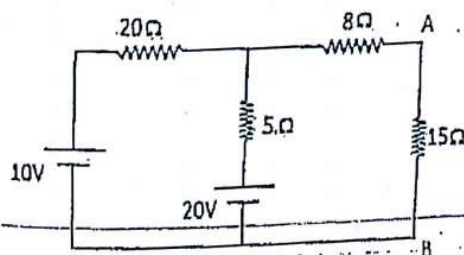
Now,

Total current flowing through 5Ω resistor is,

$$I_{5\Omega} = I_1 + I_2 = 1 \text{ A (A to B)} + 0.5 \text{ A (B to A)} = 0.5 \text{ A (A to B)}$$



7. Determine the current passing through 15Ω resistors connected across a-b terminals in the circuit using Norton's theorem. [PU 2005S]



D.C. Circuit Analysis

Solution:

Here $R_L = 15\Omega$

Calculation of I_N

From fig (i),

In mesh 1,

$$10 - 20I_1 - 5I_1 + 5I_2 - 20 = 0$$

$$25I_1 - 5I_2 = -10 \quad \text{--- (i)}$$

In mesh 2,

$$20 - 5I_2 + 5I_1 - 8I_2 = 0$$

$$5I_1 - 13I_2 = -20 \quad \text{--- (ii)}$$

On solving (i) & (ii) we get,

$$I_1 = -0.1 \text{ A} \quad I_2 = 1.5 \text{ A}$$

Here, $I_N = I_2 = 1.5 \text{ A}$

Calculation of R_{eq}

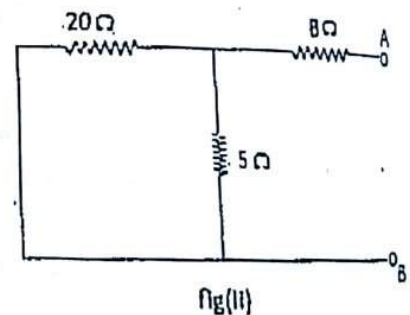
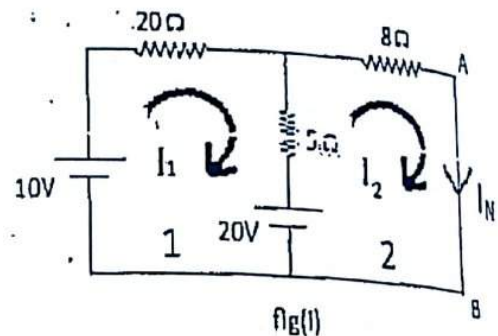
From fig (ii),

$$R_{eq} = (20 \parallel 5) + 8$$

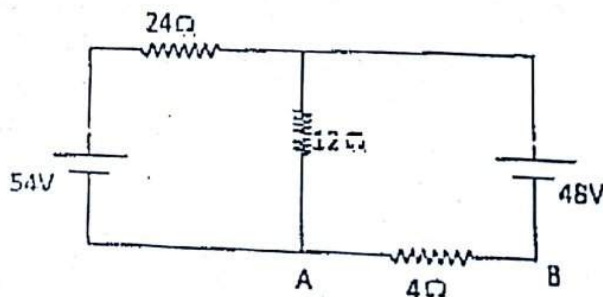
$$= 12\Omega$$

Hence, current through branch AB is,

$$I_{AB} = I_N \cdot \frac{R_{eq}}{R_{eq} + R_L} = 1.5 \cdot \frac{12}{12 + 15} = 0.667 \text{ A}$$



8. Use superposition theorem to determine the current through the 4Ω resistors shown below. [PU 20055]



Solution:

D.C. Circuit Analysis

- a) Take 54V source only, (as shown in fig.(i))

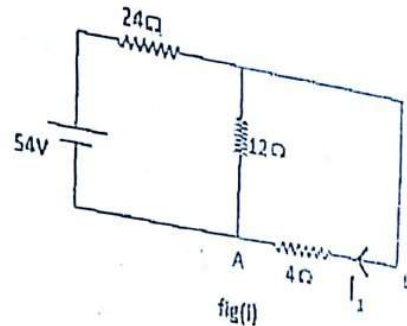
$$R_{eq} = (4 \parallel 12) + 24$$

$$= 27\Omega$$

$$I = \frac{V}{R_{eq}} = \frac{54}{27} = 2A$$

Current flowing through AB branch,

$$I_1 = I \times \frac{12}{12+4} = 2 \times \frac{12}{16} = 1.5 A \text{ (B to A)}$$



- b) Take 48V source only, (as shown in fig.(ii))

$$R'_{eq} = (12 \parallel 24) + 4$$

$$= 12\Omega$$

$$I' = \frac{V'}{R'_{eq}} = \frac{48}{12} = 4A$$

Current flowing through AB branch,

$$I_2 = 4A \text{ (A to B)}$$

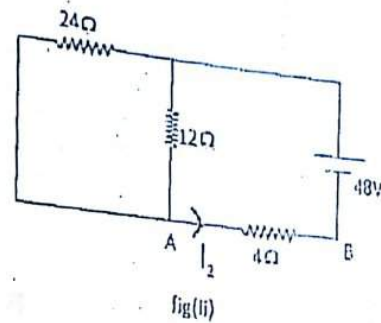
Now,

Total current flowing through 4Ω resistor is,

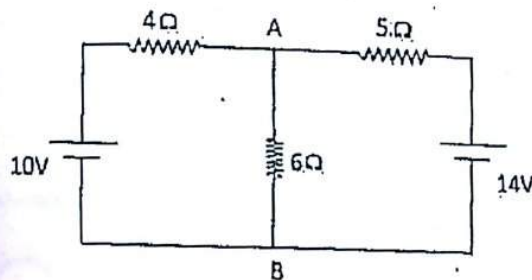
$$I_{4\Omega} = I_1 + I_2$$

$$= 1.5 \text{ (B to A)} + 4 \text{ (A to B)}$$

$$= 2.5 A \text{ (A to B)}$$



9. Use superposition theorem to determine the current through the 6Ω resistors in the figure shown below. [PU 2005F]



Solution:

- a) Take 10V source only, (as shown in fig.(i))

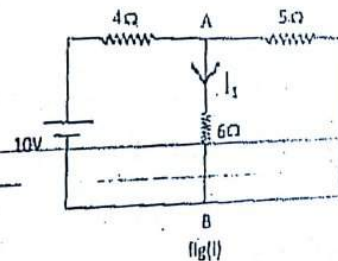
$$R_{eq} = (5 \parallel 6) + 4$$

$$= 6.72\Omega$$

$$I = \frac{V}{R_{eq}} = \frac{10}{6.72} = 1.48 A$$

Current flowing through AB branch,

$$I_1 = I \times \frac{5}{5+6} = 1.48 \times \frac{5}{11} = 0.672 A \text{ (A to B)}$$



D.C. Circuit Analysis

b) Take 14V source only, (as shown in fig. (ii))

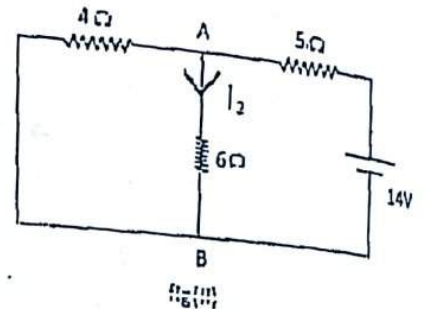
$$R'_{eq} = (4 \parallel 6) + 5$$

$$= 7.4 \Omega$$

$$I' = \frac{V'}{R'_{eq}} = \frac{14}{7.4} = 1.89 \text{ A}$$

Current flowing through AB branch,

$$I_2 = 1.89 \times \frac{4}{4+6} = 0.756 \text{ A (A to B)}$$



Now,

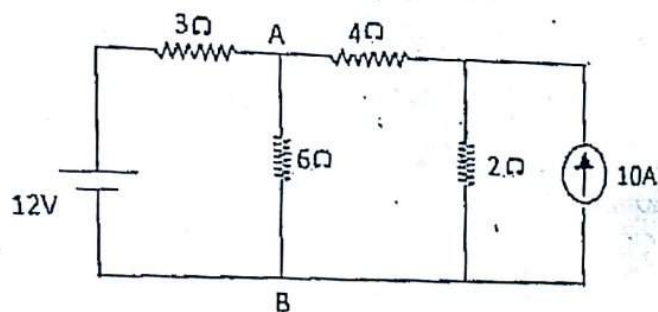
Total current flowing through AB branch is,

$$I_{AB} = I_1 + I_2$$

$$= 0.672 + 0.756$$

$$= 1.428 \text{ A (A to B)}$$

10. Using Thevenin's theorem to calculate the current flowing through 6Ω resistors, [PU 2010S]



Solution:

Here $R_L = 6 \Omega$

Calculation of V_{Th} :

From fig (i), using Mesh analysis,

In mesh 2,

$$I_2 = -10 \text{ A}$$

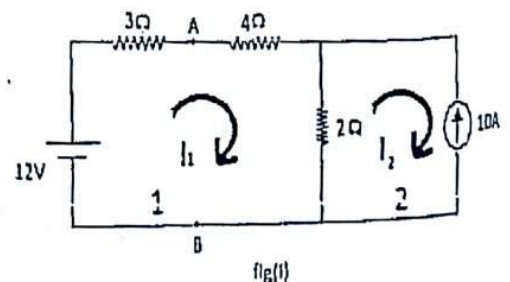
In mesh 1,

$$12 - 3I_1 - 4I_1 - 2(I_1 - I_2) = 0$$

$$12 - 9I_1 - 20 = 0$$

$$I_1 = -\{8/9\} \text{ A (i.e. Direction of current is in reverse direction)}$$

$$\text{Voltage across } 3\Omega \text{ resistor is } V_{3\Omega} = \frac{8}{9} \times 3 = \frac{8}{3} \text{ V}$$



D.C. Circuit Analysis

$$V_{Th} = 12 - \frac{6}{3} = 9.33V$$

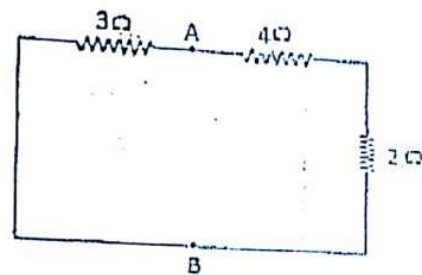
Calculation of R_{eq}

From fig (ii),

$$R_{eq} = R_{Th} = (4+2) \parallel 3 = \frac{6 \cdot 3}{6+3} = 2\Omega$$

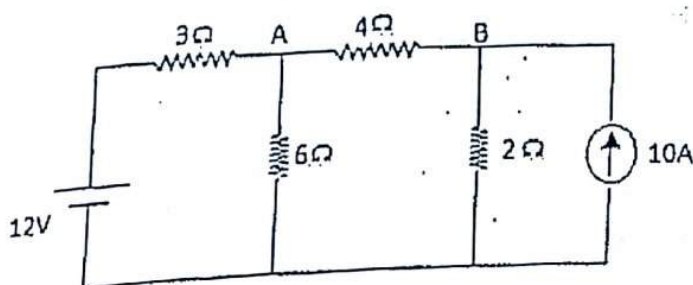
Now, current passing through the load resistance R_L is:

$$I_L = \frac{V_{Th}}{R_{eq} + R_L} = \frac{9.33}{2+6} = 1.166 A$$



fig(ii)

11. Utilizing Thevenin's theorem, find the current in 4Ω resistors in the circuit given below. [PU 2011F]



Solution:

Here $R_L = 4\Omega$

Calculation of V_{Th} :

From fig (i), using Nodal analysis,

At node 'a'

$$\frac{v_a - 12}{3} + \frac{v_a}{6} = 0$$

$$\Rightarrow 2v_a - 24 + v_a = 0$$

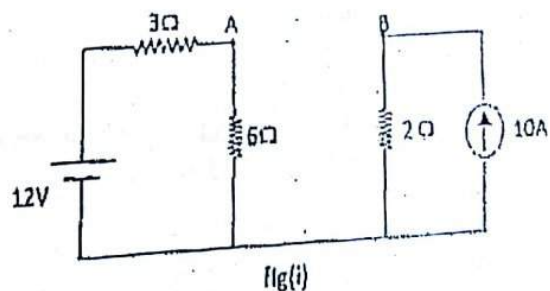
$$\Rightarrow 3v_a = 24$$

$$\Rightarrow v_a = 8V$$

At Node 'b'

$$10 - \frac{v_b}{2} = 0$$

$$V_b = 20V$$



fig(i)

D.C. Circuit Analysis

Now, $V_{Th} = V_b - V_a = 20 - 8 = 12V$

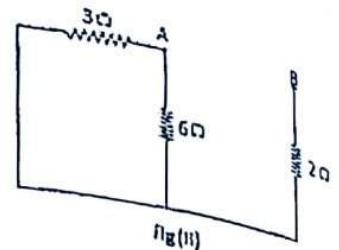
Calculation of R_{eq}

From fig (ii),

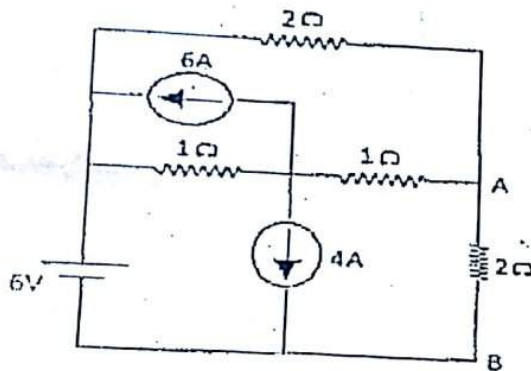
$$R_{eq} = R_{Th} = (3 \parallel 6) + 2 = 4\Omega$$

Now, current passing through the load resistance R_L is:

$$I_L = \frac{V_{Th}}{R_{eq} + R_L} = \frac{12}{4+4} = \frac{12}{8} = 1.5 A$$



12. By using Superposition theorem determine the current flowing through 2Ω resistors in the given circuit. [PU 2011F]



Solution:

- a) Take 6V source only, (as shown in fig.(i))

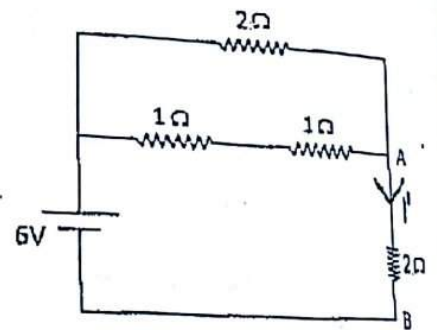
$$R_{eq} = (2 \parallel 2) + 2$$

$$= 3\Omega$$

$$I = \frac{V}{R_{eq}} = \frac{6}{3} = 2 A$$

Current flowing through AB branch,

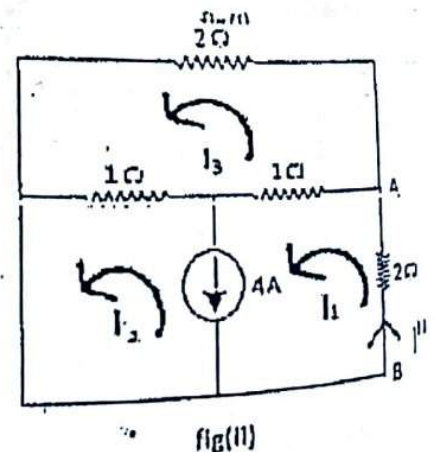
$$I' = 2 A (A \text{ to } B)$$



- b) Take 4A source only, (as shown in fig.(ii))

Using Mesh Analysis,

$$I_1 - I_2 = 4 \dots\dots\dots (i)$$



D.C. Circuit Analysis

In super mesh,

$$\begin{aligned} -2I_1 - I_1 + I_3 - I_2 + I_3 &= 0 \\ -3I_1 - I_2 + 2I_3 &= 0 \end{aligned} \quad \text{..... (ii)}$$

In mesh 3,

$$\begin{aligned} -2I_3 - I_3 + I_1 - I_3 + I_2 &= 0 \\ I_1 + I_2 - 4I_3 &= 0 \end{aligned} \quad \text{..... (iii)}$$

On solving (ii) & (iii) we get,

$$\begin{aligned} I_1 &= 0.66 \text{ A} & I_2 &= -3.33 \text{ A} & I_3 &= -0.66 \text{ A} \\ \therefore I'' &= I_1 = 0.66 \text{ A} & (B \text{ to } A) \end{aligned}$$

c) Take 4A source only, (as shown in fig.(iii).)

Using Mesh Analysis,

$$I_1 - I_2 = 4 \quad \text{..... (i)}$$

In super mesh,

$$\begin{aligned} -I_2 + I_1 - I_3 + I_1 - 2I_3 &= 0 \\ 2I_1 - I_2 - 3I_3 &= 0 \end{aligned} \quad \text{..... (ii)}$$

In mesh 1,

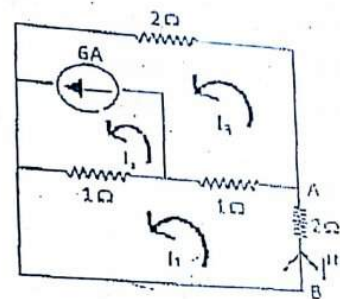
$$\begin{aligned} -2I_1 - I_1 + I_3 - I_1 + I_2 &= 0 \\ -4I_1 + I_2 + I_3 &= 0 \end{aligned} \quad \text{..... (iii)}$$

On solving (ii) & (iii) we get,

$$\begin{aligned} I_1 &= 1 \text{ A} & I_2 &= 5 \text{ A} & I_3 &= -1 \text{ A} \\ \therefore I''' &= I_1 = 1 \text{ A} & (B \text{ to } A) \end{aligned}$$

Now,

Total current flowing through AB branch is,

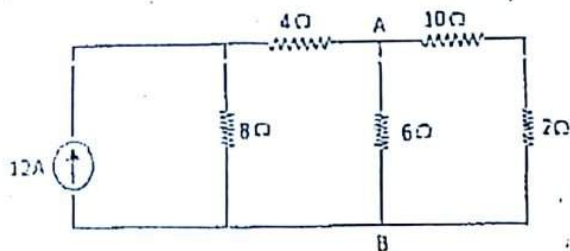


fig(iii)

D.C. Circuit Analysis

$$I_{AB} = I' - I'' - I''' \\ \approx 2 - 0.66 - 1 \\ = 0.34 \text{ A (A to B)}$$

13. Use Norton's theorem to calculate the current in 6Ω resistors in the given network. [PU 2006S, 2007S]



Solution:

Here $R_L = 6 \Omega$

Calculation of I_N

From fig (i),

$$I_1 = 12 \text{ A}$$

In mesh 2,

$$-8I_2 + 8I_1 - 4I_2 = 0$$

$$-12I_2 + 96 = 0$$

$$I_2 = 8 \text{ A}$$

In mesh 3,

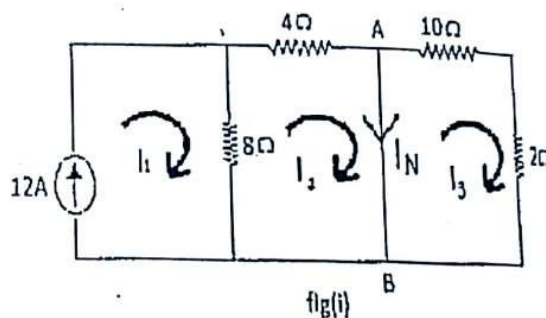
$$-10I_3 - 2I_3 = 0$$

$$I_3 = 0$$

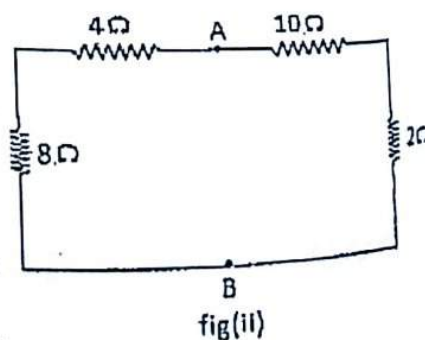
$$\text{Here, } I_N = I_2 - I_3 = 8 \text{ A}$$

Calculation of R_{eq}

From fig (ii),



fig(i)



fig(ii)

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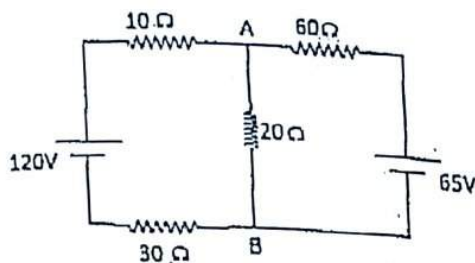
D.C. Circuit Analysis

$$R_{eq} = (12 \parallel 12) \\ = 6 \Omega$$

Hence, current through branch AB is,

$$I_{AB} = I_N \cdot \frac{R_{eq}}{R_{eq} + R_L} = 8 \cdot \frac{6}{8+6} = 4 \text{ A}$$

14. Use superposition theorem to find the current through 20Ω resistance in the following circuit. [PU 2006S]



Solution:

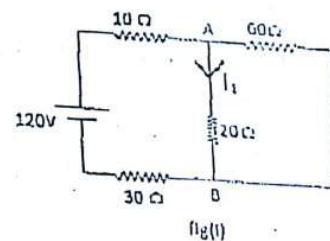
- a) Take 120V source only, (as shown in fig. (i))

$$R_{eq} = (60 \parallel 20) + 30 + 10 \\ = 55 \Omega$$

$$I = \frac{V}{R_{eq}} = \frac{120}{55} = 2.18 \text{ A}$$

Current flowing through AB branch,

$$I_1 = I \cdot \frac{60}{20+60} = 2.18 \cdot \frac{60}{80} = 1.635 \text{ A (A to B)}$$



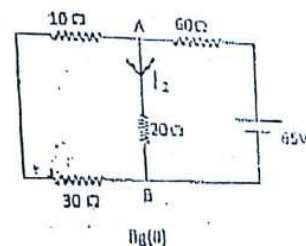
- b) Take 65V source only, (as shown in fig. (ii))

$$R'_{eq} = (40 \parallel 20) + 60 \\ = 73.34 \Omega$$

$$I' = \frac{V'}{R'_{eq}} = \frac{65}{73.34} = 0.88 \text{ A}$$

Current flowing through AB branch,

$$I_2 = 0.88 \cdot \frac{40}{40+20} = 0.58 \text{ A (A to B)}$$

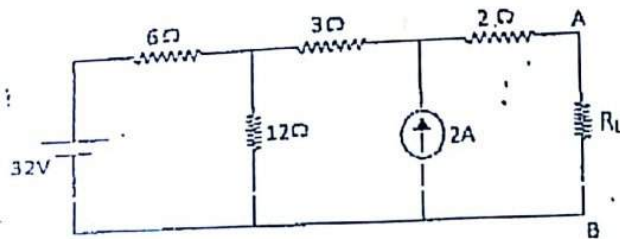


Now,

Total current flowing through AB branch is,

$$I_{AB} = I_1 + I_2 \\ = 1.635 + 0.58 \\ = 2.215 \text{ A (A to B)}$$

15. Find the value of R_L for maximum power transfer in the circuit below. [PU 2007S]



Solution:

Calculation of V_{Th} :

From fig (i), using Nodal analysis,

At node 'c'

$$\begin{aligned}
 -\left(\frac{v_c - 32}{6}\right) - \frac{v_c}{12} + \frac{v_c - v_d}{3} &= 0 \\
 \Rightarrow -2v_c + 64 - v_c - 4v_c + 4v_d &= 0 \\
 \Rightarrow -7v_c + 4v_d &= -64 \dots\dots\dots (i)
 \end{aligned}$$

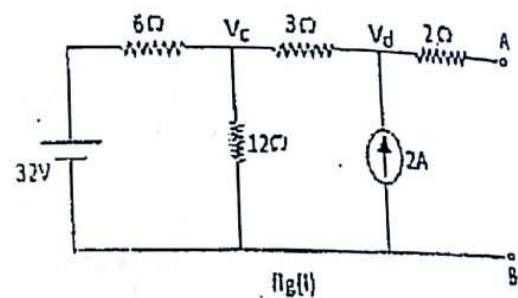
At node 'd'

$$\begin{aligned}
 -\left(\frac{v_d - v_c}{3}\right) + 2 &= 0 \\
 \Rightarrow -v_d + v_c &= -6 \\
 \Rightarrow v_c - v_d &= -6 \dots\dots\dots (ii)
 \end{aligned}$$

On Solving (i) & (ii) we get,

$$v_c = 29.33V \quad v_d = 35.33V$$

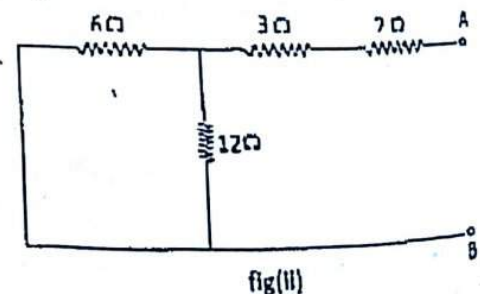
$$\text{Now, } V_{AB} = V_{Th} = v_d = 35.33V$$



Calculation of R_{eq}

From fig (ii),

$$R_{eq} = R_{Th} = (6 \parallel 12) + 5$$



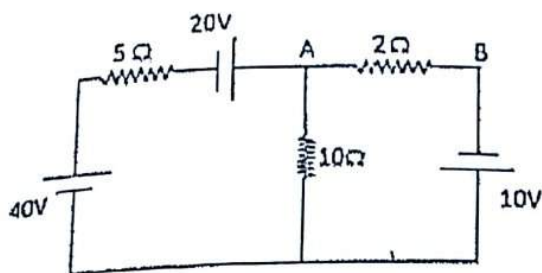
D.C. Circuit Analysis

$$= 9\Omega$$

∴ For maximum power transfer in ckt,

$$R_L = R_{Th} = 9\Omega$$

16. Determine the current flowing in 2Ω resistors by Thevenin's theorem. [PU 2009S]



Solution:

Here $R_L = 2\Omega$

Calculation of V_{Th} :

From fig (i), using Nodal analysis,

At node 'a'

$$\frac{v_a - 20 - 40}{5} + \frac{v_a}{10} = 0$$

$$\Rightarrow 2v_a - 40 - 80 + v_a = 0$$

$$\Rightarrow 3v_a = 120$$

$$\Rightarrow v_a = 40V$$

At node 'b'

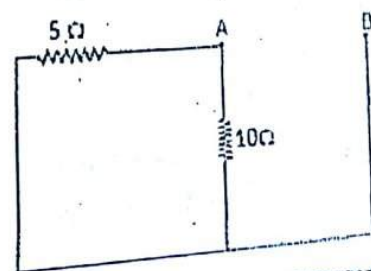
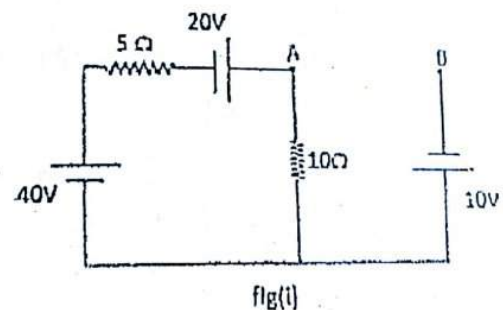
$$v_b = -10V$$

$$\text{Now, } v_{ab} = v_a - v_b = 40 - (-10) = 50V$$

Calculation of R_{eq}

From fig (ii),

$$R_{Th} = (5 \parallel 10) + \frac{5 \cdot 10}{5 + 10} = \frac{50}{15} = 3.34\Omega$$

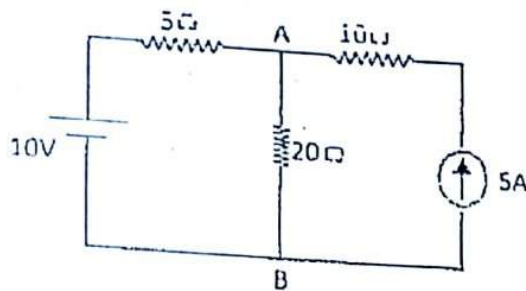


D.C. Circuit Analysis

Now, current passing through the load resistance R_L is :

$$I_L = \frac{V_{Th}}{R_{eq} + R_L} = \frac{50}{3.34 + 2} = 9.3738 \text{ A}$$

17. Apply Superposition theorem to find magnitude and direction of current through 20Ω resistors. [PU 2009S]



Solution:

- a) Take 10V source only, (as shown in fig.(I))

$$R_{eq} = 5 + 20$$

$$= 25 \Omega$$

$$I = \frac{V}{R_{eq}} = \frac{10}{25} = 0.4 \text{ A}$$

Current flowing through AB branch,

$$I_1 = I = 0.4 \text{ A (A to B)}$$

- b) Take 5A source only, (as shown in fig.(ii))

$$I_2 = 5 \times \frac{5}{5+20} = 1 \text{ A (A to B)}$$

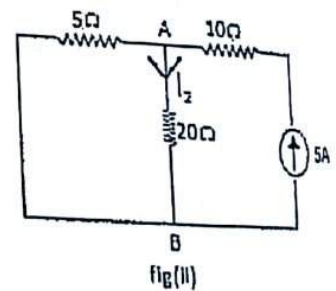
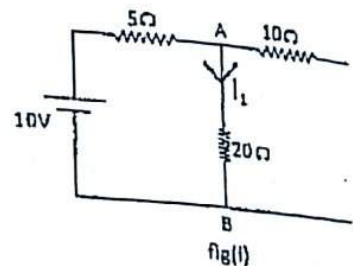
Now,

Total current flowing through AB branch is,

$$I_{AB} = I_1 + I_2$$

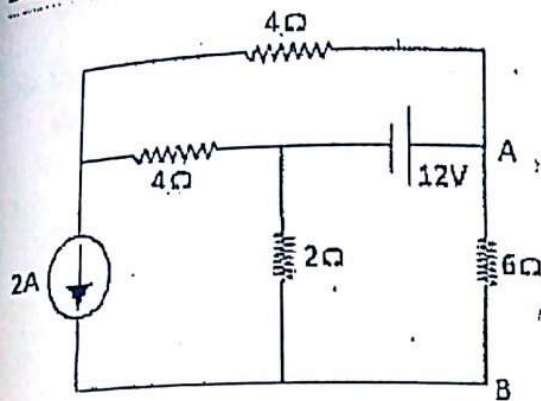
$$= 0.4 + 1$$

$$= 1.4 \text{ A (A to B)}$$



18. Use Thevenin's theorem to find the current flowing through the 6Ω resistors in the network shown below. [PU 2012S]

D.C. Circuit Analysis



Solution:

Here $R_L = 6\Omega$

Calculation of V_{Th} :

From fig (i),

$$V_{Th} = 12 - (2 \times 2)$$

$$= 12 - 4$$

$$= 8V$$

Calculation of R_{eq}

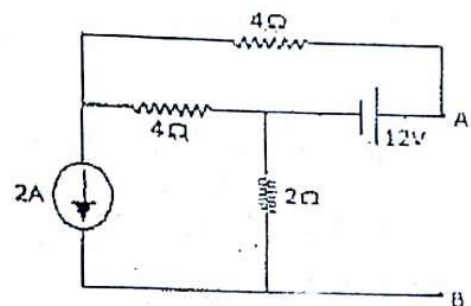
From fig (ii),

Clearly,

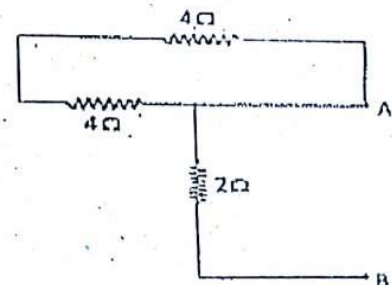
$$R_{eq} = R_{Th} = 2\Omega$$

Now, current passing through the load resistance R_L is:

$$I_L = \frac{V_{Th}}{R_{eq} + R_L} = \frac{8}{2 + 6} = 1A$$



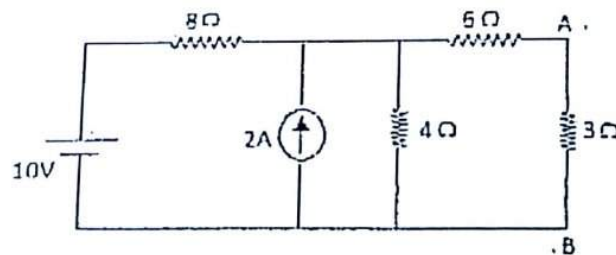
fig(i)



fig(ii)

19. Use Thevenin's theorem to calculate power consumed by 3Ω resistors in the network shown below. [PU 2013F]

D.C. Circuit Analysis



Solution:

Here $R_L = 3\Omega$

Calculation of V_{Th} :

From fig (I),

Since V_b is reference node,

$$\therefore V_b = 0$$

At node 'a':

$$\frac{v_a - 10}{8} + \frac{v_a - 0 - v_b}{4} = 0$$

$$V_a - 10 + 2V_a - 16 - 0 = 0$$

$$3V_a = 26$$

$$V_a = 8.667V$$

$$\therefore V_{Th} = V_{ab} = V_a - V_b = 8.667 - 0 = 8.667V$$

Calculation of R_{eq}

From fig (ii),

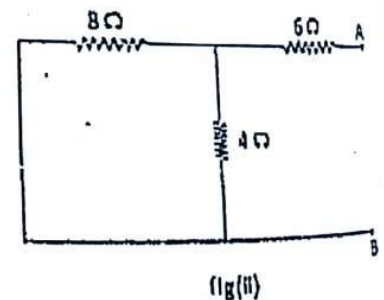
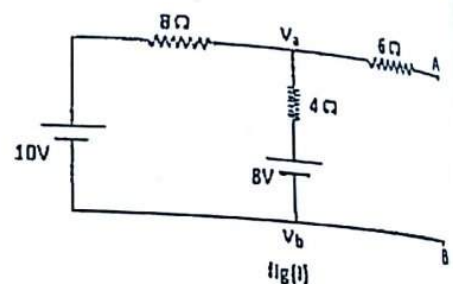
$$R_{Th} = (8 \parallel 4) + 6 = 8.667\Omega$$

Now, current passing through the load resistance R_L is:

$$I_L = \frac{V_{Th}}{R_{eq} + R_L} = \frac{8.667}{8.667 + 3} = 0.7428A$$

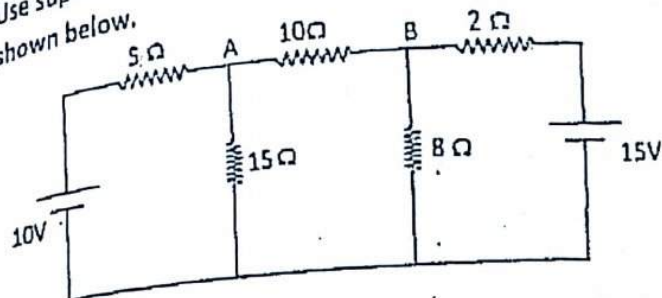
And, power consumed is:

$$P = I_L^2 \times R_L = 0.7428^2 \times 3 = 1.6555W$$



D.C. Circuit Analysis

20. Use superposition theorem to calculate the current through 10Ω resistor of the circuit shown below. [PU 2013F]



Solution:

a) Take 10V source only, (as shown in fig.(i))

$$R_{eq} = [(2 \parallel 8) + 10] \parallel 15 + 5$$

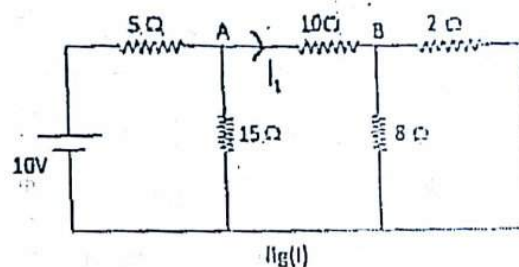
$$= 11.54\Omega$$

$$I = \frac{V}{R_{eq}} = \frac{10}{11.54} = 0.866 \text{ A}$$

Current flowing through AB branch,

$$I_1 = 0.866 \times \frac{15}{15 + (10 \parallel 8 \parallel 2)} = 0.866 \times \frac{15}{15 + 11.6}$$

$$= 0.488 \text{ A (A to B)}$$



b) Take 15V source only, (as shown in fig.(ii))

$$R'_{eq} = [(5 \parallel 15) + 10] \parallel 8 + 2$$

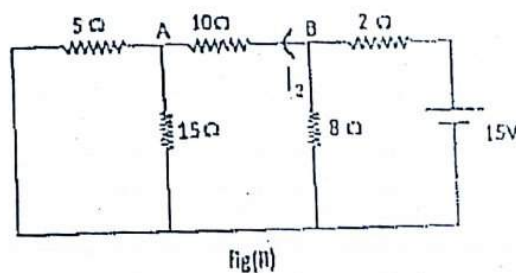
$$= 7.05\Omega$$

$$I' = \frac{V'}{R'_{eq}} = \frac{15}{7.05} = 2.12 \text{ A}$$

Current flowing through AB branch,

$$I_2 = 2.12 \times \frac{8}{8 + (10 \parallel 5 \parallel 15)} = 2.12 \times \frac{8}{8 + 13.75} = 0.779 \text{ A}$$

(B to A)



Now,

Total current flowing through 10Ω resistor is,

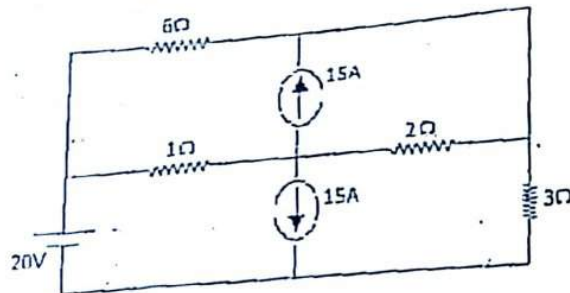
$$I_{0.5\Omega} = I_1 + I_2$$

$$= 0.488 \text{ (A to B)} + 0.779 \text{ (B to A)}$$

$$= 0.291 \text{ A (B to A)}$$

21. Use superposition theorem to calculate the current through 3Ω of the ckt shown in fig below. [PU 2013S]

D.C. Circuit Analysis



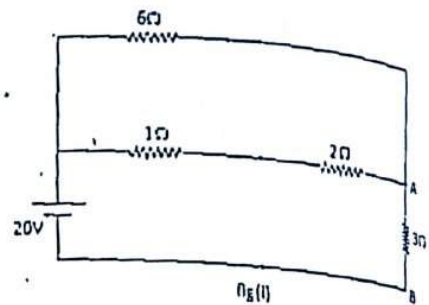
Solution:

- a) Take 20V source only, (as shown in fig.(i))

$$R_{eq} = (6 \parallel 3) + 3 = 5\Omega$$

$$I = \frac{V}{R_{eq}} = \frac{20}{5} = 4 \text{ A}$$

Current flowing through AB branch,
 $I' = 4 \text{ A (A to B)}$



- b) Take lower 15A source only, (as shown in fig.(ii))
 Using Mesh Analysis,

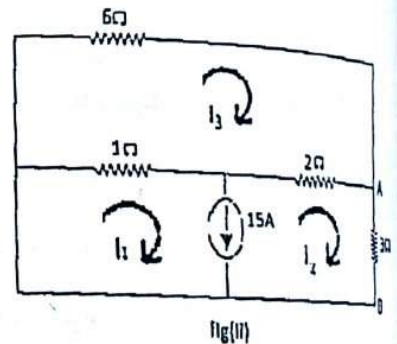
$$I_1 - I_2 = 15 \dots\dots\dots (i)$$

In super mesh,

$$\begin{aligned} -1(I_1 - I_3) - 2(I_2 - I_3) - 3I_2 &= 0 \\ -I_1 + I_3 - 2I_2 + 2I_3 - 3I_2 &= 0 \\ -I_1 - 5I_2 + 3I_3 &= 0 \dots\dots\dots (ii) \end{aligned}$$

In mesh 3,

$$\begin{aligned} -6I_3 - 2(I_3 - I_2) - 1(I_3 - I_1) &= 0 \\ -6I_3 - 2I_3 + 2I_2 - I_3 + I_1 &= 0 \\ I_1 + 2I_2 - 9I_3 &= 0 \dots\dots\dots (iii) \end{aligned}$$



On solving (i), (ii) & (iii) we get,

$$I_1 = 13 \text{ A} \qquad I_2 = -2 \text{ A} \qquad I_3 = 1 \text{ A}$$

$$\therefore I'' = I_2 = 2 \text{ A (B to A)}$$

D.C. Circuit Analysis

c) Take upper 15A source only, (as shown in fig(iii))
Using Mesh Analysis,

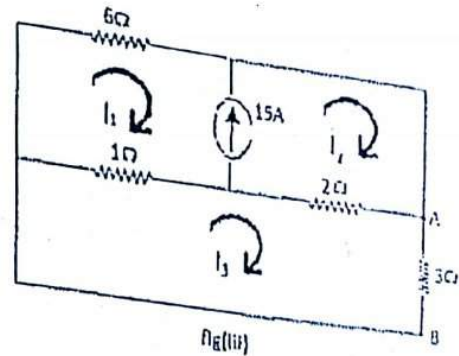
$$I_2 - I_1 = 15 \dots\dots\dots (i)$$

In super mesh,

$$\begin{aligned} -1(I_1 - I_3) - 6I_1 - 2(I_2 - I_3) &= 0 \\ -I_1 + I_3 - 6I_1 - 2I_2 + 2I_3 &= 0 \\ -7I_1 - 2I_2 + 3I_3 &= 0 \dots\dots\dots (ii) \end{aligned}$$

In mesh 3,

$$\begin{aligned} -1(I_3 - I_1) - 2(I_3 - I_2) - 3I_3 &= 0 \\ -I_3 + I_1 - 2I_3 + 2I_2 - 3I_3 &= 0 \\ I_1 + 2I_2 - 6I_3 &= 0 \dots\dots\dots (iii) \end{aligned}$$



On solving (i), (ii) & (iii) we get,

$$I_1 = -2 \text{ A} \quad I_2 = 13 \text{ A} \quad I_3 = 4 \text{ A}$$

$$\therefore I''' = I_3 = 4 \text{ A} \quad (\text{A to B})$$

Now,

Total current flowing through AB branch is,

$$\begin{aligned} I_{AB} &= I' + I'' + I''' \\ &= 4 \text{ (A to B)} + 2 \text{ (B to A)} + 4 \text{ (A to B)} \\ &= 6 \text{ A (A to B)} \end{aligned}$$

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Chapter-3 Single phase AC circuit Analysis

Comparison of AC with DC:

Alternating current (AC)	Direct current (DC)
- Both voltage and current changes periodically. - Low cost of production. - AC voltages can be lowered or raised by using transformers. - AC ckt current can be decreased by using choke or capacitor. - AC can be converted into DC by using converter. - AC cannot be used directly for electroplating, electrotyping, etc. - Generally its frequency is 50Hz. - It is more dangerous than DC at the same voltage rating.	- Both voltage and current remain constant. - Higher cost of production. - No such provision can be made. - DC ckt current can be decreased by using resistance. - DC can be converted into AC by using Inverter. - DC can be used directly for electroplating, electrotyping, etc. - Its frequency is zero. - It is less dangerous than AC at the same voltage rating.

Sinusoidal waveform: The sine wave is a very common type of alternating current and alternating voltage. It is known as sinusoidal waveform. The electric supply provided by the power companies is in the form of sinusoidal voltage and current.

Generation of alternating voltage

- Alternating voltages can be generated by rotating a magnetic field within a stationary coil or rotating a stationary coil in a magnetic field.
- The value of voltage generated depends upon the number of turns in a coil, strength of field and speed at which the coil or magnetic field rotates.
- Faraday's laws are applicable for generation of alternating voltage.

Faraday's laws of electromagnetic Induction:

1. First law: According to first law of electromagnetic induction "if there is change in magnetic flux linked coil, an emf induced in it which lasts so long as change in magnetic flux continues".

2. Second law: According to second law of electromagnetic induction "magnitude of emf induced in coil is directly proportional to the rate of change of magnetic flux linked with coil (flux linkage)", i.e. $E \propto \frac{d\phi}{dt}$

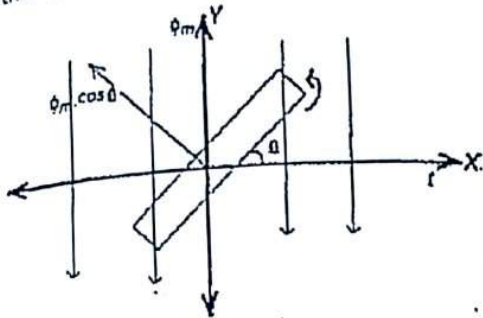
$$\text{Or, } E = -N \frac{d\phi}{dt} \text{ (where } d\phi = \phi_1 - \phi_2 \text{)}$$

A.C. Circuit Analysis

(Where -ve sign indicates that induced emf opposes the further change in magnetic flux.)

Equation of alternating voltage and current:

Consider a rectangular coil having 'N' turns and rotating in a uniform magnetic field with an angular velocity ω radian/second. Let time be measured from X-axis. The maximum flux ϕ_m is linked with the coil when its plane coincides with the X-axis. In time 't' seconds the coil rotates through an angle $\theta = \omega t$.



In this deflected position the component of flux which is perpendicular to the plane of coil is

$$\phi = \phi_m \cos \omega t$$

According to Faraday's law, $E = -N \frac{d\phi}{dt} = -N \frac{d(\phi_m \cos \omega t)}{dt}$

$$\therefore E = \omega N \phi_m \sin \theta \quad (\text{since } \theta = \omega t)$$

➤ When coil has turned through 90° then $\sin \theta = 1$

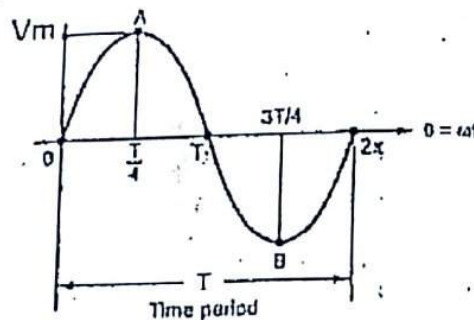
$$\therefore E_{\max} = \omega N \phi_m$$

$$\therefore E = E_{\max} \sin \theta$$

Similarly we can obtain equation for alternating current, $I = I_m \sin \omega t = I_m \sin \theta$

A.C. Terminology:

- 1) **Cycle:** One complete set of change in value and direction of alternating quantity is called cycle.
- 2) **Time period or periodic time (T):** It is time taken to complete one cycle. For e.g.: 50 Hz ac has a time of 1/50 second.



- 3) Frequency (f): It is the number of cycles occurred in one second. The frequency of an ac is reciprocal of the time period. i.e. $f = 1/T$.
- 4) Instantaneous value: The value of an alternating quantity at particular Instant (time) in a cycle is called instantaneous value. It is given by

$$i = I_m \sin \omega t$$

$$V = V_m \sin \omega t \quad \text{where } I_m = \text{peak current and } V_m = \text{peak voltage}$$

- 5) Peak value (or amplitude): The maximum value of an alternating current or voltage during the cycle is known as its peak value. It is denoted by I_m, V_m .
- 6) Average (or mean) value: The average value of an alternating quantity is defined as sum of all instantaneous values divided by number of values taken over that time or interval. It is denoted by I_{av}, V_{av} . i.e.

$$I_{av} = \frac{I_1 + I_2 + I_3 + \dots + I_n}{n} \quad (\text{This is known as mid ordinate method})$$

- Also the average value of any cycle of wave form is area under the wave form divided by time period (i.e. length of cycle), i.e.

$$I_{av} = \frac{1}{T} \int_0^T i dt \quad \text{and } V_{av} = \frac{1}{T} \int_0^T V dt \quad (\text{Integration method})$$

Note:

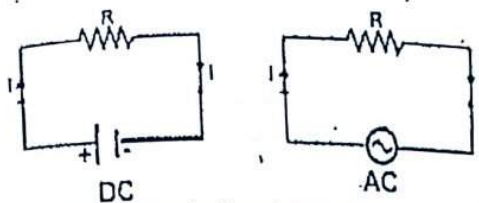
Since the area under the sine wave over a complete cycle is zero, because positive and negative area are equal and opposite. So in such case average value of an ac is calculated only over half cycle, i.e.

$$I_{av} = \frac{1}{\pi} \int_0^{\pi} i dt$$

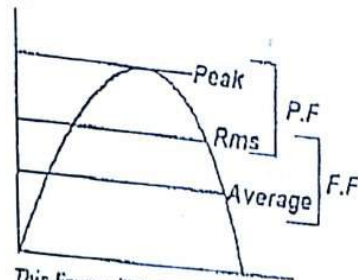
- 7) Effective or root mean square value (rms): The rms value of an ac is defined as steady dc current which when allowed to flow through a given circuit for a given time produces the

A.C. Circuit Analysis

same amount of heat as produced by alternating current, when allowed to flow through the same circuit for the same time. i.e.



Here same amount of heat produced in same time.



This figure will help to remember peak factor n form factor

Average power due to dc = power due to ac

$$\frac{1}{T} \int_0^T R i^2 dt = I^2 R$$

$$\Rightarrow I^2 = \frac{1}{T} \int_0^T i^2 dt$$

$$I_{rms} = I_{eff} = \sqrt{\frac{1}{T} \int_0^T i^2 dt}$$

Similarly,

$$V_{rms} = \sqrt{\frac{1}{T} \int_0^T v^2 dt}$$

$$\text{Form factor (F.F.)} = \frac{\text{Rms value}}{\text{Average value}}$$

$$\text{Peak factor (P.F.)} = \frac{\text{Peak (max.) value}}{\text{Rms value}}$$

Note: Peak factor is also known as crest or amplitude factor.

8) **Phase:** Phase of an alternating quantity is the fraction of the time period or cycle that has elapsed since it last passed from the chosen zero position on origin. The phase at 't' from the chosen origin is given by 't/T', where T is time period.

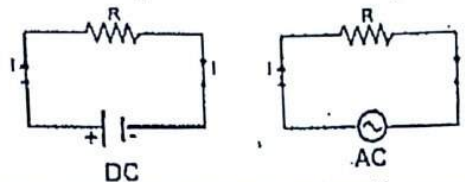
9) **Phase angle (ϕ):** It is equivalent of phase in radians or degrees. Thus phase angle is $\frac{2\pi t}{T}$.

10) **Phase difference:** It is time difference or angle difference between two sine waves since they last passed through their zero values.

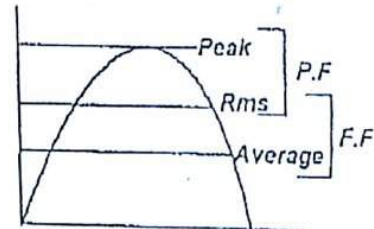
- Two quantities are **in phase**, when each pass through their zero values at same instant and also attain their maximum value at same instant in a given cycle. i.e.
- Two quantities are **out of phase**, if they reach their maximum or minimum value at different time but always has equal phase angle between them. i.e.

A.C. Circuit Analysis

same amount of heat as produced by alternating current, when allowed to flow through the same circuit for the same time. i.e.



Here same amount of heat produced in same time.



This figure will help to remember peak factor n form factor

Average power due to dc = power due to ac

$$\frac{1}{T} \int_0^T R i^2 dt = I^2 R$$

$$\Rightarrow I^2 = \frac{1}{T} \int_0^T i^2 dt$$

$$I_{rms} = I_{eff} = \sqrt{\frac{1}{T} \int_0^T i^2 dt}$$

Similarly,

$$V_{rms} = \sqrt{\frac{1}{T} \int_0^T V^2 dt}$$

$$\text{Form factor (F.F.)} = \frac{\text{Rms value}}{\text{Average value}}$$

$$\text{Peak factor (P.F.)} = \frac{\text{Peak (max.) value}}{\text{Rms value}}$$

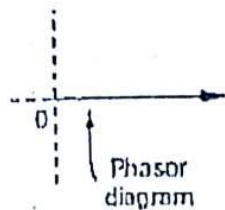
Note: Peak factor is also known as crest or amplitude factor.

- 8) **Phase:** Phase of an alternating quantity is the fraction of the time period or cycle that has elapsed since it last passed from the chosen zero position on origin. The phase at 't' from the chosen origin is given by 't/T', where T is time period.
- 9) **Phase angle (ϕ):** It is equivalent of phase in radians or degrees. Thus phase angle is $\frac{2\pi t}{T}$.
- 10) **Phase difference:** It is time difference or angle difference between two sine waves since they last passed through their zero values.
 - Two quantities are in phase, when each pass through their zero values at same instant and also attain their maximum value at same instant in a given cycle. i.e.
 - Two quantities are out of phase, if they reach their maximum or minimum value at different time but always has equal phase angle between them. i.e.

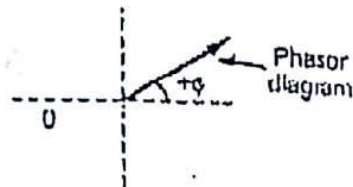
A.C. Circuit Analysis

Note: A plus (+) sign is used, when phase difference denotes lead or leading; while a minus sign (-) denotes lag or lagging.

- 11) **Peak to peak value:** The full voltage between positive and negative peaks of the wave form (i.e. the sum of magnitude of positive and negative peaks) is called peak to peak value.
- 12) **Phasor diagram:** It is one in which different alternating quantities (sinusoidal) of the same frequency are represented by phasors with their correct phase relationship, i.e.



(a) Zero phase



(b) Positive phase

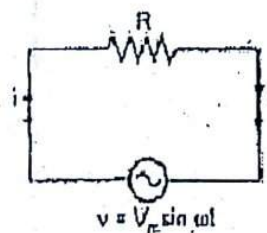


(c) Negative phase

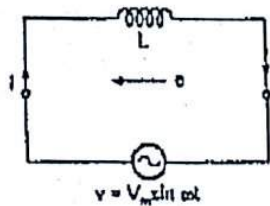
Note: The angle between the two phasors is the phase difference between the two emfs. For draw the phasor diagram, the following points should be taken:

- It is necessary to include X and Y axes.
- Normally phasor are drawn to represent rms values.
- Reference phasor is drawn horizontally.
- The phase is measured with respect to reference direction, i.e. w.r.t +ve X axis.
- Phasor are assumed to rotate in anti-clockwise direction.

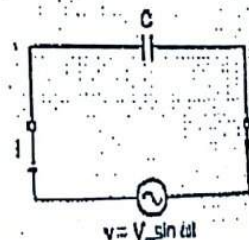
Steady state response of circuits:



a) Pure resistive circuit



b) Purely Inductive circuit

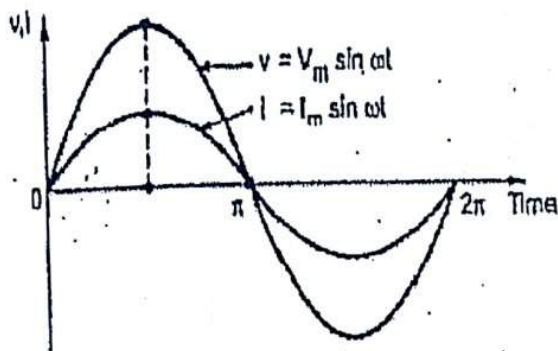


c) Purely capacitive circuit

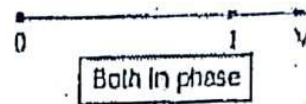
1. AC through purely resistive circuit:

- Circuit contains pure ohmic resistor only.
- In purely resistive circuit alternating voltage and alternating current are in phase with each other and hence, both of these can be represented by sine waves.

A.C. Circuit Analysis



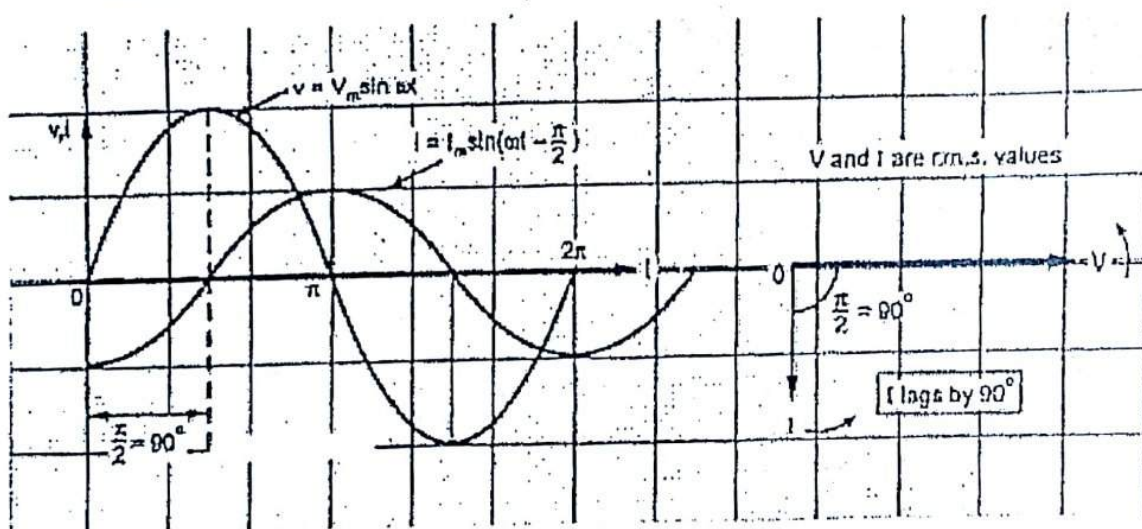
[Fig: wave form]



[Fig: phasor diagram]

2. AC through purely inductive circuit:

- In an ac circuit containing inductance only; current lags voltage by $\frac{\pi}{2}$ radians or 90 degree.



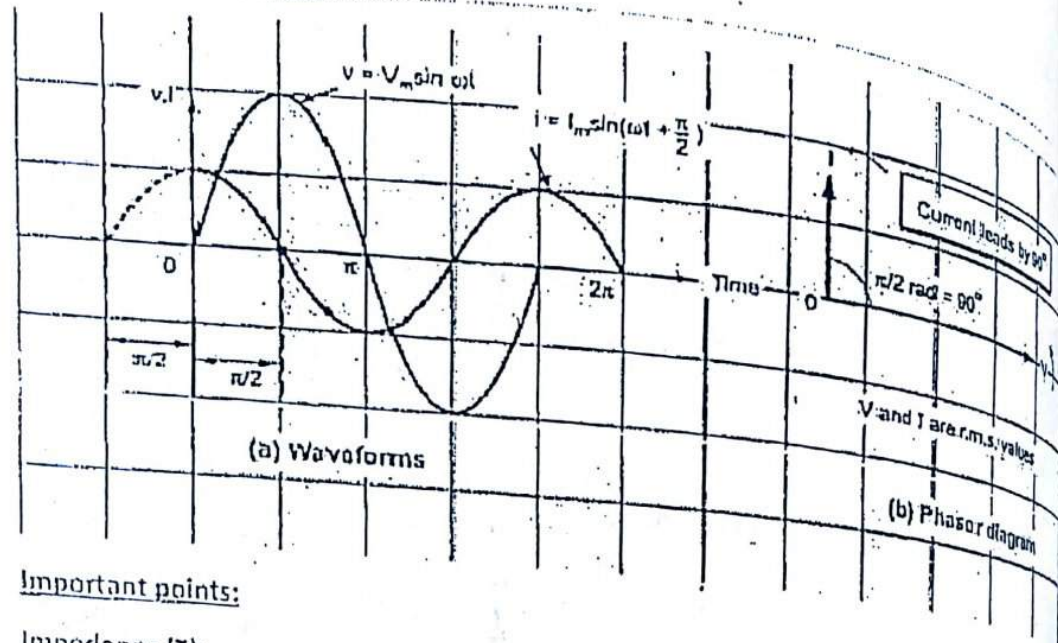
[Fig: wave form]

[Fig: phasor diagram]

3. AC through purely capacitive circuit:

- In this case current leads the applied voltage by $\frac{\pi}{2}$ radians or 90 degree.

A.C. Circuit Analysis



Important points:

Impedance (Z):

The total resistance offered by AC circuit to the flow of alternating current is called Impedance of circuit. It is denoted by Z. for e.g. the impedance of RLC series circuit is given by

$$Z = \sqrt{R^2 + (X_L - X_C)^2}$$

Inductive reactance (X_L):

The resistance offered by inductor to the flow of alternating current is called inductive reactance. It is denoted by X_L and given by $X_L = \omega L = 2\pi f L$.

Capacitive reactance (X_C):

The resistance offered by capacitor to the flow of alternating current is called capacitive reactance. It is given by

$$X_C = \frac{1}{\omega C} = \frac{1}{2\pi f C}$$

Admittance (Y):

The admittance of circuit is defined as the reciprocal of its Impedance. If Z is Impedance of circuit then admittance of circuit is given by $Y = \frac{1}{Z}$.

1. Voltage triangle
2. Impedance triangle

A.C. Circuit Analysis

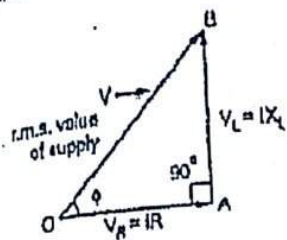


Fig. (a) Voltage triangle

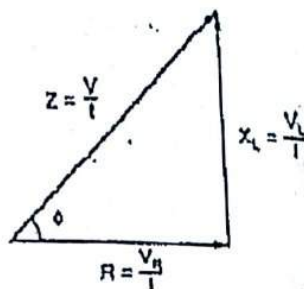


Fig. (b) Impedance triangle

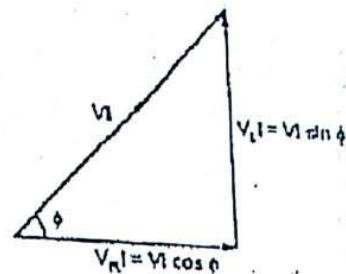


Fig. (c) Power triangle

3. Power triangle:

Unit

$P = \text{Active power (True or real power)} = VI \cos \phi$ kW

$Q = \text{Reactive power} = VI \sin \phi$ kVAR

$S = \text{Apparent power} = VI$ kVA

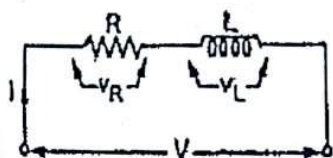
It is also called average power.

Note:

Active component (or wattful component) $= I \cos \phi$

Reactive component (or wattless/ idle component) $= I \sin \phi$

4. AC through Series R-L circuit:



$$\text{Impedance of circuit (Z)} = \sqrt{R^2 + X_L^2}$$

In rectangular form, $Z = R + jX_L$

In polar form, $Z = |Z| \angle \phi$

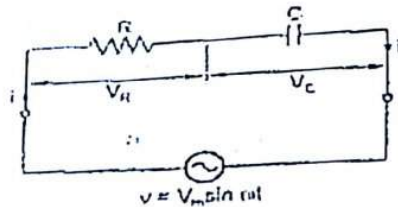
$$\text{where } |Z| = \sqrt{R^2 + X_L^2}$$

$$\tan \phi = \frac{X_L}{R} = \frac{\omega L}{R}$$

$$\phi = \tan^{-1} \left(\frac{\omega L}{R} \right)$$

5. AC through series R-C circuit:

A.C. Circuit Analysis



Impedance of circuit (Z) = $\sqrt{R^2 + X_C^2}$

In rectangular form, $Z = R - jX_C$

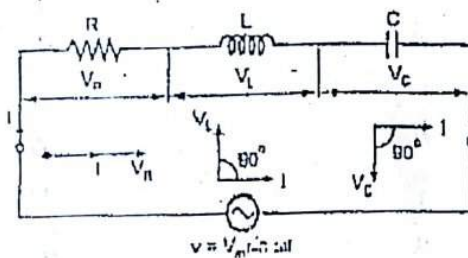
In polar form, $Z = |Z| \angle -\phi$

where $|Z| = \sqrt{R^2 + X_C^2}$

$\tan \phi = (-\frac{X_C}{R})$

$\phi = \tan^{-1}(-\frac{X_C}{R})$

6. AC through R-L-C series circuit:



[Fig: R-L-C series ckt]

In this case impedance of circuit (Z) = $\sqrt{R^2 + (X_L - X_C)^2}$

In rectangular form, $Z = R + j(X_L - X_C)$

$\phi = \tan^{-1} \frac{(X_L - X_C)}{R}$

Power in ac circuit:

Let $V = V_m \sin \omega t$ and $i = i_m \sin(\omega t + \phi)$ then

➤ Instantaneous power (P) = $V i = V_m i_m \sin \omega t \sin(\omega t + \phi)$

➤ Average power, $P_{av} = V_{rms} \cdot i_{rms} \cdot \cos \phi$

Where, $V_{rms} = \frac{V_m}{\sqrt{2}}$ and $i_{rms} = \frac{i_m}{\sqrt{2}}$

Note:

➤ For purely resistive circuit, $\phi = 0$ so $P_{av} = V_{rms} \cdot i_{rms}$ (i.e. maximum)

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➤ For purely inductive and capacitive circuit, $\phi = \pm 90^\circ$ so $P_{av} = 0$. (i.e. minimum)

[Power factor ($\cos\phi$) and significance of power factor: [2005F, 2008F, 2010F, 2012F, 2013S]

Power factor (P.f) is the cosine of angle of lead or lag. It is also the ratio of active power (P) to apparent power (S). Then

$$P.f = \cos\phi = \frac{\text{active power (in watts)}}{\text{apparent power (in volt ampere)}} = \frac{VI \cos\phi}{VI}$$

➤ Power factor is also equal to ratio of resistance (R) to the impedance (Z) of the circuit. i.e

$$\text{Power factor} = \frac{\text{Resistance (R)}}{\text{Impedance (Z)}} = 0 \text{ to } 1 \quad \cos\phi$$

Significance of power factor:

The power factor of an A.C. system should be as high as possible so that current and applied voltage are brought as near in phase as possible.

$$\text{Since } P = VI \cos\phi$$

$$\text{Or, } I = \frac{P}{V \cos\phi}$$

From this equation, it is seen that V is normally constant, hence for a given power P, the load current I varies inversely as the load power factor. Thus a given load takes more current at low power factor than it does at higher power factor; higher the power factor, lower the current and hence

- Requires small size conductor for transmission and distribution,
- Lower the line losses higher will be the system efficiency,
- As the voltage dropping cables decreases, voltage regulation increases.

Resonance in series RLC circuit:

[2004S, 2005S, 2009S, 2013F]

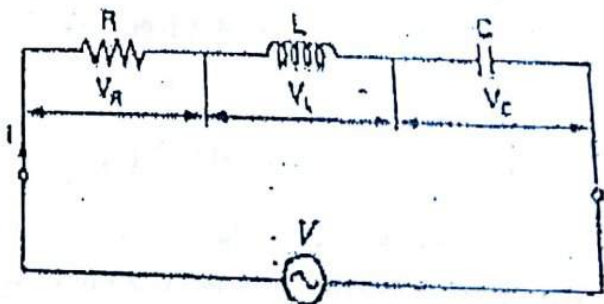
A series RLC circuit is said to be in resonance, when its net reactance (X) is zero. And frequency at which it happens is known as resonant frequency (f_0). Thus at resonance

$$X = X_L - X_C = 0$$

$$\text{Or, } X_L = X_C$$

$$\text{Or, } \omega L = \frac{1}{\omega C}$$

$$\text{Or, } 2\pi f_0 L = \frac{1}{2\pi f_0 C}$$



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Or, $f_0 = \frac{1}{2\pi\sqrt{LC}}$ = resonant frequency.

Note: At resonance:

- Net reactance, $X_L - X_C = 0$ so its Impedance $Z = \sqrt{R^2 + (X_L - X_C)^2} = \sqrt{R^2 + 0^2} = R$
i.e. $Z = R$ (i.e. circuit impedance is minimum)
- Circuit current is maximum i.e. $I = \frac{V}{Z} = \frac{V}{R}$
- Voltage drop across inductor is equal to voltage drop across capacitor, i.e. $V_L = V_C$
- Power factor of circuit is unity, i.e. $\cos\phi = \frac{R}{Z} = \frac{R}{R} = 1$.
- Hence power = VI (i.e. power dissipated is maximum)

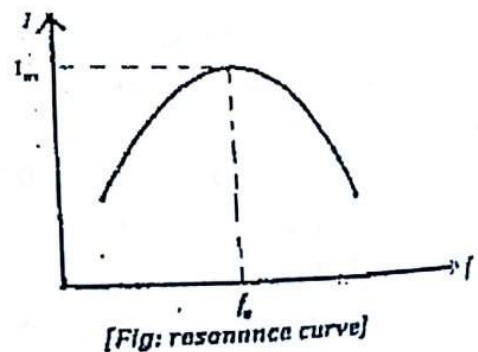
Variation of resistance, inductive reactance, capacitive reactance, impedance and current with frequency in RLC circuit series: [2013F]

- i. Resistance: it is independent of frequency; hence it is represented by straight line in graph.
- ii. Inductive reactance (X_L): since $X_L = 2\pi fL$ so X_L is directly proportional to f , (i.e. X_L linearly increases with frequency). Hence it is straight line passing through origin.
- iii. Capacitive reactance (X_C): since $X_C = \frac{1}{2\pi fC}$ so X_C is inversely proportional to f . Its graph is asymptote to horizontal axis.
- iv. Impedance (Z): we know, $Z = \sqrt{R^2 + (X_L - X_C)^2}$, at low frequencies Z is large because X_C is large. Also at high frequencies, Z is again large because X_L is large.
- v. Current (I): it is reciprocal of circuit Impedance so when Z is low I is high and vice versa.

Resonance curve:

[2009S]

The graphical relation between magnitude of current and frequency of applied emf is called resonance curve. From figure it is clear that current increases as the frequency increases. It becomes maximum at particular frequency; this frequency is called resonant frequency.



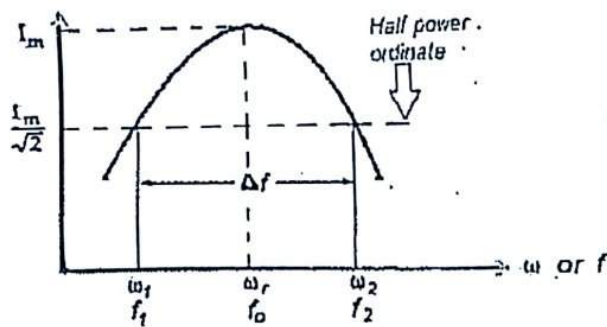
Bandwidth of series RLC circuit:

[2005S, 2008S, 2010F]

The bandwidth of series RLC circuit is defined as the range of frequency for which the power dissipated in the resistance is equal to or greater than power dissipated at resonance. Also

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bandwidth is the difference in the frequency of bands. i.e. $\Delta f = f_1 - f_2$



Where f_1 and f_2 are the frequencies at which power delivered to the circuit is half the power delivered at resonance.

$$\text{Power at resonance} = I_m^2 R$$

Let I_1 and I_2 be the currents at frequencies f_1 and f_2 respectively. Then by definition

$$I_1^2 R = I_2^2 R = \frac{1}{2} I_m^2 R$$

$$\text{Therefore, } I_1 = I_2 = \frac{I_m}{\sqrt{2}} \quad (1)$$

$$\text{Let } I_1 = I_2 = I \text{ then } I = \frac{I_m}{\sqrt{2}}$$

$$\begin{aligned} \text{Also for RLC circuit, } I &= \frac{V}{Z} = \frac{V}{\sqrt{R^2 + (X_L - X_C)^2}} \\ &= \frac{\frac{V}{R}}{\sqrt{1 + \frac{(X_L - X_C)^2}{R^2}}} = \frac{I_m}{\sqrt{1 + \frac{(X_L - X_C)^2}{R^2}}} \quad (2) \end{aligned}$$

Comparing equations (1) and (2), we get

$$\sqrt{2} = \sqrt{1 + \frac{(X_L - X_C)^2}{R^2}}$$

Squaring on both sides, we get

$$2 = 1 + \frac{(X_L - X_C)^2}{R^2} \quad \text{i.e. } R^2 = (X_L - X_C)^2$$

$$\text{Or, } \omega L - \frac{1}{\omega C} = \pm R \quad (3)$$

Taking negative sign only, we get

$$X_L = -R \quad \text{where } X_L = \omega_1 L = \frac{1}{\omega_1 C}$$

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Therefore, $\omega_1 L - \frac{1}{\omega_1 C} = -R$

Multiplying both sides by $\frac{\omega_1}{L}$, we have

$$\omega_1^2 - \frac{1}{LC} = -\frac{R}{L} \omega_1$$

Or, $\omega_1^2 + \frac{R}{L} \omega_1 - \frac{1}{LC} = 0$

The roots of equation are $\omega_1 = \frac{-\frac{R}{L} \pm \sqrt{(\frac{R}{L})^2 - 4 \cdot 1 \cdot (-\frac{1}{LC})}}{2 \cdot 1}$

$$\therefore \omega_1 = -\frac{R}{2L} \pm \sqrt{\left(\frac{R}{2L}\right)^2 + \frac{1}{LC}} \quad \dots\dots\dots (4)$$

Similarly taking positive sign from equation (3), we get

$$\omega_2 L - \frac{1}{\omega_2 C} = R$$

Multiplying both sides by $\frac{\omega_2}{L}$, we get

$$\omega_2^2 - \frac{R}{L} \omega_2 - \frac{1}{LC} = 0$$

$$\therefore \omega_2 = \frac{\left(\frac{R}{L}\right) \pm \sqrt{\left(\frac{R}{L}\right)^2 - 4 \cdot 1 \cdot \left(-\frac{1}{LC}\right)}}{2 \cdot 1}$$

Or, $\omega_2 = \frac{R}{2L} \pm \sqrt{\left(\frac{R}{2L}\right)^2 + \frac{1}{LC}} \quad \dots\dots\dots (5)$

Since negative frequency is meaningless so taking positive sign from equation (4) and (5) and subtract (4) and (5), we get

$$\begin{aligned} \omega_2 - \omega_1 &= \frac{R}{2L} + \sqrt{\left(\frac{R}{2L}\right)^2 + \frac{1}{LC}} - \left(-\frac{R}{2L} + \sqrt{\left(\frac{R}{2L}\right)^2 + \frac{1}{LC}}\right) \\ &= \frac{2R}{2L} = \frac{R}{L} \end{aligned}$$

Hence, $\Delta\omega = \frac{R}{L}$

Or, $2\pi\Delta f = \frac{R}{L}$

Therefore, $\Delta f = \frac{R}{2\pi L}$

Note: $f_1 = f_0 - \frac{R}{2\pi L}$

$$f_2 = f_0 + \frac{R}{2\pi L}$$

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Quality factor of series RLC ckt :

Quality factor 'Q' is defined as the ratio of resonant frequency to bandwidth of that circuit. i.e.

$$Q\text{-factor} = \frac{\text{resonant frequency}}{\text{bandwidth}}$$

$$\text{Or, } Q = \frac{\frac{1}{2\pi\sqrt{LC}}}{\frac{R}{2\pi L}} = \frac{1}{R} \sqrt{\frac{L}{C}}$$

Therefore,

$$Q = \frac{1}{R} \sqrt{\frac{L}{C}}$$

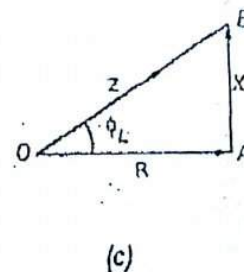
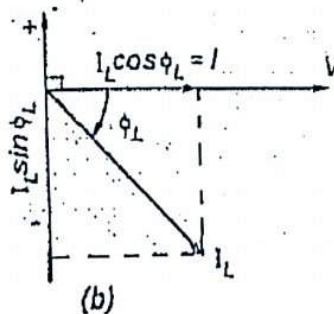
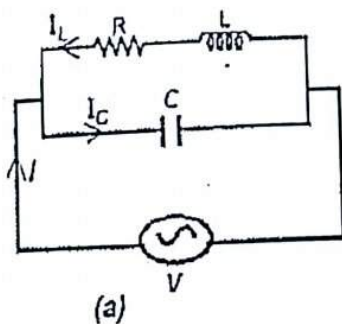
➤ It is also the ratio of Inductive reactance of coil (X_L) to its reactance (R). i.e. $Q = \frac{X_L}{R}$

Since at resonance, $X_L = X_C$ so $Q = \frac{X_C}{R}$.

Resonance in parallel RLC circuit:

[2012F]

Consider a practical case of parallel ac circuit, consisting of resistance (R) in series with an Inductor (L), both connected parallel to a capacitor (C) across ac voltage (V) as shown in figure. Such a circuit is said to be in resonance, when the reactive component of supplied current I is zero, and the corresponding frequency at which this happens, is called resonant frequency (f_0).



Reactive component of $I_L = I_L \sin \phi_L$

Net reactive component of circuit I is $I_C - I_L \sin \phi_L$

For resonance, net reactive component of current is zero. So $I_C - I_L \sin \phi_L = 0$

$$\text{Or, } I_C = I_L \sin \phi_L \dots \dots \dots (1)$$

$$\text{But } I_C = \frac{V}{X_C}, I_L = \frac{V}{Z} \text{ and } \sin \phi_L = \frac{X_L}{Z}$$

Therefore equation (1) becomes, $\frac{V}{X_C} = \frac{V}{Z} \times \frac{X_L}{Z}$

$$\text{Or, } X_L \cdot X_C = Z^2$$

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$$\text{Or, } \omega L \cdot \frac{1}{\omega C} = \left(\sqrt{R^2 + X_L^2} \right)^2$$

$$\text{Or, } R^2 + (2\pi f_0 L)^2 = \frac{L}{C}$$

$$\text{Or, } (2\pi f_0 L)^2 = \frac{L}{C} - R^2$$

$$\text{Hence } f_0 = \frac{1}{2\pi} \sqrt{\frac{1}{LC} - \frac{R^2}{L^2}}$$

If R is negligible (i.e. $R=0$) then $f_0 = \frac{1}{2\pi} \sqrt{\frac{1}{LC}}$

Bandwidth of parallel RLC circuit:

$$\text{Here, } Y = \frac{1}{Z_{eq}} = \frac{1}{R} + \frac{1}{X_L} + \frac{1}{X_C}$$

$$\text{Or, } Y = \frac{1}{R} + \frac{1}{j\omega L} + j\omega C$$

$$\text{Or, } Y = \frac{1}{R} - \frac{j}{\omega L} + j\omega C$$

$$\text{Or, } Y = \frac{1}{R} + j \left(\omega C - \frac{1}{\omega L} \right)$$

$$\therefore Z = \frac{1}{Y} = \frac{1}{\frac{1}{R} + j \left(\omega C - \frac{1}{\omega L} \right)}$$

At half power points, $P_1 = P_2 = \frac{1}{2} P$

$$\text{Or, } \frac{V_1^2}{R} = \frac{V_2^2}{R} = \frac{1}{2} \frac{V_m^2}{R}$$

$$\text{Or, } V_1 = V_2 = \frac{V_m}{\sqrt{2}}$$

$$\text{Or, } \frac{V_m}{\sqrt{2}} = I \times Z$$

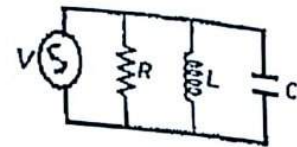
$$\text{Or, } \frac{V_m}{\sqrt{2}} = I \times \frac{1}{\sqrt{\left(\frac{1}{R} \right)^2 + \left(\omega C - \frac{1}{\omega L} \right)^2}}$$

$$\text{Or, } \frac{R}{\sqrt{2}} = \frac{1}{\sqrt{\left(\frac{1}{R} \right)^2 + \left(\omega C - \frac{1}{\omega L} \right)^2}}$$

Squaring and solving, we get

$$R = \pm \frac{1}{\left(\omega C - \frac{1}{\omega L} \right)}$$

$$\text{After doing some calculation, } \omega = \pm \frac{1}{RC} \pm \sqrt{\left(\frac{1}{2RC} \right)^2 + \frac{1}{LC}}$$



(since $\frac{1}{j} = \frac{j}{j \cdot j} = -j$)

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$$\text{Since } \frac{1}{2RC} \ll \frac{1}{LC} \text{ so } \omega = \pm \frac{1}{2RC} \pm \sqrt{\frac{1}{LC}}$$

Since negative frequency is meaningless, so

$$\omega = \pm \frac{1}{2RC} + \sqrt{\frac{1}{LC}}$$

$$\text{Therefore, } \omega_1 = \omega_r + \frac{1}{2RC} \quad \text{and} \quad \omega_2 = \omega_r - \frac{1}{2RC} \quad \text{where } \omega_r = \sqrt{\frac{1}{LC}}$$

$$\Delta\omega = \omega_1 - \omega_2 = \frac{1}{RC}$$

$$\text{Or, } 2\pi\Delta f = \frac{1}{RC}$$

Hence bandwidth of parallel RLC circuit is

$$\Delta f = \frac{1}{2\pi RC}$$

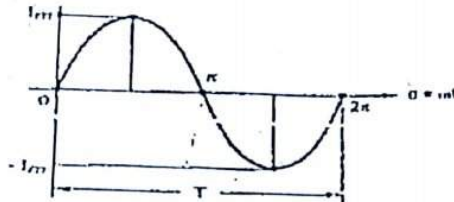
Points to Note:

1. The average value I_{av} of an alternating quantity over a half-cycles may be found by dividing the half cycles area by the length of its base. The area itself may be found by using the equation of the alternating quantity (if given) or geometrically.
2. The procedure for finding the rms value is as under:
 - i. Find the area under the curve of the square of the alternating quantity.
 - ii. Divide the area by length of the base.
 - iii. Take the square root of the result.
3. If the given sine waveform contains both DC and AC currents, then
 - i. Average value of the given waveform equals the sum of the dc current and the average of the alternating current.
 - ii. The rms value of a given wave-form containing AC and DC is given by:

$$I_{rms} = \sqrt{I_{dc}^2 + I_{rms}^2} \text{ where, } I_{dc} \text{ is the dc component and } I_{rms} \text{ is the r.m.s value of ac- component.}$$

Numericals asked in final exams:

1. Find the average, rms value, form factor and peak factor of ac over a complete cycle,



Or

Find the average value, effective value, form factor and peak factor of the sinusoidal waveform.
[2008F, 2009F, 2007S]

In sine wave I_{av} is taken only over half cycle because in complete cycle $I_{av} = 0$ so

$$I_{av} = \frac{1}{T} \int_0^T I dt$$

$$\begin{aligned} \text{Or, } I_{av} &= \frac{1}{\pi} \int_0^{\pi} I_m \sin \theta d\theta \\ &= \frac{I_m}{\pi} [-\cos \theta]_0^{\pi} \\ &= \frac{I_m}{\pi} [-\cos \pi - (-\cos 0)] \\ &= \frac{2I_m}{\pi} \end{aligned}$$

$\therefore I_{av} = 0.637 I_m$ i.e. $I_{av} = 63.7\%$ of peak current.

$$I_{rms} = \sqrt{\frac{1}{T} \int_0^T i^2 dt}$$

$$\text{Or, } I_{rms} = \sqrt{\frac{1}{\pi} \int_0^{\pi} (I_m \sin \theta)^2 d\theta}$$

$$\text{Or, } I_{rms} = \sqrt{\frac{I_m^2}{\pi} \int_0^{\pi} \frac{1 - \cos 2\theta}{2} d\theta}$$

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$$\text{Or, } I_{rms} = \sqrt{\frac{I_m^2}{2\pi} \left[\theta - \frac{\sin 2\theta}{2} \right]_0^\pi}$$

$$\text{Or, } I_{rms} = \sqrt{\frac{I_m^2}{2\pi} \left[\pi - \frac{\sin 2\pi}{2} - 0 + \frac{\sin 0}{2} \right]} = \sqrt{\frac{I_m^2}{2\pi} \times \pi}$$

$$= \frac{I_m}{\sqrt{2}} = 0.707 I_m$$

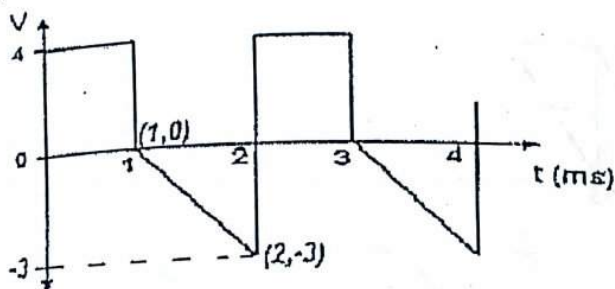
Therefore, $I_{rms} = 70.7\%$ of peak current.

$$\text{Form factor} = \frac{\text{Rms value}}{\text{Average value}} = \frac{0.707 I_m}{0.637 I_m} = 1.11$$

$$\text{Peak factor} = \frac{\text{Peak value}}{\text{Rms value}} = \frac{I_m}{0.707 I_m} = 1.41$$

2. Find the average value, effective value and form factor of the periodic waveform shown in figure. (Appropriate method)

[2007S]



Solution:

0 to 1, $V = 4$ i.e. $V_1 = 4$

1 to 2, $V = ?$ i.e. $V_2 = ?$

$T = 0$ to 2

Now for V_2

$$y - y_1 = \frac{y_2 - y_1}{x_2 - x_1} (x - x_1)$$

$$\text{Or, } y - 0 = \frac{-3 - 4}{2 - 1} (x - 1)$$

$$\text{Or, } y = -3x + 3$$

Here in x-axis there is time taken and in y-axis voltage is taken so

$$V = -3t + 3$$

$$V_{av} = \frac{1}{T} \int_0^T V dt$$

$$V_{av} = \frac{1}{2} \left[\int_0^1 4 dt + \int_1^2 (-3t + 3) dt \right]$$

$$= \frac{1}{2} \left\{ 4[t]_0^1 + \left[-\frac{3t^2}{2} + 3t \right]_1^2 \right\}$$

A.C. Circuit Analysis

$$= \frac{1}{2} \left[4 - 6 + 6 + \frac{3}{2} - 3 \right] = \frac{5}{4} = 1.25V$$

$$V_{rms} = \sqrt{\frac{1}{T} \int_0^T v^2 dt}$$

$$= \sqrt{\frac{1}{2} \left[\int_0^1 4^2 dt + \int_1^2 (-3t + 3)^2 dt \right]}$$

$$= \sqrt{\frac{1}{2} \left[16(1 - 0) + \int_1^2 (9t^2 - 18t + 9) dt \right]}$$

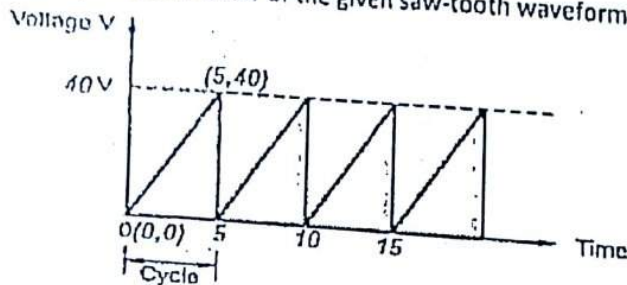
$$= \sqrt{\frac{1}{2} \left[16 \right] + \frac{1}{2} \left[9 \frac{t^3}{3} - 18 \frac{t^2}{2} + 9t \right]_1^2}$$

$$= \sqrt{8 + \frac{3}{2}}$$

$$= 3.08 \text{ volt}$$

3. Compute form factor of the given saw-tooth waveform.

[2012F]



Solution:

$$T = 5$$

$$y - 0 = \frac{40 - 0}{5 - 0} (x - 0)$$

$$\text{Or, } y = 8x$$

Here time is taken in x-axis and voltage is taken in y-axis so $V = 8t$

$$\text{Now average value, } V_{av} = \frac{1}{5} \int_0^5 8t dt$$

$$= \frac{1}{5} \left[\frac{8t^2}{2} \right]_0^5$$

$$= \frac{1}{5} \left[\frac{8 \cdot 5^2}{2} - \frac{8 \cdot 0^2}{2} \right]$$

$$= \frac{1}{5} [100] = 20V$$

$$\text{Also } V_{rms} = \sqrt{\frac{1}{5} \int_0^5 (8t)^2 dt} = \sqrt{\frac{1}{5} \left[\frac{64t^3}{3} \right]_0^5}$$

$$= \sqrt{\frac{1}{5} \left[\frac{64 \cdot 5^3}{3} - \frac{64 \cdot 0^3}{3} \right]} = \sqrt{\frac{1}{5} \left[\frac{64 \times 125}{3} \right]}$$

$$\text{Therefore, } V_{rms} = 23.09V$$

A.C. Circuit Analysis

Hence Form factor (F.F) = $\frac{\text{Rms voltage}}{\text{Average voltage}} = \frac{23.09}{20} = 1.15$

4. The output voltage wave of a Full wave rectifier is shown below. Find its average and root mean square (RMS) value. [2008S, 2006F, PU internal (2010)]

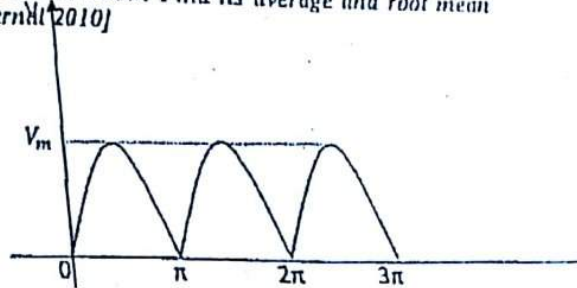
Solution:

Here, $V_{av} = \frac{1}{\pi} \int_0^{\pi} V_m \sin \theta \, d\theta$

$$= \frac{V_m}{\pi} [-\cos \theta]_0^{\pi}$$

$$= \frac{V_m}{\pi} [-\cos \pi - (-\cos 0)]$$

$$= \frac{2V_m}{\pi}$$



Therefore, $V_{av} = 0.637V_m$

Also, $V_{rms} = \sqrt{\frac{1}{\pi} \int_0^{\pi} (V_m \sin \theta)^2 \, d\theta}$

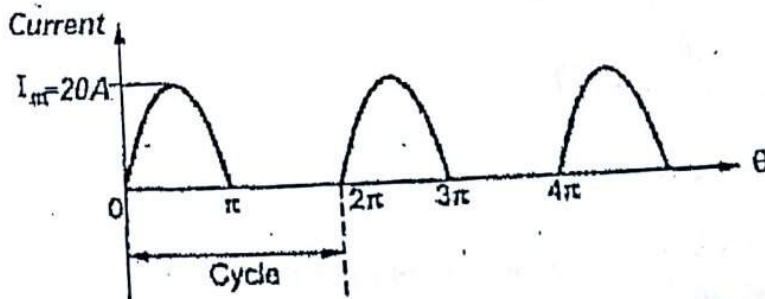
Or, $V_{rms} = \sqrt{\frac{V_m^2}{\pi} \int_0^{\pi} \frac{1 - \cos 2\theta}{2} \, d\theta}$

Or, $V_{rms} = \sqrt{\frac{V_m^2}{2\pi} \left[\theta - \frac{\sin 2\theta}{2} \right]_0^{\pi}}$

Or, $V_{rms} = \sqrt{\frac{V_m^2}{2\pi} \left[\pi - \frac{\sin 2\pi}{2} - 0 + \frac{\sin 0}{2} \right]} = \sqrt{\frac{V_m^2}{2\pi} \times \pi}$

$$= \frac{V_m}{\sqrt{2}} = 0.707V_m$$

5. Find rms value, average value, peak factor and form factor of half wave rectified wave whose maximum value is 20A. [2005F, PU internal 2013]



Or

Determine the average value and form factor of half wave rectified alternating current represented by $i = 20 \sin \omega t$. [2011F]

A.C. Circuit Analysis

Solution:

Here, 0 to π ; $I_m = 20A$

π to 2π ; $I = 0$

But period of wave, $T = 2\pi$

$$\begin{aligned}\text{Therefore, } I_{av} &= \frac{1}{2\pi} \left[\int_0^\pi I_m \sin \theta d\theta + \int_\pi^{2\pi} 0 d\theta \right] \\ &= \frac{1}{2\pi} \left[\int_0^\pi 20 \sin \theta d\theta \right] \\ &= \frac{1}{2\pi} 20 \times [-\cos \theta]_0^\pi \\ &= \frac{10}{\pi} [-\cos \pi - (-\cos 0)] \\ &= \frac{10}{\pi} [-(-1) - (-1)] = \frac{10}{\pi} \times 20 \\ &= 6.36A\end{aligned}$$

$$\begin{aligned}\text{Also, } I_{rms} &= \sqrt{\frac{1}{2\pi} \int_0^\pi (I_m \sin \theta)^2 d\theta + \int_\pi^{2\pi} 0^2 d\theta} \\ &= \sqrt{\frac{1}{2\pi} \int_0^\pi 20^2 \sin^2 \theta d\theta + 0} \\ &= \sqrt{\frac{1}{2\pi} \times 400 \int_0^\pi \left(\frac{1-\cos 2\theta}{2}\right) d\theta} \\ &= \sqrt{\frac{200}{\pi} \left[\theta - \frac{\sin 2\theta}{2}\right]_0^\pi} \\ &= \sqrt{\frac{100}{\pi} \left[\pi - \frac{\sin 2\pi}{2} - 0 + \frac{\sin 0}{2}\right]} = \sqrt{\frac{100}{\pi} [\pi - 0 - 0 + 0]} \\ &= \sqrt{100} = 10A\end{aligned}$$

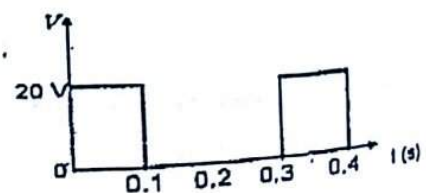
$$\text{Form factor (F.F)} = \frac{\text{Rms value}}{\text{Average value}} = \frac{10}{6.36} = 1.57$$

$$\text{Peak factor (P.F)} = \frac{\text{Peak (max.) value}}{\text{Rms value}} = \frac{20}{10} = 2$$

6. Compute the average and effective values of the square voltage wave shown in figure below. [2006S, 2013F]

Solution:

Since $0 < t < 0.1$ i.e. for the time interval 0 to 0.1 second,



A.C. Circuit Analysis

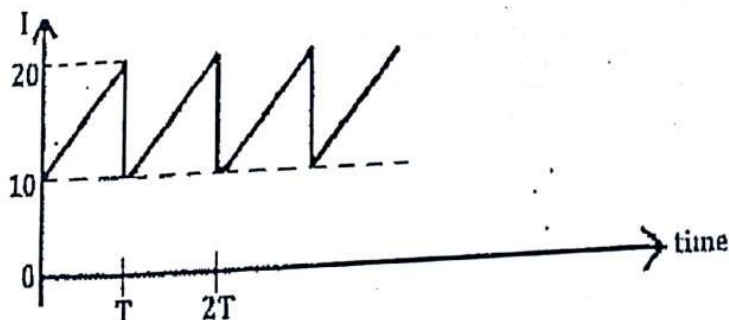
$V_m = 20V$. Similarly for 0.1 to 0.3, $V = 0$. Also time period of voltage wave is 0.3 second.

$$\begin{aligned} V_{av} &= \frac{1}{T} \int_0^T V dt = \frac{1}{0.3} \left[\int_0^{0.1} 20 dt + \int_{0.1}^{0.3} 0 dt \right] \\ &= \frac{1}{0.3} [20t]_0^{0.1} + 0 \\ &= \frac{20}{0.3} [0.1 - 0] \\ &= \frac{0.1 \times 20}{0.3} = 6.67V \end{aligned}$$

Also,

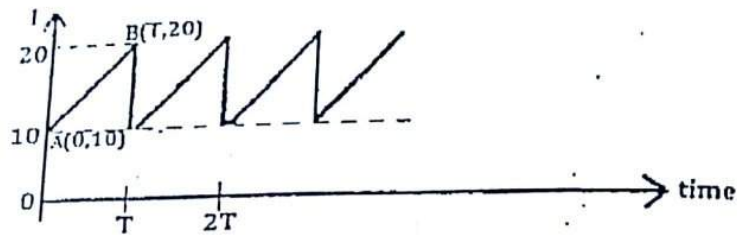
$$\begin{aligned} V_{rms} &= \sqrt{\frac{1}{T} \int_0^T V^2 dt} = \sqrt{\frac{1}{0.3} \int_0^{0.1} 20^2 dt} \\ &= \sqrt{\frac{1}{0.3} [400t]_0^{0.1}} \\ &= \sqrt{\frac{1}{0.3} [400 \times 0.1 - 400 \times 0]} \\ &= \sqrt{\frac{40}{0.3}} = \sqrt{133.33} \\ &= 11.54V \end{aligned}$$

7. Determine the rms and average value of waveform shown below. [20055]



Solution:

Method-1:



Equation of AB is; $I - 10 = \frac{20-10}{T-0} (t - 0)$ {since, $y - y_1 = \frac{y_2 - y_1}{x_2 - x_1} (x - x_1)$ }

$$\therefore I = 10 + (10/T)t$$

$$\begin{aligned} \text{Now, } I_{av} &= \frac{1}{T} \int_0^T \{10 + (10/T)t\} dt \\ &= \frac{1}{T} \left[10t + \frac{10}{T} \times \frac{t^2}{2} \right]_0^T = \frac{1}{T} \left[10T + \frac{5}{T} \times T^2 \right] \\ &= 15 \end{aligned}$$

$$\begin{aligned} I_{rms} &= \sqrt{\frac{1}{T} \int_0^T (10 + \frac{10}{T}t)^2 dt} \\ &= \sqrt{\frac{1}{T} \int_0^T (100 + \frac{200}{T}t + \frac{100}{T^2}t^2) dt} = \sqrt{\frac{1}{T} \left[100t + \frac{200t^2}{2T} + \frac{100t^3}{3T^2} \right]_0^T} \\ &= \sqrt{\frac{700}{3}} = 15.27A \end{aligned}$$

Method 2: area method

$$\begin{aligned} I_{av} &= 10 + \frac{\text{Area under full-cycle}}{\text{Base}} \\ &= 10 + \frac{\frac{1}{2} \times T \times 10}{T} = 15A \\ I_{rms} &= 10 + \sqrt{\frac{\frac{1}{3} \times T \times 10^2}{T}} = 15.77A \end{aligned}$$

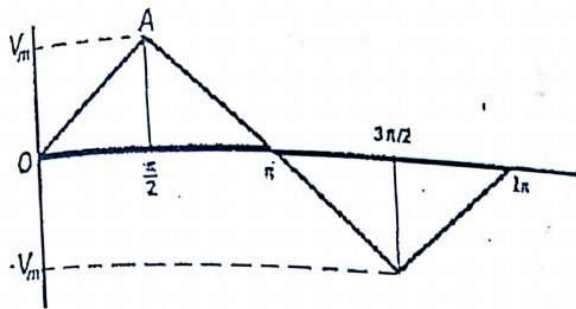
Some other important questions:

Find rms(effective) and average(mean) value of following waveforms:

Here we solve these type of question by area method, but we can also solve these question by using direct formula as above.

8.

A.C. Circuit Analysis



Solution:

For average and rms values; we need first-quarter of cycle (i.e. upto $\frac{\pi}{2}$) only.

$$\text{Now, } V_{av} = \frac{\text{Area under the wave for the first quarter-cycle}}{\text{Base}}$$

$$= \frac{\frac{1}{2} \times \frac{\pi}{2} \times V_m}{\frac{\pi}{2}} = 0.5V_m$$

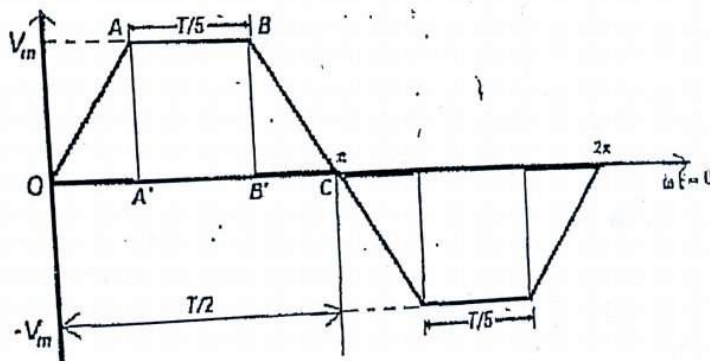
(since area of triangle = $\frac{1}{2}$ base $\times$ height)

$$V_{rms}^2 = \frac{\text{Area under squared wave}}{\text{Base}}$$

$$= \frac{\frac{1}{3} \times \frac{\pi}{2} \times V_m^2}{\frac{\pi}{2}} \quad (\text{Note: In case of triangle the area under squared wave} = \frac{1}{3} \text{ base} \times \text{height}^2)$$

$$\therefore V_{rms} = \sqrt{\frac{V_m^2}{3}} = \frac{V_m}{\sqrt{3}} = 0.577V_m$$

9. Find average and effective value of the trapezoidal voltage wave, shown below.
[PU Internal]



Solution:

$$\text{Linear part} = A'B' = T/5$$

$$\therefore OA' + B'C = \left(\frac{T}{2} - \frac{T}{5}\right) = 3T/10$$

$$\therefore OA' = B'C = 3T/20$$

A.C. Circuit Analysis

$$\text{Now, } V_{av} = \frac{\text{Area under the curve OABC}}{\text{Half-period}} = \frac{\left(\frac{1}{2} \times \frac{3T}{20} \times V_m\right) + \left(\frac{T}{5} \times V_m\right) + \left(\frac{1}{2} \times \frac{3T}{20} \times V_m\right)}{\frac{T}{2}} = \frac{V_m \times T \left(\frac{3}{40} + \frac{1}{5} + \frac{3}{40}\right)}{\frac{T}{2}}$$

$$\therefore V_{av} = 0.7V_m$$

$$V_{rms} = \sqrt{\frac{\text{Area under the squared curve of half cycle}}{\text{Half-period}}}$$

$$= \sqrt{\frac{\left(\frac{1}{2} \times \frac{3T}{20} \times V_m^2\right) + \left(\frac{T}{5} \times V_m^2\right) + \left(\frac{1}{2} \times \frac{3T}{20} \times V_m^2\right)}{\frac{T}{2}}}$$

$$= \sqrt{2V_m^2 \left(\frac{1}{20} + \frac{1}{5} + \frac{1}{20}\right)} = 0.7746V_m$$

(Note: area under squared wave of triangle = $\frac{1}{2} \times \text{base} \times \text{height}^2$ and area under squared wave of rectangle = $\text{base} \times \text{height}^2$)

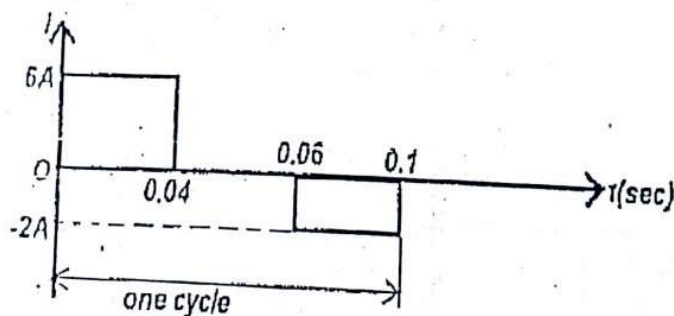
10. A varying current with a periodic waveform as shown in figure below flows through an 8Ω resistor. Determine the

- Average value
- Rms value
- Heat dissipated in 5 minute.

[2013S]

Solution:

Given, $R = 8\Omega$, $t = 5\text{min} = 5 \times 60 = 300\text{second}$



$$I_{av} = \frac{\text{Area under complete cycle}}{\text{Base}}$$

$$= \frac{0.04 \times 6 + 0.06 \times (-2)}{0.1} = 1.6 \text{ Ampere}$$

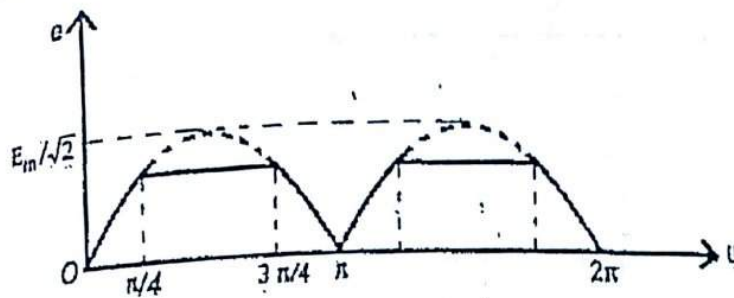
$$I_{rms} = \sqrt{\frac{\text{Area under squared wave}}{\text{Base}}}$$

A.C. Circuit Analysis

$$= \sqrt{\frac{0.04 \times 6^2 + 0 + 0.04 \times (-2)^2}{0.1}} = 4 \text{ Ampere}$$

$$\text{Heat dissipated in 5 minute} = I_{\text{rms}}^2 R t = 4^2 \times 8 \times 300 = 38400 \text{ Joule}$$

11. Calculate the average, and rms value of clipped full-rectified sine wave, shown below.



Solution:

$$E_{\text{av}} = \frac{\text{Area under half-cycle wave}}{\text{Half-period}}$$

$$= \frac{\int_0^{\pi/4} E_m \sin \theta d\theta + \frac{E_m}{\sqrt{2}} \left(\frac{3\pi}{4} - \frac{\pi}{4} \right) + \int_{3\pi/4}^{\pi} E_m \sin \theta d\theta}{\pi}$$

$$= \frac{2 \int_0^{\pi/4} E_m \sin \theta d\theta + \frac{E_m}{\sqrt{2}} \times \frac{\pi}{2}}{\pi}$$

$$= \frac{E_m}{\pi} \left[[-2 \cos \theta]_0^{\pi/4} + \frac{\pi}{2\sqrt{2}} \right]$$

$$= 0.54 E_m$$

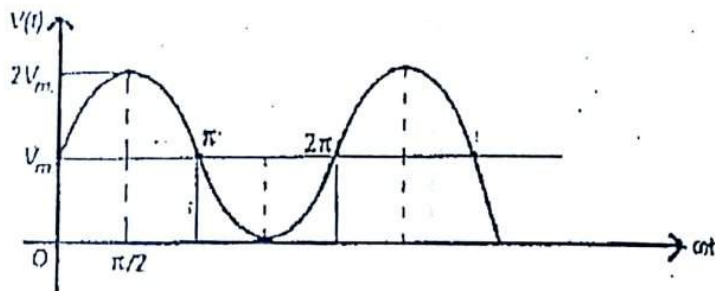
$$E_{\text{rms}} = \sqrt{\frac{[2 \int_0^{\pi/4} E_m^2 \sin^2 \theta d\theta + \frac{E_m^2}{2} \left(\frac{3\pi}{4} - \frac{\pi}{4} \right)]}{\pi}}$$

$$= \frac{E_m}{\sqrt{\pi}} \sqrt{\int_0^{\pi/4} (1 - \cos 2\theta) d\theta + \frac{\pi}{4}} = \frac{E_m}{\sqrt{\pi}} \sqrt{\left[\theta - \frac{\sin 2\theta}{2} \right]_0^{\pi/4} + \frac{\pi}{4}}$$

$$= \frac{E_m}{\sqrt{\pi}} \sqrt{\frac{\pi}{4} - \frac{1}{2} + \frac{\pi}{4}} = 0.584 E_m$$

12. Calculate the average and rms value in which the ac waveform is rising on d.c. with magnitude V_m .

A.C. Circuit Analysis



Solution:

Here the wave equation for given waveform is:

For d.c component; $V_{dc} = V_m$

For a.c component; $V_{ac} = V_m \sin \omega t$

Hence total wave equation is:

$$V = V_{dc} + V_{ac} = V_m + V_m \sin \omega t$$

$$= \frac{1}{2\pi} [V_m(\omega t) + V_m(-\cos \omega t)] \Big|_0^{2\pi}$$

$$= \frac{1}{2\pi} [V_m(2\pi - 0) + V_m(-\cos 2\pi + \cos 0)]$$

$$\therefore V_{av} = V_m$$

$$V_{rms} = \sqrt{\frac{1}{2\pi} \int_0^{2\pi} [V_m + V_m \sin \omega t]^2 d\omega t}$$

$$= \sqrt{\frac{1}{2\pi} \int_0^{2\pi} [V_m^2 + 2V_m^2 \sin \omega t + V_m^2 \sin^2 \omega t] d\omega t}$$

$$= \sqrt{\frac{1}{2\pi} \int_0^{2\pi} [V_m^2 + 2V_m^2 \sin \omega t + V_m^2 \frac{1 - \cos 2\omega t}{2}] d\omega t}$$

$$= \sqrt{\frac{1}{2\pi} [V_m^2 \{2\pi\} + 2V_m^2 \{0\} + \frac{V_m^2}{2} \{2\pi - 0\}]} = \sqrt{\frac{3}{2}} V_m$$

13. Find the root mean square value of the resultant current in a wire which simultaneously carries a current of $2\sqrt{2}$ A and sinusoidal current of peak value $2\sqrt{2}$ A [PU 2011 F]

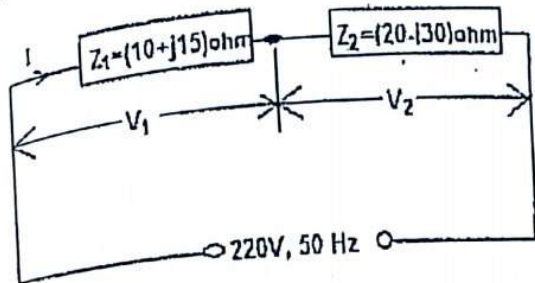
Solution:

$$I_{rms} = \sqrt{I_{dc}^2 + I_{rms}^2} = \sqrt{(2\sqrt{2})^2 + \left(\frac{I_m}{\sqrt{2}}\right)^2} = \sqrt{8 + \left(\frac{2\sqrt{2}}{\sqrt{2}}\right)^2} = \sqrt{8 + 4} = \sqrt{12} = 2\sqrt{3} \text{ A}$$

Another type of questions

1. For the circuit below, find: (i) current, (ii) V_1 and V_2 and (iii) power factor. Draw the phasor diagram.

A.C. Circuit Analysis



Solution:

(i) For a given series combination,

$$Z = Z_1 + Z_2 = (10 + j15) + (20 - j30) = (30 - j15)\Omega$$

$$\therefore Z = 33.54 \angle -26.565^\circ$$

$$\text{Now, } I = \frac{V}{Z} = \frac{220 \angle 0^\circ}{33.54 \angle -26.565^\circ} = 6.559 \angle 26.565^\circ \text{ A}$$

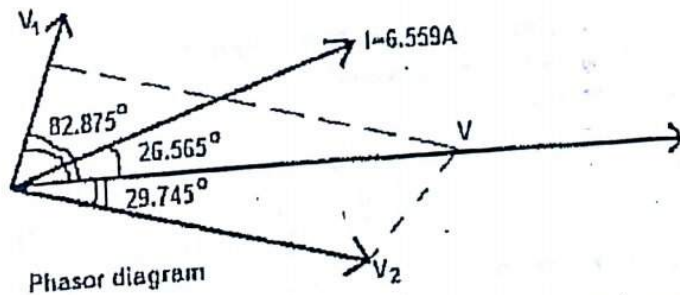
$$(ii) V_1 = IZ_1 = (6.559 \angle 26.565^\circ) \times (10 + j15) = (6.559 \angle 26.565^\circ)(18.03 \angle 56.31^\circ)$$

$$= 118.24 \angle 82.875^\circ \text{ Volts}$$

$$V_2 = IZ_2 = (6.559 \angle 26.565^\circ) \times (20 - j30) = (6.559 \angle 26.565^\circ)(36.05 \angle -56.31^\circ)$$

$$= 236.49 \angle -29.745^\circ \text{ Volts}$$

$$(iii) \text{ P.f.} = \cos \phi = \cos 26.565^\circ = 0.89 \text{ (leading)}$$



2. An Inductor of 0.5H inductance, 90 ohms resistance is connected in parallel with a 20μF capacitor. Find: (i) total current; (ii) power factor of the circuit; (iii) total power taken from source. A voltage of 230V, 50Hz is maintained across the circuit. Draw the vector diagram also.

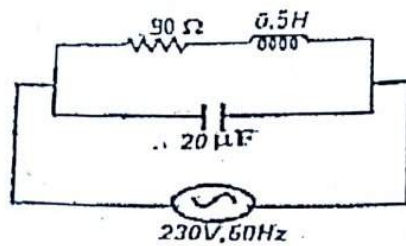
Solution:

$$\text{Given, } R = 90\Omega; X_L = 2\pi \times 50 \times 0.5 = 157.1\Omega; X_C = \frac{1}{2\pi \times 50 \times 20 \times 10^{-6}} = 159.1\Omega$$

$$\therefore Z_1 = (90 + j157.1)\Omega = 181.05 \angle 60.2^\circ \Omega$$

$$Z_2 = (0 - j159.1)\Omega = 159.1 \angle -90^\circ \Omega$$

A.C. Circuit Analysis

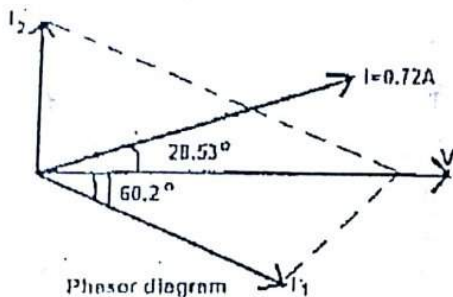


$$\therefore Z_p = \frac{Z_1 Z_2}{Z_1 + Z_2} = \frac{101.05 \angle 60.2^\circ \times 159.1 \angle -90^\circ}{101.05 \angle 60.2^\circ + 159.1 \angle -90^\circ} = 320.06 \angle -28.53^\circ \Omega$$

$$i. \quad I = \frac{V}{Z} = \frac{230}{320.06 \angle -28.53^\circ} = 0.72 \angle 28.53^\circ A$$

$$ii. \quad P.f = \cos \phi = \cos 28.53^\circ = 0.879 \text{ (leading)}$$

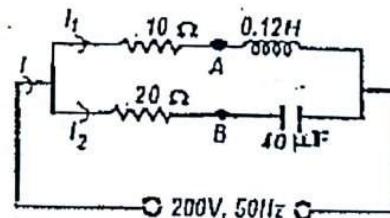
$$iii. \quad \text{Power} = VI \cos \phi = 230 \times 0.72 \times 0.879 = 145.6W$$



3. Two circuits A and B are connected in parallel across a 200V, 50Hz mains. Circuit A consists of a resistance 10Ω and an Inductance of $0.12H$ connected in series. Circuit B consists of a resistance 20Ω in series with capacitor of $40\mu F$.

Calculate:

- Currents in each branch
- Source current
- Power factor. Also draw phasor diagram.



Solution:

$$\text{Here, } Z_A = R_A + jX_L = 10 + j(2\pi \times 50 \times 0.12) = (10 + j37.7)\Omega = 39 \angle 75.15^\circ \Omega$$

$$Z_B = R_B - jX_C = 20 - j\left(\frac{1}{2\pi \times 50 \times 40 \times 10^{-6}}\right) = (20 - j79.57)\Omega = 82 \angle -75.9^\circ \Omega$$

$$\therefore Z_{eq} = \frac{Z_A Z_B}{Z_A + Z_B} = \frac{(10 + j37.7)(20 - j79.57)}{(10 + j37.7) + (20 - j79.57)} = 36.85 + j50 = 62 \angle 53.7^\circ \Omega$$

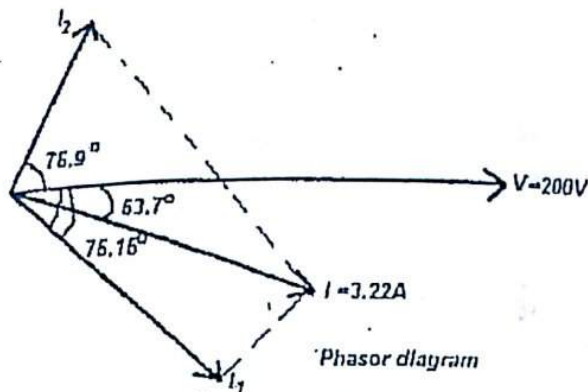
$$\text{➤ Source current, } I = \frac{V}{Z} = \frac{200 \angle 0^\circ}{62 \angle 53.7^\circ} = 3.22 \angle -53.7^\circ A$$

$$\text{➤ Branch currents, } I_1 = \frac{V}{Z_A} = \frac{200 \angle 0^\circ}{39 \angle 75.15^\circ} = 5.13 \angle -75.15^\circ A$$

A.C. Circuit Analysis

$$\text{And } I_2 = \frac{V}{Z_B} = \frac{200 \angle 0^\circ}{82 \angle -75.9^\circ} = 2.44 \angle 75.9^\circ \text{ A}$$

$$\text{P.f.} = \cos 53.7^\circ = 0.59 \text{ (lag)}$$



4. Two impedance of $(20 + j 5 \Omega)$ and $(30 + j 8 \Omega)$ are connected in series across a 200V, 50Hz supply. Find current, active power, apparent power and reactive power and power factor. Also draw the phasor diagram. [2008F, 2008S, 2012S]

Solution:

$$\text{Given that, } Z_1 = (20 + j 5 \Omega) = 20.61 \angle 14.03^\circ$$

$$Z_2 = (30 + j 8 \Omega) = 30.04 \angle 14.93^\circ$$

$$V = 200\text{V}$$

$$f = 50\text{Hz}$$

Since impedances are connected in series so total impedance $(Z_T) = Z_1 + Z_2$

$$\text{Or, } Z_T = 20 + j 5 + 30 + j 8$$

$$= (50 + j 13) \Omega$$

$$= 51.66 \angle 14.57^\circ$$

$$\begin{aligned} \text{Total current (I)} &= \frac{\text{total voltage (V)}}{\text{total impedance (Z}_T\text{)}} \\ &= \frac{200 \angle 0^\circ}{51.66 \angle 14.57^\circ} = 3.87 \angle -14.57^\circ \end{aligned}$$

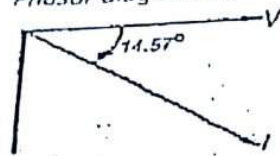
$$\begin{aligned} \text{Active power} &= VI \cos \phi \\ &= 200 \times 3.87 \times \cos 14.57^\circ \\ &= 749.10 \text{ watt} \end{aligned}$$

$$\begin{aligned} \text{Reactive power} &= VI \sin \phi \\ &= 200 \times 3.87 \times \sin 14.57^\circ \\ &= 193.79 \text{ VAR} \end{aligned}$$

$$\begin{aligned} \text{Apparent power} &= VI \\ &= 200 \times 3.87 = 774 \text{ VA} \end{aligned}$$

$$\text{Power factor} = \cos \phi = 0.97$$

➤ Phasor diagram is :



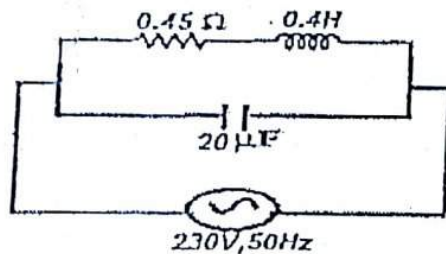
(-ve sign indicates lagging)

5. A coil having resistance of 45Ω and an inductance of $0.4H$ is connected in parallel with a capacitor having capacitance $20 \mu F$ across a $230 V, 50 Hz$ system. Calculate:

- Total current
- Branch currents
- Active and reactive power of entire circuit
- Draw the phasor diagram

[20045]

Solution:



Here, $R = 45 \Omega$, $L = 0.4H$, $C = 20 \mu F = 20 \times 10^{-6} F$

Now,

$$X_L = 2\pi fL$$

$$= 2\pi \times 50 \times 0.4 = 125.66 \text{ ohm}$$

Therefore, $Z_1 = R + jX_L$

$$= 45 + 125.66j = 133.44 \angle 70.29^\circ$$

Also, $X_C = \frac{1}{2\pi fC} = \frac{1}{2\pi \times 50 \times 20 \times 10^{-6}} = 159.15 \Omega$

Therefore, $Z_2 = 0 - jX_C$

$$= 0 - 159.15j = 159.15 \angle -90^\circ$$

Since Z_1 is parallel with Z_2 ,

So $Z_{eq} = \frac{Z_1 \times Z_2}{Z_1 + Z_2} = \frac{(45 + 125.66j) \times (0 - 159.15j)}{(45 + 125.66j) + (0 - 159.15j)}$

$$= 361.90 + 110.67j$$

$$= 378.45 \angle 17^\circ$$

i. Total current $(I) = \frac{V}{Z} = \frac{230 \angle 0^\circ}{378.45 \angle 17^\circ}$

$$= 0.608 \angle -17^\circ$$

A.C. Circuit Analysis

ii. Branch currents i.e.

$$I_1 = \frac{V}{Z_1} = \frac{230 \angle 0^\circ}{133.44 \angle 70.29^\circ} = 1.72 \angle -70.29^\circ$$

$$I_2 = \frac{V}{Z_2} = \frac{230 \angle 0^\circ}{159.15 \angle -90^\circ} = 1.445 \angle 90^\circ$$

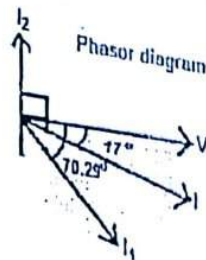
iii. Active power = $V \cos \phi$

$$= 230 \times 0.608 \times \cos 17^\circ = 133.72 \text{ watt}$$

Reactive power = $V \sin \phi$

$$= 230 \times 0.608 \times \sin 17^\circ = 40.89 \text{ VAR}$$

iv. Phasor diagram is:



6. Define resonance and deduce the expression for resonance frequency. A series RLC circuit has the following parameters; $R = 15\Omega$, $L = 150\text{mh}$, $C = 120\mu\text{F}$. Compute the resonance frequency, quality factor and bandwidth for that circuit. [2004S]

Solution:

Solution of first part is already done.

Given, $R = 15\Omega$; $L = 150\text{mh} = 150 \times 10^{-3}\text{H}$; $C = 120\mu\text{F} = 120 \times 10^{-6}\text{F}$

We know that for RLC series circuit:

$$> \text{Resonant frequency } (f_0) = \frac{1}{2\pi\sqrt{LC}}$$

$$= \frac{1}{2\pi\sqrt{150 \times 10^{-3} \times 120 \times 10^{-6}}} = 37.53 \text{ Hz}$$

$$> Q\text{-factor} = \frac{1}{R} \sqrt{\frac{L}{C}} = \frac{1}{15} \sqrt{\frac{150 \times 10^{-3}}{120 \times 10^{-6}}} = 2.35$$

$$> \text{Bandwidth } (\Delta f) = \frac{R}{2\pi L} = \frac{15}{2\pi \times 150 \times 10^{-3}} = 15.92 \text{ Hz}$$

7. Two impedance of $(5+j10)\Omega$ and $(10-j5)\Omega$ are connected in series across a 200 V, 50 Hz supply. Find current, active power, apparent power and reactive power and power factor. Also draw a phasor diagram. [2005S]

Solution:

Here, $Z_1 = (5 + j10)\Omega$

$$Z_2 = (10 - j5)\Omega$$

Since impedances are in series so,

$$Z_{eq} = Z_1 + Z_2$$

$$= 5 + j10 + 10 - j5 = (15 + j5)\Omega$$

$$= 5\sqrt{10} \angle 18.43^\circ \Omega$$

A.C. Circuit Analysis

$$V = 200 \angle 0^\circ \text{ and } f = 50 \text{ Hz}$$

$$\text{Current } (I) = \frac{V}{Z} = \frac{200 \angle 0^\circ}{5\sqrt{10} \angle -18.43^\circ} = 12.64 \angle 18.43^\circ \text{ A}$$

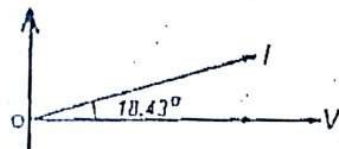
$$\begin{aligned} \text{Active power} &= VI \cos \phi \\ &= 200 \times 12.64 \times \cos 18.63^\circ = 2398.34 \text{ watt} \\ &= 2.398 \text{ kW} \end{aligned}$$

$$\begin{aligned} \text{Apparent power} &= VI = 200 \times 12.64 = 2528 \text{ VA} \\ &= 2.528 \text{ kVA} \end{aligned}$$

$$\begin{aligned} \text{Reactive power} &= VI \sin \phi \\ &= 200 \times 12.64 \times \sin 18.43^\circ \\ &= 799.21 \text{ VAR} \end{aligned}$$

$$\text{Power factor} = \cos \phi = \cos 18.63^\circ = 0.94$$

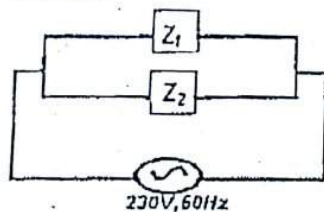
Phasor diagram



8. An Impedance of $(6-j8)\Omega$ is Connected in parallel with an Impedance of $(8+j6)\Omega$. The impedance are fed from a 220V, 50 Hz supply. Find current through each branch, total impedance, total admittance, active power, reactive power, apparent power and power factor.

[2005F]

Solution:



$$\text{Here, } Z_1 = (6-j8)\Omega = 10 \angle -53^\circ$$

$$Z_2 = (8+j6)\Omega = 10 \angle 36.86^\circ$$

Since Z_1 is parallel with Z_2 ,

$$\begin{aligned} \text{So } Z_{eq} &= \frac{Z_1 \times Z_2}{Z_1 + Z_2} = \frac{(6-j8) \times (8+j6)}{(6-j8) + (8+j6)} \\ &= 7-j \\ &= 7.07 \angle -8.13^\circ \end{aligned}$$

$$\begin{aligned} \text{Total current } (I) &= \frac{V}{Z_{eq}} = \frac{220 \angle 0^\circ}{7.07 \angle -8.13^\circ} \\ &= 31.11 \angle 8.13^\circ \text{ Ampere} \end{aligned}$$

A.C. Circuit Analysis

- > Current in branch Z_1 is, $I_1 = \frac{V}{Z_1} = \frac{220 \angle 0^\circ}{10 \angle -53^\circ} = 22 \angle 53^\circ \text{ A}$
- Current in branch Z_2 is, $I_2 = \frac{V}{Z_2} = \frac{220 \angle 0^\circ}{10 \angle 36.86^\circ} = 22 \angle -36.86^\circ \text{ A}$
- > Here total impedance, $Z_T = 7.07 \angle -8.13^\circ$
- Therefore, total admittance (Y) = $\frac{1}{Z_T} = \frac{1}{7.07 \angle -8.13^\circ} = 0.141 \angle 8.13^\circ$
- > Active power = $V \cos \phi$
 $= 220 \times 31.11 \times \cos 8.13^\circ = 6775.41 \text{ watt}$
 $= 6.775 \text{ kW}$
- > Reactive power = $V \sin \phi$
 $= 220 \times 31.11 \times \sin 8.13^\circ$
 $= 965.5 \text{ VAR}$
- > Apparent power = $VI = 220 \times 31.11 = 6844.2 \text{ VA}$
 $= 6.844 \text{ kVA}$
- > Power factor = $\cos \phi = \cos 8.13^\circ = 0.989$

9. A 50Ω resistor, 5 mH inductor and $15 \mu\text{F}$ capacitor are connected in series across a 1 V source. Find resonant frequency, quality factor, bandwidth, current and voltage across capacitor at resonance. [20055]

Solution:

Given that, $R = 50 \Omega$

$$L = 5 \text{ mH} = 5 \times 10^{-3} \text{ H}$$

$$C = 15 \mu\text{F} = 15 \times 10^{-6} \text{ F}$$

$$V = 1 \text{ V}$$

$$\begin{aligned} \text{a) Resonant frequency } (f_0) &= \frac{1}{2\pi\sqrt{LC}} \\ &= \frac{1}{2\pi\sqrt{5 \times 10^{-3} \times 15 \times 10^{-6}}} = 581.15 \text{ Hz} \end{aligned}$$

$$\text{b) Q-factor } (Q) = \frac{1}{R} \sqrt{\frac{L}{C}} = \frac{1}{50} \sqrt{\frac{5 \times 10^{-3}}{15 \times 10^{-6}}} = 0.36$$

$$\text{c) Bandwidth } (\Delta f) = \frac{R}{2\pi L} = \frac{50}{2\pi \times 5 \times 10^{-3}} = 1591.54 \text{ Hz}$$

$$\text{d) At resonance } Z = R \quad (\text{since at resonance } X_L - X_C = 0)$$

$$\begin{aligned} \text{Therefore current across capacitor at resonance } (I) &= \frac{V}{R} \\ &= \frac{1}{50} = 0.02 \text{ A} \end{aligned}$$

$$\begin{aligned} \text{e) Voltage across capacitor at resonance } (V_C) &= I \times X_C = I \times \frac{1}{2\pi f C} \\ &= 0.02 \times 0.018 \end{aligned}$$

A.C. Circuit Analysis

$$= 3.6 \times 10^{-4} \text{ A}$$

10. An AC voltage of $(80+j60)$ V applied to a circuit results current of $(4+j10)$ A. Find:
- The Impedance of the circuit stating whether it is inductive or capacitive.
 - Power consumed
 - Phase angle
 - Reactive power

[2007F, 2010S]

Solution:

$$\text{Here, } V = (80+j60) \text{ V} = 100 \angle 36.86^\circ$$

$$I = (4+j10) \text{ A} = 10.77 \angle 68.19^\circ$$

$$\begin{aligned} \text{Now, impedance of circuit (Z)} &= \frac{V}{I} = \frac{100 \angle 36.86^\circ}{10.77 \angle 68.19^\circ} \\ &= 9.28 \angle -31.33^\circ \\ &= 7.92 - j4.82 \end{aligned}$$

Hence it is capacitive circuit.

(Since negative sign indicates capacitive ckt)

- > Power consumed, $P = VI \cos \phi$

$$= 100 \times 10.77 \times \cos(68.19^\circ - 36.86^\circ)$$

$$= 919.95 \text{ watt}$$
- > Phase angle, $\phi = 68.19^\circ - 36.86^\circ = 31.32^\circ$
- > Reactive power $= VI \sin \phi$

$$= 100 \times 10.77 \times \sin 31.32^\circ$$

$$= 559.84 \text{ VAR}$$

11. An ac series circuit has a resistance of 10 Ohm and an inductance of 0.2 H and capacitance of 60 μf . If the applied voltage is 200 V, calculate resonant frequency and power at resonance.

[2006S, 2007F]

Solution:

$$\text{Here, } R = 10 \text{ Ohm, } L = 0.2 \text{ H, } C = 60 \times 10^{-6} \text{ F, } V = 200 \text{ V}$$

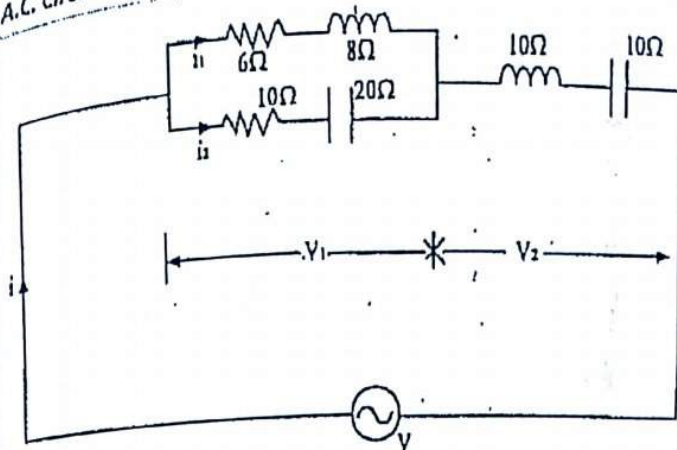
$$\begin{aligned} \text{Now, resonant frequency (} f_r \text{)} &= \frac{1}{2\pi\sqrt{LC}} \\ &= \frac{1}{2\pi\sqrt{0.2 \times 60 \times 10^{-6}}} = 45.94 \text{ Hz} \end{aligned}$$

$$\begin{aligned} \text{Power at resonance} &= \frac{V^2}{R} \\ &= \frac{200^2}{10} = 4000 \text{ watt} \end{aligned}$$

12. In the circuit given below, total current $I = (20+j0)$ A. Calculate (i) branch current I_1 and I_2 , (ii) voltage V_1 , V_2 and V . (iii) power factor of entire circuit. (iv) Active and reactive powers in the entire circuit. (v) Draw the phasor diagram.

[2009S]

A.C. Circuit Analysis



Solution:

Given, $I = 20 \angle 0^\circ = 20 \angle 0^\circ$

$Z_1 = R + jX_L = (6 + j8) \Omega$

$Z_2 = R - jX_C = (10 - j20) \Omega$

$Z_3 = jX_L - jX_C = (j10 - j10) \Omega = 0$

i. Now, branch currents is given by

$$I_1 = I \times \frac{Z_2}{Z_1 + Z_2} = (20 \angle 0^\circ) \times \frac{(10 - j20)}{(6 + j8) + (10 - j20)} = \frac{200 - 400j}{16 - 12j} = (20 - j10) \text{ A}$$

$$= 22.36 \angle -26.56^\circ$$

$$I_2 = I \times \frac{Z_1}{Z_1 + Z_2} = (20 \angle 0^\circ) \times \frac{(6 + j8)}{(6 + j8) + (10 - j20)} = \frac{120 + 160j}{16 - 12j} = j10 \text{ A} = 10 \angle 90^\circ$$

ii. Since Z_1 parallel to Z_2 so their equivalent impedance is

$$Z_{eq} = \frac{Z_1 \times Z_2}{Z_1 + Z_2}$$

$$= \frac{(6 + j8) \times (10 - j20)}{(6 + j8) + (10 - j20)} = 10 + j5 = 11.18 \angle 26.56^\circ$$

Now Z_{eq} and Z_3 are in series so total impedance is

$$Z = Z_3 + Z_{eq} = 10 + j5 \quad (= R + jX_L)$$

Therefore, $V = IZ = 20 \times 11.18 \angle 26.56^\circ = 223.6 \angle 26.56^\circ$

$V_1 = I_1 Z_1 = I_2 Z_2$ (Since voltage remains same in parallel ckt)

$$= (20 - j10) \times (6 + j8) = 200 + j100 = 223.6 \angle 26.56^\circ \text{ Volt}$$

$V_2 = I Z_2 = (20 + j0) \times 0 = 0$

iii. Power factor $= \frac{R}{Z} = \frac{10}{11.18} = 0.89$

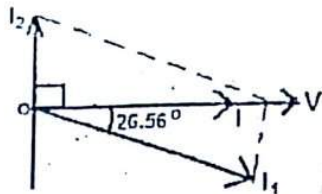
iv. Active power $= VI \cos \phi = 223.6 \times 20 \times 0.89 = 4 \text{ kW}$

Reactive power $= VI \sin \phi = 223.6 \times 20 \times \sin 26.56^\circ = 2 \text{ kVAR}$

Apparent power $= VI = 223.6 \times 20 = 4.472 \text{ kVA}$

v. Phasor diagram is:

A.C. Circuit Analysis



13. A coil of resistance 40Ω and inductive reactance of 25Ω is connected in series to $220V$, $50Hz$ supply. Calculate:

- Active and reactive components.
- Total power of circuit.

[2013F]

Solution:

Given, $R = 40\Omega$, $X_L = 25\Omega$, $V = 220V$, $f = 50Hz$

$$\therefore Z = R + jX_L = 40 + j25 = 47.17 \angle 32^\circ$$

$$\text{Now, } I = \frac{V}{Z} = \frac{220 \angle 0^\circ}{47.17 \angle 32^\circ} = 4.663 \angle -32^\circ \text{ Ampere}$$

$$a. \text{ Active component} = I \cos \phi = 4.663 \times \cos 32^\circ = 3.954A$$

$$\text{Reactive component} = I \sin \phi = 4.663 \times \sin 32^\circ = 2.471A$$

$$b. \text{ Total power of circuit} = V I \cos \phi = 220 \times 4.663 \times \cos 32^\circ = 869.98W$$

Some other important questions:

14. In series RL circuit, $R = 20\text{ ohm}$ while $L = 60\text{ mH}$. The input current lags supply voltage by 60° . Obtain the value of applied frequency.

Solution:

Here, given

$$R = 20\Omega, L = 60\text{ mH} = 60 \times 10^{-3} H$$

In RL series ckt current lags voltage by ϕ . Therefore $\phi = 60^\circ$

$$\text{We know, } \tan \phi = \frac{X_L}{R} = \frac{2\pi f L}{R}$$

$$\tan 60^\circ = \frac{2\pi \times f \times 60 \times 10^{-3}}{20}$$

$$\text{Therefore, } f = \frac{20\sqrt{3}}{2\pi \times 60 \times 10^{-3}} = 91.93\text{ Hz}$$

Hence applied frequency = 91.93 Hz

15. In series RLC ckt $R = 3\Omega$, $X_L = 6\Omega$, $X_C = 2\Omega$. What is the voltage applied across the combination if the series current is $10 \angle -143^\circ$ Ampere.

Solution:

Given, $R = 3\Omega$, $X_L = 6\Omega$, $X_C = 2\Omega$, $10 \angle -143^\circ$ Ampere

Applied voltage (V) = ?

A.C. Circuit Analysis

We know, $Z = \sqrt{R^2 + (X_L - X_C)^2} = \sqrt{3^2 + (6 - 2)^2} = 5 \text{ ohm}$

And $\phi = \tan^{-1} \left(\frac{X_L - X_C}{R} \right) = \tan^{-1} \left(\frac{6-2}{3} \right) = 53.13^\circ$

Therefore $V = IZ = 10 \angle -143^\circ \times 5 \angle 53.13^\circ = 50 \angle -89.86^\circ \text{ volt}$

16. An impedance of $(6-j6)\Omega$ is connected in parallel with another impedance $(4+j4)\Omega$. If the applied voltage be $20 \angle 10^\circ \text{V}$, find the branch currents and total current.

Solution:

Given, $Z_1 = (6-j6)\Omega = 6\sqrt{2} \angle -45^\circ$

$Z_2 = (4+j4)\Omega = 4\sqrt{2} \angle 45^\circ$

$V = 20 \angle 10^\circ \text{ volt}$

Z_1 and Z_2 are connected in parallel so their equivalent impedance is,

$Z = \frac{Z_1 \cdot Z_2}{Z_1 + Z_2} = \frac{(6-j6)(4+j4)}{(6-j6)+(4+j4)} = \frac{6\sqrt{2} \angle -45^\circ \cdot 4\sqrt{2} \angle 45^\circ}{2\sqrt{2} \angle -11.3^\circ} = 4.7 \angle 11.3^\circ$

Therefore, total current $(I) = \frac{V}{Z} = \frac{20 \angle 10^\circ}{4.7 \angle 11.3^\circ} = 4.24 \angle -1.3^\circ \text{A}$

Also, branch currents:

$I_1 = \frac{V}{Z_1} = \frac{20 \angle 10^\circ}{6\sqrt{2} \angle -45^\circ} = 2.35 \angle 55^\circ \text{ Ampere}$

$I_2 = \frac{V}{Z_2} = \frac{20 \angle 10^\circ}{4\sqrt{2} \angle 45^\circ} = 3.53 \angle -35^\circ \text{ Ampere}$

17. A series RLC circuit has $R = 2 \text{ ohm}$, $L = 2 \text{ mH}$, $C = 10 \mu\text{F}$. Calculate quality factor, bandwidth, resonant frequency and half power frequencies f_1 and f_2 .

Solution:

Given, $R = 2\Omega$, $L = 2 \text{ mH} = 2 \times 10^{-3} \text{H}$, $C = 10 \mu\text{F} = 10 \times 10^{-6} \text{F}$

> For RLC series ckt, resonant frequency

$f_0 = \frac{1}{2\pi\sqrt{LC}} = \frac{1}{2\pi\sqrt{2 \times 10^{-3} \times 10 \times 10^{-6}}} = 1125.96 \text{ Hz}$

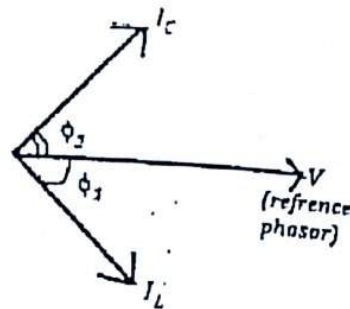
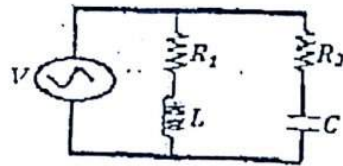
> Quality factor $(Q) = \frac{1}{R} \sqrt{\frac{L}{C}} = \frac{1}{2} \sqrt{\frac{2 \times 10^{-3}}{10 \times 10^{-6}}} = 5\sqrt{2}$

> Bandwidth $(\Delta f) = \frac{R}{2\pi L} = \frac{2}{2\pi \times 2 \times 10^{-3}} = 159.23 \text{ Hz}$

> Half frequencies, $f_1 = f_0 - \frac{R}{4\pi L} = 1125.96 - 79.57 = 1046.38 \text{ Hz}$

$f_2 = f_0 + \frac{R}{4\pi L} = 1125.96 + 79.57 = 1205.53 \text{ Hz}$

18. Find the condition that the currents in two parallel branches of the circuit shown below; shall remain in quadrature, when both R_1 and R_2 are varied simultaneously.



Solution:

Let ω , i_L and i_C be respectively the angular frequency, the current through R_1 - L branch and the current through R_2 - C branch. Let i_L lags by ϕ_1 and i_C leads V by ϕ_2 . Then,

$$\tan \phi_1 = \frac{\omega L}{R_1} \text{ and } \tan \phi_2 = \frac{1}{\omega C R_2}$$

Since, i_L and i_C are in quadrature. So, $\phi_1 + \phi_2 = 90^\circ$
(Quadrature means perpendicular to each other)

$$\text{Now, } \tan(\phi_1 + \phi_2) = \frac{\tan \phi_1 + \tan \phi_2}{1 - \tan \phi_1 \tan \phi_2} = \tan 90^\circ = \infty$$

$$\text{Or, } \tan \phi_1 \cdot \tan \phi_2 = 1$$

$$\text{Or, } \frac{\omega L}{R_1} \cdot \frac{1}{\omega C R_2} = 1$$

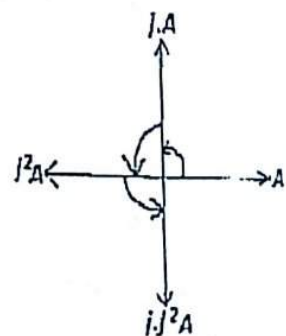
$\therefore R_1 \cdot R_2 = \frac{L}{C}$ Which is the required condition for two branch currents i_L and i_C are in quadrature when both R_1 and R_2 are varied simultaneously.

19. 'j' operator and its significance:

[PEC Internal]

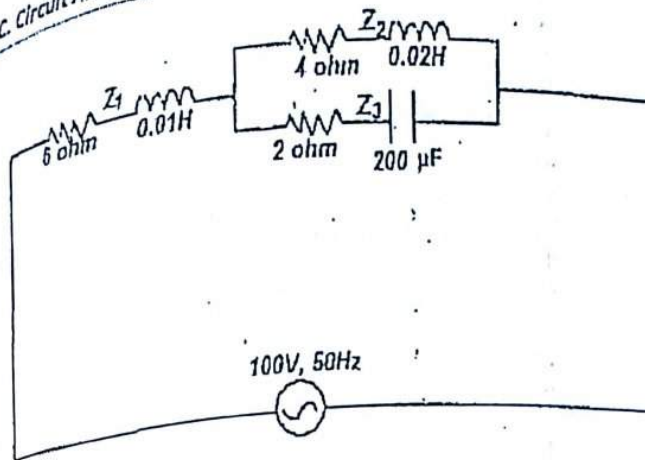
The letter 'j' is a symbol of an operator just as other (+, -, *, /) operator. It is used to indicate the anticlockwise rotation of vector through 90° . Its value is $j = \sqrt{-1}$.

Double operation of 'j' rotates the vector A by 180° anticlockwise in direction. Triple operation of 'j' rotates the vector A by 270° in anticlockwise direction. We know that $j^2 = -1$, hence when operator 'j' is operated on vector A we get, the new vector j.A which is 90° displaced from A in anticlockwise direction. Similarly, when 'j' is operated on j.A then we get new vector $j^2 \cdot A$ which is 90° displaced from j.A in anticlockwise direction. It means when we apply operator 'j' twice on vector A we get new vector $j^2 \cdot A = -A$. (rectangular to polar, polar to rectangle).



20. Determine the current in the circuit below, and also find the power consumed as well as power factor.

A.C. Circuit Analysis



Solution:

$$Z_1 = 6 + j(2\pi \times 50 \times 0.01) = (6 + j3.14) = 6.77 \angle 27.62^\circ \Omega$$

$$Z_2 = 4 + j(2\pi \times 50 \times 0.02) = (4 + j6.28) = 7.44 \angle 57.51^\circ \Omega$$

$$Z_3 = 2 - j \left(\frac{1}{2\pi \times 50 \times 200 \times 10^{-6}} \right) = (2 - j15.92) = 16.04 \angle -82.84^\circ \Omega$$

$$\therefore Z_p = \frac{Z_2 Z_3}{Z_2 + Z_3} = \frac{(4 + j6.28)(2 - j15.92)}{(4 + j6.28) + (2 - j15.92)} = (8.85 + j5.69) \Omega$$

$$\text{Now, } Z_{eq} = Z_1 + Z_p = (6 + j3.14) + (8.85 + j5.69) = (14.85 + j8.83) \Omega = 17.28 \angle 30.74^\circ \Omega$$

$$\triangleright I = \frac{V}{Z_{eq}} = \frac{100 \angle 0^\circ}{17.28 \angle 30.74^\circ} = 5.79 \angle -30.74^\circ \text{ A}$$

$$\triangleright \text{Power factor} = \cos(-30.74^\circ) = 0.859 \text{ (lagging)}$$

$$\triangleright \text{Power} = V \cos \phi = 100 \times 5.79 \times 0.8594 = 497.6 \text{ W}$$



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Polyphase System

1. Introduction

Systems with more than one phase are generally termed poly phase. The 'word' poly means many or several. Hence polyphase system means a system which has two or more separate sources of voltage. The two phase, three phase, six phase, twelve phase etc. are examples of polyphase systems. But three phase systems are the most common in used. The three voltages are sinusoidal and equal in magnitude but differ in phase by 120° . The three voltages do not reach their maximum values at the same time but with time interval of $\frac{T}{3}$ seconds or $1/3$ of a cycle or phase angle of 120° . For most application three phase system is preferred to a single phase system.

Advantages of polyphase (three phase) over single phase: [2009F, 2011F, 2013F]

1. For a given size of frame, a polyphase generator (or motors) has greater output than that of single phase.
2. A polyphase transmission line requires much less conductors than a single phase line.
3. Polyphase motors have uniform torque; whereas single phase motors passes pulsating torque (except commutating motors).
4. Polyphase motors are self-starting; whereas single phase motors are not.
5. Efficiency of polyphase generator is higher than that of single phase generators.
6. Polyphase motors are more compact; whereas single phase motors requires more space for conveying the same power.
7. Power in polyphase never falls to zero; whereas it may fall to zero in single phase system.

Differences between 3 phase & single phase system:

Single phase system	Polyphase system
1. It requires more copper for transmitting power	1. It requires less copper for transmitting power
2. Power may fall to zero value	2. power never fall to zero
3. Single phase system have fluctuating torque	3. It has uniform torque
4. Single phase motor require more space	4. 3 phase motor are more compact
5. Single phase motors are costly	5. Polyphase motors are a low in cost
6. Less efficiency	6. High efficiency
7. Output of single phase generators is low	7. Output of polyphase machine is high
8. They are not self-starting	8. They are self-starting

Generation of 3 phase voltage:

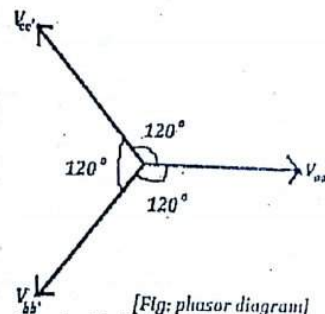
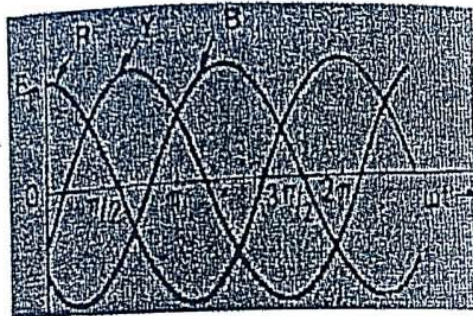
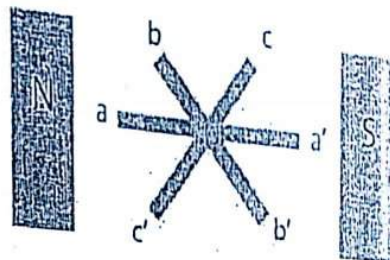
For a single phase generator we have only one armature winding if the number of armature windings is increased then it becomes poly phase generator & it produces as many independent voltage wave as the no. of windings or phases. These windings are displaced from one another

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by equal angles the values of these angles being determined by the no. of phases. In general the displacement between different phases is $\frac{360^\circ}{n}$ where n is the no. of phase or windings.

When three identical coils are placed with their axis at 120° apart from one another & rotated in a uniform magnetic field a sinusoidal voltage is generated across each coil. This is the basis of 3 phase generation.

The figure shows a 2 pole alternator. It has 3 sets of coils aa' , bb' & cc' , symmetrically mounted on a rotor such that their axis are 120° apart from each other. When, the rotor is rotated in a anticlockwise direction at a constant angular velocity ω rad/sec, a sinusoidal voltage is generated across each coil. Since the 3 coils rotate at the same velocity the generated voltages have the same frequency also since the coils are identical the generated voltages have the same magnitude but there is a phase difference of 120° between these voltages.



[Fig: phasor diagram]

$$V_{aa'} = V_m \sin \omega t$$

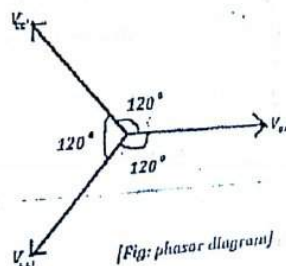
$$V_{bb'} = V_m \sin(\omega t - 120^\circ)$$

$$V_{cc'} = V_m \sin(\omega t - 240^\circ) \text{ or } V_{cc'} = V_m \sin(\omega t + 120^\circ)$$

Phase sequence:

The order in which the phase voltages reach as obtained their maximum values is called phase sequence. There may be either abc phase sequence or acb phase sequence.

If the anticlockwise rotation of the coil is considered voltage across coil aa' attain its maximum value at first and then bb' & cc' respectively. In this case the phase sequence is abc . If the rotation is reversed the phase sequence is acb .



[Fig: phasor diagram]

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For abc phase sequence,

$$V_{aa'} = V_m \sin \omega t = V \angle 0^\circ$$

$$V_{bb'} = V_m \sin(\omega t - 120^\circ) = V \angle -120^\circ$$

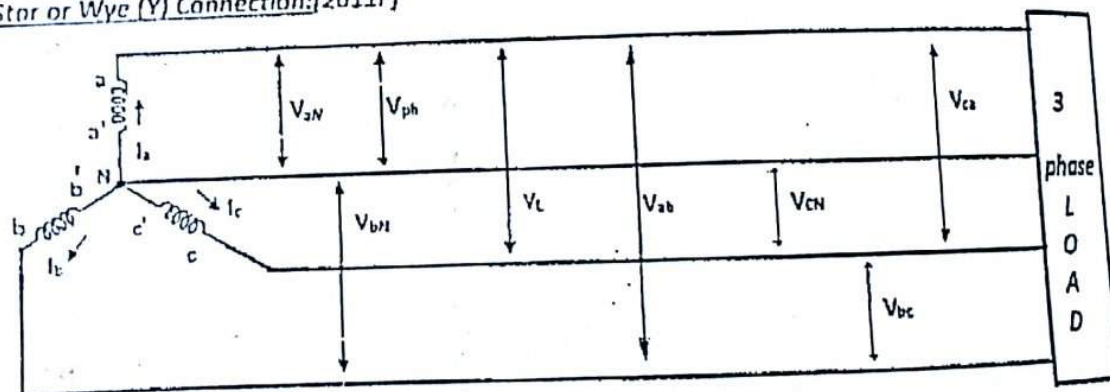
$$V_{cc'} = V_m \sin(\omega t - 240^\circ) = V \angle -240^\circ$$

Three phases may be marked as 123 or abc or they may be given 3 colors. The colors used commercially are red, yellow & blue. In this case sequence is RGB and if reversed it is RBY.

Interconnection of 3 phases:

If three armature coils of the 2 phase alternator are not connected then each phase or circuit could need two conductors which will make six conductors for 3 phases. It means that each transmission cable could contain six conductors which will make the whole system complicated and expensive. Hence, the 3 phases are generally interconnected which results in saving of conductors. The general methods of interconnection are star & delta connection.

Star or Wye (Y) Connection: [2011F]



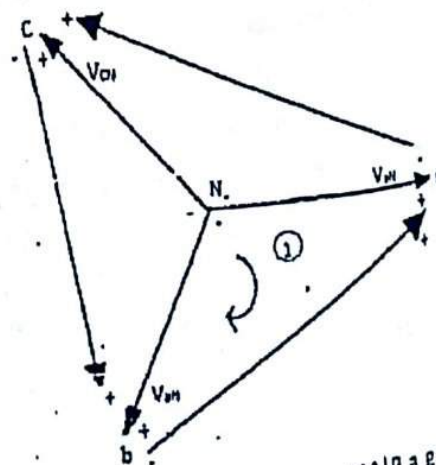
In this connection, the similar ends a' , b' , c' of three coils are joined together at point N as shown in figure. The point N is known as star or neutral point. If the conductor is not attached from the star point to the load, the system is called Y or star connected three phase three wire generator. If the neutral point is connected to the load the system then it is called Y or star connected three phase four wire generator.

The voltage induced in each winding is called the phase voltage (V_{ph}) and currents in each winding is called phase current (I_{ph}). However the voltage available between any pair of terminals is called line voltage (V_L) and current flowing in each line is called line current (I_L).

Let abc be the phase sequence then

$$V_{aN} = V_{bN} = V_{cN} = V_{ph}$$

$$V_{aN} = V_{ph} \angle 0^\circ$$



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$$V_{bN} = V_{ph} \angle 120^\circ$$

$$V_{cN} = V_{ph} \angle 240^\circ$$

$$\text{Also, } V_{ab} = V_{bc} = V_{ca} = V_L$$

By Kirchhoff's law (in loop abN),

$$V_{aN} + V_{ab} - V_{bN} = 0 \text{ (take a cycle from N to a, a to b and b to N in loop)}$$

$$\therefore V_{ab} = V_{bN} - V_{aN}$$

$$= V_{ph} \angle 120^\circ - V_{ph} \angle 0^\circ$$

$$= V_{ph} [\cos(-120^\circ) + j\sin(-120^\circ)] - V_{ph} [\cos 0^\circ + j\sin 0^\circ]$$

$$= V_{ph} [\cos 120^\circ - j\sin 120^\circ] - V_{ph} [1 + j0]$$

$$= V_{ph} [-1/2 - j\sqrt{3}/2] - V_{ph}$$

$$= -1/2 V_{ph} - V_{ph} - j\sqrt{3}/2 V_{ph}$$

$$= -3/2 V_{ph} - j\sqrt{3}/2 V_{ph}$$

$$= -V_{ph} [3/2 + j\sqrt{3}/2]$$

$$= -\sqrt{3} V_{ph} [\sqrt{3}/2 + j/2]$$

$$= -\sqrt{3} V_{ph} [-\cos 30^\circ - j\sin 30^\circ]$$

$$= \sqrt{3} V_{ph} [\cos 210^\circ + j\sin 210^\circ]$$

$$= \sqrt{3} V_{ph} \angle 210^\circ$$

$$V_L = \sqrt{3} V_{ph}$$

$$\text{Therefore, } V_{ab} = \sqrt{3} V_{ph} \angle 210^\circ$$

$$V_{bc} = \sqrt{3} V_{ph} \angle 90^\circ$$

$$V_{ca} = \sqrt{3} V_{ph} \angle 30^\circ$$

Thus, we conclude that the magnitude of the line voltage for a star connection system is $\sqrt{3}$ times the phase voltage. It is clear that current flowing through lines must be the same current flowing through the phases.

$$\text{i.e. } I_L = I_{ph}$$

Power:

The total active or true power in the circuit is the sum of the three phase powers.

Hence, total active power = 3 × phase power

$$\text{Or, } P = 3 \times V_{ph} I_{ph} \cos \phi$$

$$\text{Now } V_{ph} = V_L / \sqrt{3} \text{ and } I_{ph} = I_L$$

Hence, in terms of line values, the above expression becomes

For three phase system:

$$\text{Active power (P)} = \sqrt{3} V_L I_L \cos \phi$$

In watt

$$\text{Reactive power (Q)} = \sqrt{3} V_L I_L \sin \phi$$

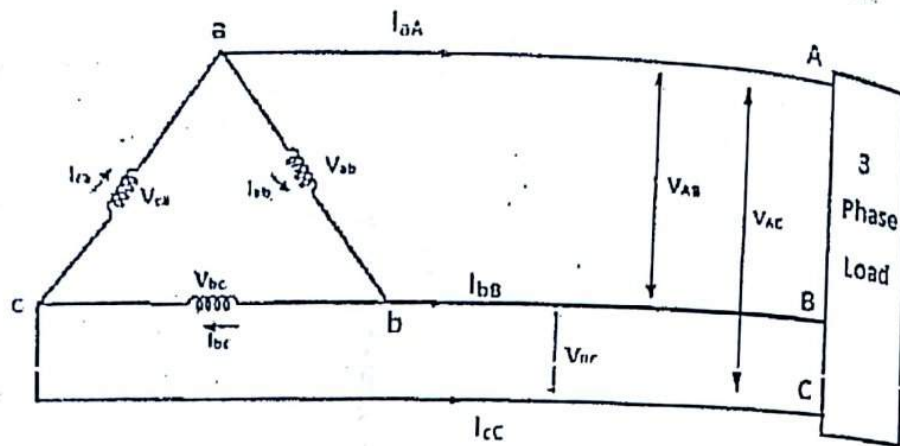
In VAR

$$\text{Apparent power (S)} = \sqrt{3} V_L I_L$$

In VA

Delta connection (Δ):

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In this form of connection the dissimilar ends of the 3 phase windings are joined together i.e. starting end of the phase is connected to the finishing ends of the other. The 3 phase load is connected to the source to allow line currents to be drawn from the source. From the circuit the phase voltages are V_{ab}, V_{bc}, V_{ca} and line voltages are V_{AB}, V_{BC}, V_{CA} . It is obvious that the line voltages are equal in magnitude. Hence $V_L = V_{\text{phase}}$ (for delta connection)

From the above ckt we have the phase currents; I_{ab}, I_{bc}, I_{ca} and line currents are; I_{aA}, I_{bB}, I_{cC} . Now applying Kirchhoff's current law at nodes a, b & c we have,

$$I_{aA} = I_{ca} - I_{ab} \quad (1)$$

$$I_{bB} = I_{ab} - I_{bc} \quad (2)$$

$$I_{cC} = I_{bc} - I_{ca} \quad (3)$$

Let the nature of the load is such that the phase current can be expressed as:

$$I_{ab} = I_p \angle 0^\circ$$

$$I_{bc} = I_p \angle -120^\circ$$

$$I_{ca} = I_p \angle -240^\circ$$

Now taking equation (1)

$$I_{aA} = I_p \angle -240^\circ - I_p \angle 0^\circ$$

$$= I_p [\cos(-240^\circ) - j\sin 240^\circ] - I_p$$

$$= I_p [-1/2 - (j\sqrt{3}/2)] - I_p$$

$$= I_p [-1 - 1/2 + j\sqrt{3}/2]$$

$$= \sqrt{3} I_p (-\sqrt{3}/2 + j.1/2)$$

$$= \sqrt{3} I_p (\cos 150^\circ + j\sin 150^\circ)$$

$$= \sqrt{3} I_p \angle 150^\circ$$

Similarly,

$$I_{bB} = \sqrt{3} I_p \angle -30^\circ$$

$$I_{cC} = \sqrt{3} I_p \angle -90^\circ$$

Key points:

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In star connection	In delta connection
$V_L = \sqrt{3} V_{ph}$ $I_L = I_{ph}$	$I_L = \sqrt{3} I_{ph}$ $V_L = V_{ph}$

Key points:

Power Triangle for Three Phase Load

Total apparent power

$$S = 3 \times \text{Apparent power per phase}$$

$$S = 3 V_{ph} I_{ph} = 3 \frac{V_L}{\sqrt{3}} I_L = 3 V_L \frac{I_L}{\sqrt{3}}$$

$$S = \sqrt{3} V_L I_L \text{ volt-amperes (VA) or kVA}$$

Total active power

$$P = \sqrt{3} V_L I_L \cos \phi \text{ watts (W) or kW}$$

Total reactive power

$$Q = \sqrt{3} V_L I_L \sin \phi \text{ reactive volt-amperes (VAR) or kVAR}$$

Hence power triangle is as shown in the Fig. 8.18.

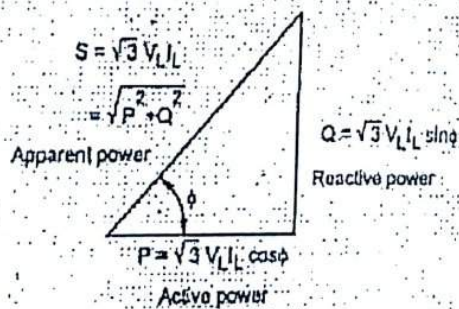
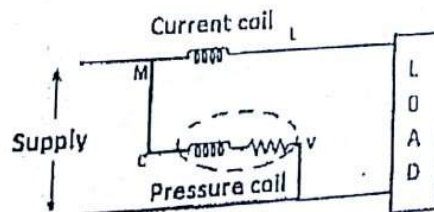


Fig: Power triangle

Power measurement in 3 phase:

Watt meters are the instruments which measure electric power developed by a device or a circuit or power delivered to any devices. A wattmeter is hardly ever required in dc circuits since the power can be determined from voltmeter and ammeter reading. However, in ac circuit a wattmeter is required since the power depends not only on the voltage and current but also on the phase shift between them. The method of connecting a wattmeter is as shown in figure.

The coil which is connected in series with the load is known as current coil (C.C). The other coil that is connected across the load is called pressure coil (P.C) or voltage coil. For measuring power in 3-phase circuit, two-



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wattmeter method is usually employed:

Two wattmeter method for measuring 3-phase power:
[2004S, 2007F, 2008F, 2009F, 2009S, 2010S, 2011F, 2012F]

For measuring power in a three phase circuit two wattmeter methods is generally preferred. This method is easier for measuring power in star or delta connected balanced or unbalanced load of a 3 phase, 3 wire system. The current coils of two wattmeter's are inserted in any two lines and the pressure coils of both are connected across the third line then instantaneous total power is the sum or difference of the two wattmeter reading i.e.

$$W = W_1 + W_2$$

Theory: Let us consider the connection diagram for star connected load (see the figure as shown below)

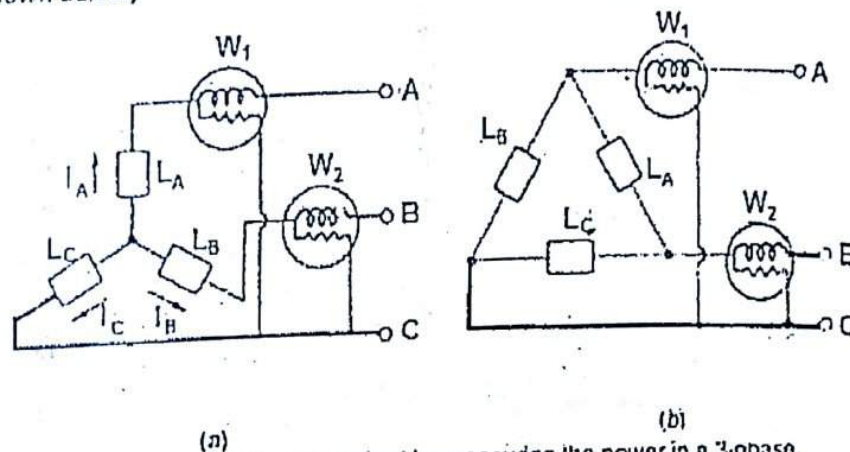


Fig: Two-wattmeter method for measuring the power in a 3-phase.
(a) Star-connected circuit; (b) delta-connected circuit.

For wattmeter W_1 the instantaneous current through its current coil is i_A and instantaneous voltage across its pressure coils is $e_{AC} = (e_A - e_C)$.

Therefore power read by it $= i_A(e_A - e_C)$ ————— (i)

For wattmeter W_2 the instantaneous current through its current coil is i_B and the voltage across the pressure coils is $e_{BC} = (e_B - e_C)$.

Therefore, instantaneous power read by it $i_B(e_B - e_C)$ ————— (ii)

Therefore, total instantaneous power read by the two wattmeters W_1 and W_2 is:

$$W_1 + W_2 = i_A(e_A - e_C) + i_B(e_B - e_C)$$

$$= i_A e_A - i_A e_C + i_B e_B - i_B e_C = i_A e_A + i_B e_B - e_C(i_A + i_B) \text{ ————— (iii)}$$

$$\text{But } i_A + i_B + i_C = 0$$

$$\text{Or, } i_A + i_B = -i_C \text{ ————— (iv)}$$

From equations (iii) and (iv), we get

$$W_1 + W_2 = i_A e_A + i_B e_B + i_C e_C = P_A + P_B + P_C$$

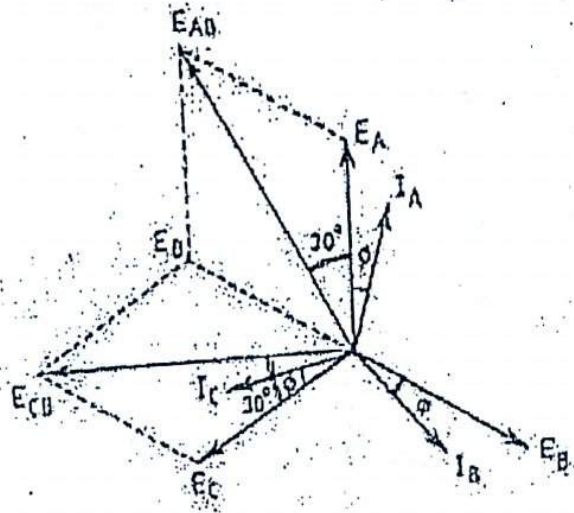
$$= \text{total instantaneous power} = \sqrt{3}EI \cos \phi$$

where P_A, P_B and P_C are the respective power absorbed by load currents i_A, i_B and i_C .

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Power factor angle:

The phasor diagram for the balanced star connected 3 phase load is shown in the figure below



[Fig: Phasor diagram for two wattmeter method of measuring power in a 3 phase circuit]

For wattmeter W_1 , the rms current through its current coil is I_A and the rms voltage across the pressure coil is E_{AB} .

For wattmeter W_2 the rms current through its current coil is I_C and the rms voltage across the pressure coil is E_{CB} .

However, the magnitude of each voltage is $\sqrt{3} E$, where E is the line to neutral voltage.

From the phasor diagram, it is clear that:

- (i) The phase angle between E_{AB} and I_B is $30^\circ + \phi$;
- While, (ii) The phase angle between E_{CB} and I_C is $30^\circ - \phi$

Where ϕ is the power factor angle of load.

$$W_1 + W_2 = \sqrt{3} E I_A \cos(30^\circ + \phi) + \sqrt{3} E I_C \cos(30^\circ - \phi)$$

$$\text{But } I_A = I_B = I_C = I$$

$$W_1 + W_2 = \sqrt{3} E I \cos(30^\circ + \phi) + \sqrt{3} E I \cos(30^\circ - \phi)$$

$$= \sqrt{3} E I [2 \cos 30^\circ \cos \phi]$$

$$= \sqrt{3} E I [2 \times (\sqrt{3}/2) \cos \phi]$$

$$= \sqrt{3} E I \cos \phi \quad \text{--- (vi)}$$

$$\text{Now } W_1 - W_2 = \sqrt{3} E I [\cos(30^\circ + \phi) - \cos(30^\circ - \phi)]$$

$$= \sqrt{3} E I [-2 \sin 30^\circ \sin \phi]$$

$$= \sqrt{3} E I (-2)(1/2) \sin \phi$$

$$= -\sqrt{3} E I \sin \phi \quad \text{--- (vii)}$$

From equations (vi) and (vii)

$$\frac{W_1 - W_2}{W_1 + W_2} = \frac{-\sqrt{3} E I \sin \phi}{\sqrt{3} E I \cos \phi} = -\tan \phi$$

$$\therefore \tan \phi = \frac{\sqrt{3}(W_2 - W_1)}{W_1 + W_2}$$

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From this equation, the power factor, $\cos \phi$, can be calculated.

Numericals:

Note: In most of the cases:

- If line or phase impedance is not specified then given impedance is always taken as phase impedance.
- Line voltage is greater than or equal to 400, and phase voltage is less than 400. Generally line voltage is 400 or 410V, and phase voltage is 230 V.

Numericals:

1. A 3 phase star connected system supplies a balanced star connected load. The current in each phase is 5A. What is the current in neutral wire?

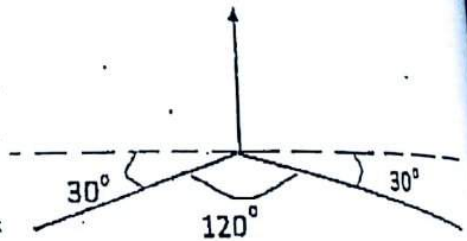
Solution:

Resolving the current in vertical and horizontal

components. We get horizontal

component, $I_H = 5\cos 90^\circ + 5\cos 30^\circ + 5\cos 210^\circ = 0$

Vertical component (I_V) = $5\sin 90^\circ + 5\sin 120^\circ + 5\sin 210^\circ = 0$.



$$\therefore \text{Current in neutral wire} = \sqrt{0^2 + 0^2} = 0$$

2. A balanced 3 phase load draws 8kW at a lagging power factor of 0.8. If the line voltage applied is 440 V, find the reactive power and line current.

Solution:

Active Power (P) = 8KW = 8000 W

Power factor = $\cos \phi \therefore \phi = 36.86^\circ$

Line voltage (V_L) = 440 V Reactive power = ? I_L = ?

We know that, active power;

$$P = \sqrt{3} V_L I_L \cos \phi \text{ or } 8000 = \sqrt{3} \times 440 \times I_L \times \cos \phi$$

$$\therefore I_L = 13.12 \text{ A also,}$$

$$\text{Reactive power} = \sqrt{3} V_L I_L \sin \phi = \sqrt{3} \times 440 \times 13.12 \times \sin 36.8 = 5998.617 \text{ VAR} = 5.998 \text{ kVAR}$$

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3. A balanced 3 phase load draws 7.5 kVA at power factor 0.6 leading. If the line voltage applied is 440 V, find the active power and line current.

Solution:

$$P_{VA} = 7.5 \text{ kVA} = 7.5 \times 10^3 \text{ VA} = 7500 \text{ VA}$$

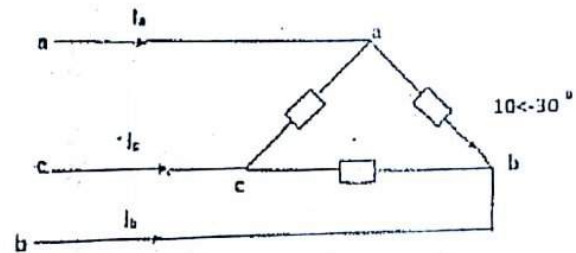
$$P_f = \cos \phi = 0.6 (\text{leading}), \therefore \phi = 53.13^\circ, V_L = 440 \text{ V}, P_{\text{active}} = ?, I_L = ?$$

$$\text{Now, } \sqrt{3} V_L I_L = 7500 \therefore I_L = \frac{7500}{\sqrt{3} \times 440} = 9.84 \text{ A}$$

$$\text{Hence, } P_{\text{active}} = \sqrt{3} V_L I_L \cos \phi = \sqrt{3} \times 9.84 \times 440 \times 0.6 = 4500 \text{ W} = 4.5 \text{ kW}$$

4. A 3 ϕ , 220 V supply is applied to a balanced delta (Δ) connected 3 phase load. The phase current being $I_{ab} = 10 \angle -30^\circ$, find I_a compute the total power received by the delta connected.

Find the value of the resistance portion of the phase impedance.



Solution:

$$V_{\text{phase}} = V_{\text{line}} = 220 \text{ V}$$

$$I_{\text{phase}} = I_{ab} = 10 \angle -30^\circ \text{ A}, I_a = ?, \text{Power (P)} = ?, I_p = 10 \angle -30^\circ \text{ A}$$

$$\text{Line current (I}_a\text{)} = \sqrt{3} I_p = \sqrt{3} \times 10 \angle -30^\circ \text{ A} = 17.32 \angle -30^\circ \text{ A}$$

$$\text{Total Power (P)} = \sqrt{3} V_L I_L \cos \phi = \sqrt{3} \times 220 \times 17.32 \times \cos 30 = 5715.77 \text{ W} = 5.715 \text{ kW}$$

$$\text{Now, Impedance (Z}_p\text{)} = V_p / I_p = 220 / (10 \angle -30^\circ) = 19.05 + j11 = 22 \angle 30^\circ \Omega (= R + jX_L)$$

$$\therefore \text{Resistance portion} = 19.05 \Omega$$

5. Three equal impedance each having a resistance of 8Ω and inductive reactance of 6Ω are connected in series in (i) Star (ii) Delta across 3-phase 400V system. Find (i) Phase current (ii)

Line current (iii) Power consumed in each case. [PU 2008F]

Solution:

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$$Z_{ph} = (8+j6) \Omega = 10\Omega$$

In star connection:

$$V_L = \sqrt{3} V_{ph} = \frac{V_L}{\sqrt{3}} = 400/\sqrt{3} = 230.94 \text{ V}$$

$$i) I_{ph} = V_{ph}/Z_{ph} = 230.94/10 = 23.094 \text{ A}$$

$$ii) I_L = I_{ph} = 23.094 \text{ A (Since in star connection line current is equal to phase current)}$$

$$iii) \text{Power} = \sqrt{3} V_L I_L \cos \phi = \sqrt{3} \times 400 \times 23.094 \times \cos 36.86^\circ = 12801.165 \text{ watts} = 12.801 \text{ kW}$$

For delta connection:

$$V_L = V_{ph} = 400 \text{ V}, Z_{ph} = (8+j6) \Omega = 10\Omega$$

$$i) I_{ph} = \frac{V_{ph}}{Z_{ph}} = 400/10 = 40 \text{ A}$$

$$ii) \text{Since } I_L = \sqrt{3} I_{ph} = 40\sqrt{3} \text{ A}$$

$$iii) \text{Power} = \sqrt{3} V_L I_L \cos \phi = \sqrt{3} \times 400 \times 40\sqrt{3} \times \cos 36.86^\circ = 38404.97 \text{ watts} = 38.404 \text{ kW}$$

6. A Δ -connected balanced 3ϕ load is supplied from a 3ϕ , 400V supply. The line current is 20A and the power taken by the load is 10,000 W. Find

- i) Impedance in each branch ii) The line current, power factor and power consumed if the same load is connected in star. [PU 2009F]

Solution:

(Given, $I_L = 20 \text{ A}$; $P = 10000 \text{ W}$ (active power))

In delta connection; $I_L = \sqrt{3} I_{ph}$ and $V_L = V_{ph} = 400 \text{ V}$, $I_{ph} = 20/\sqrt{3} = 11.54 \text{ A}$

$$i) \text{Impedance of circuit is; } Z = \frac{V_{ph}}{I_{ph}} = \frac{400}{11.54} = 34.67 \Omega$$

$$ii) \text{Since } P = \sqrt{3} V_L I_L \cos \phi \text{ or } 10000 = \sqrt{3} \times 400 \times 20 \times \cos \phi, \text{Power factor} = \cos \phi = 0.72$$

7. Three equal Impedance each $(30+j40)\Omega$ are connected in star and a balanced three phase 400V supply system is connected across this load. Find (i) the current flowing through the load, (ii) the power consumed in each phase and (iii) the power factor of the load. [2009S]

Solution:

$$Z_{ph} = 30+j40 = 50\Omega, V_L = 400 \text{ V}$$

In star connection:

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$$V_L = \sqrt{3} V_{ph} \text{ or } V_{ph} = 400 / \sqrt{3} = 230.94 \text{ V}$$

$$i) I_{ph} = V_{ph} / Z_{ph} = 230.94 / 50 = 4.61 \text{ A}$$

$$ii) P = \sqrt{3} V_L I_L \cos \phi$$

$$= \sqrt{3} 400 \times 4.61 \times \cos 53.13^\circ = 1920 \text{ W}$$

$$iii) \text{Power factor} = \cos 53.13^\circ = 0.6$$

8. A three phase balanced load connected in star draws a total power of 20 kW at 0.8 pf lag when connected to a 3 phase, 400 V, 50 Hz supply. Calculate Resistance of coil. Inductance of coil [2013 F]

Solution:

$$\text{Power (P)} = 20 \text{ kW} = 20000 \text{ W}$$

$$\text{Power factor} = 0.8 = \cos \phi; \text{ so } \phi = 36.86^\circ (\text{lagging})$$

$$f = 50 \text{ Hz}; V_L = 400 \text{ V}$$

$$V_L = \sqrt{3} V_{ph}$$

$$V_{ph} = 400 / \sqrt{3} = 230.94 \text{ V}, I_{ph} = I_L (\text{for star})$$

$$P = \sqrt{3} V_L I_L \cos \phi \text{ or } 20000 = \sqrt{3} \times 230.94 \times I_L \times 0.8; I_L = 62.5 \text{ A}$$

$$Z_{ph} = V_{ph} / I_{ph} = \frac{230.94}{62.5 \angle -36.86^\circ} = 2.956 + j2.21 = R + jX_L$$

$$i. \text{ Therefore resistance of coil} = 2.956 \Omega$$

$$ii. \text{ Since } X_L = 2\pi f L \text{ so inductance of coil (L)} = \frac{2.21}{2\pi \times 50} = 7.03 \times 10^{-3} \text{ H}$$

9. Three equal impedances of $10 \angle 30^\circ \Omega$ are connected in wye (Y) across a 400 V, three phase supply. What is the total power taken from the supply? If the same impedances are now connected in delta across the same supply. What will be the power taken from the supply? [PU 2008 S]

Solution:

$$Z_{ph} = 10 \angle 30^\circ \Omega, V_L = 400 \text{ V}$$

$$V_L = \sqrt{3} V_{ph} \text{ or } V_{ph} = \frac{400}{\sqrt{3}} = 230.94 \text{ V}$$

$$\text{Now, } I_{ph} = \frac{V_{ph}}{Z_{ph}} = \frac{230.94}{10} = 23.09 \text{ A}$$

Polyphase System

In star connection $I_L = I_{ph} = 23.09 \text{ A}$

$$\therefore \text{Power} = \sqrt{3} V_L I_L \cos \phi = 13.854 \text{ KW}$$

Now,

In delta connection:

$$V_L = V_{ph} = 400 \text{ V}$$

$$Z_{ph} = 10 \angle 30^\circ \Omega \quad \phi = 30^\circ$$

$$I_{ph} = \frac{V_{ph}}{Z_{ph}} = 40 \text{ A}$$

$$\therefore \text{In delta, } I_L = \sqrt{3} I_{ph} = 40\sqrt{3} \text{ A}$$

$$\therefore \text{Power} = \sqrt{3} V_L I_L \cos \phi = 41.56 \text{ KW}$$

10. A balance delta-connected load of $(3-j3) \Omega$ per phase is fed by a 400 V 3-phase system. Find line current, phase current, p.f. total-volt-ampere, active and reactive power. [PU 2005 S]

→ Solution:

$$Z_{ph} = (3-j3) \Omega = 4.24 \Omega; \quad V_L = 400 \text{ V}$$

$$\text{In delta connection, } V_L = V_p = 400 \text{ V}$$

$$I_{ph} = \frac{V_{ph}}{Z_{ph}} = 400/4.24 = 94.33 \text{ A}$$

$$\text{In delta-connected: } I_L = \sqrt{3} I_{ph} = 163.38 \text{ A}$$

- (i) Line current = 163.38 A
- (ii) Phase current (I_{ph}) = 94.33 A
- (iii) Power factor = $\cos \phi = 0.707$
- (iv) Active power = $\sqrt{3} V_L I_L \cos \phi = 80.04 \text{ KW}$
- (v) Reactive Power = $\sqrt{3} V_L I_L \sin \phi = 80.04 \text{ KVAR}$
- (vi) Total volt-ampere = Apparent power = $\sqrt{3} V_L I_L = 113.15 \text{ KVA}$

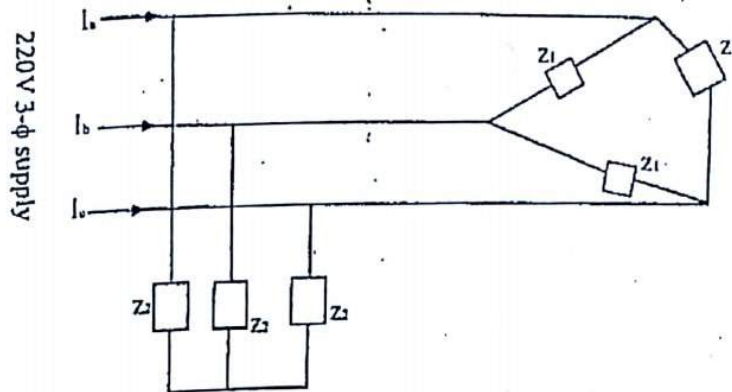
11. Two three phase loads are connected as shown below to a three phase, 220V supply

where, $Z_1 = (20 + j30) \text{ ohm}$ and $Z_2 = (30 + j60) \text{ ohm}$. Calculate:

- i) Line currents I_a , I_b and I_c
- ii) Current in each delta and wye phase and the line currents drawn by each three phase load and
- iii) Total power drawn from supply

[PU 2011 F]

Polyphase System



Solution:

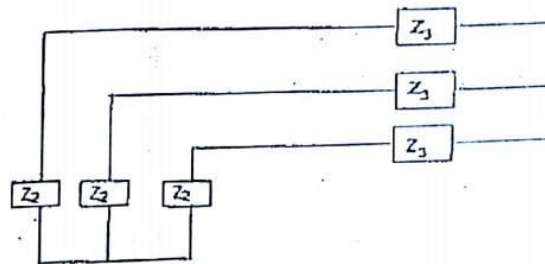
Convert delta into star,

$$Z_3 = \frac{Z_1 \times Z_1}{Z_1 + Z_1 + Z_1} = \frac{Z_1}{3} = \frac{(20 + j30)}{3} \Omega$$

Now figure becomes,

Since Z_2 and Z_3 are parallel so equivalent impedance is:

$$Z_{eq} = \frac{(30 + j60) \times \frac{(20 + j30)}{3}}{30 + j60 + \frac{(20 + j30)}{3}} = 10.20 \angle 57.39^\circ \Omega$$



$$\therefore I_{ph} = \frac{V_{ph}}{Z_{ph}} = \frac{220 \angle 0^\circ}{10.20 \angle 57.39^\circ} = 21.56 \angle -57.39^\circ A$$

$$i) I_a = 21.56 \angle -57.39^\circ A, I_b = 21.56 \angle -57.39^\circ - 120^\circ = 21.56 \angle -177.39^\circ A$$

$$ii) I_c = 21.56 \angle -57.39^\circ - 240^\circ A = 21.56 \angle -297.39^\circ A$$

$$iii) V_L = \sqrt{3} V_{ph} = 220 \sqrt{3} V$$

$$\text{Power (P)} = \sqrt{3} V_L I_L \cos \phi = \sqrt{3} \times \sqrt{3} \times 220 \times 21.56 \cos 57.4^\circ = 7666.5 \text{ watts}$$

12. Three equal impedances having resistance 8Ω and inductive reactance 6Ω are connected

in delta and connected to 230V, 3 phase wire 3 phase Y(wye) connected source. Then calculate

i. phase and line current of load. ii. Power factor. iii. Power consumed [PU, 2013 spring]

Polyphase System

Solution:

$$R=8 \Omega$$

$$Z=8+j6=10 \Omega$$

$$V_{ph}=230 V$$

$$I_L=\sqrt{3} I_{ph}=\sqrt{3} \times (V_{ph}/Z_{ph})=23\sqrt{3} A$$

$$\text{Power factor } = R/Z = 8/10 = 0.8$$

$$\text{Power} = 3 V_{ph} I_{ph} \cos \phi = 3 \times 230 \times 23 \times 0.8 = 12696 W$$

13. The star connected stator of a 3 phase, 50 Hz alternator supplies a balanced delta-connected load. Each phase of the load consists of a coil of resistance 15Ω and inductance $36 mH$, and the phase voltage generated by the alternator is $231 V$. Calculate a) the phase and line currents b) the load power factor c) the power delivered to the load. [PU 2013 fall or question]

Solution:

$$\text{Frequency } (f) = 50 \text{ Hz}$$

$$Z = 15 + j36 \times 10^{-3} = 15 \Omega$$

$$V_{ph} = 231 V$$

$$I_{ph} = V_{ph}/Z_{ph} = 231/15 = 15.4 \Omega$$

$$I_L = \sqrt{3} I_{ph} = \sqrt{3} \times 15.4 = 26.67 A$$

$$\text{Three line currents are, } I_R = 26.37 \angle -0.13^\circ, I_Y = 26.37 \angle -120.13^\circ, I_B = 26.37 \angle -240.13^\circ$$

$$\cos \phi = R/Z = 15/15 = 1$$

$$P = \sqrt{3} V_L I_L \cos \phi = \sqrt{3} \times 231 \times 26.67 \times 1 = 10670.76 \text{ watts}$$

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Polyphase System

14. Two wattmeter connected to measure the power output to a 3 phase circuit indicate 15 kW and 1.5 kW respectively, The later reading being obtained after reversing the current coil connections calculate the power factor of the load.

Solution:

Reading of first wattmeter, $W_1 = 15 \text{ kW}$

Reading of second wattmeter, $W_2 = -1.5 \text{ kW}$

Total power of the load $= 15 + [-1.5] \text{ kW} = 13.5 \text{ kW}$

$$\text{We know that, } \tan \phi = \sqrt{3} \times \frac{W_1 - W_2}{W_1 + W_2} = \sqrt{3} \left[\frac{15 - (-1.5)}{15 + (-1.5)} \right] = 2.117$$

$$\therefore \phi = 64.7^\circ; \quad \text{Hence power factor load, } \cos \phi = 0.427$$

15. The input power to a 3 phase load is 10 kW at 0.8 power factor. Two wattmeters are connected to measure the power, find the individual reading of the wattmeters.

Solution:

Let, $W_1 = \text{higher reading}$, $W_2 = \text{lower reading}$

$$W_1 + W_2 = 10 \text{ kW} \quad \text{--- (1)}$$

$$\cos \phi = 0.8 \quad \therefore \phi = 36.8^\circ$$

$$\therefore \tan \phi = \tan(36.8^\circ) = 0.75$$

$$\tan \phi = \sqrt{3} \times \frac{W_1 - W_2}{W_1 + W_2} \Rightarrow 0.75 = \sqrt{3} \frac{W_1 - W_2}{W_1 + W_2}$$

$$\Rightarrow W_1 - W_2 = \frac{0.75}{\sqrt{3}} \times (W_1 + W_2) = \frac{0.75}{\sqrt{3}} \times 10 \text{ kW} = 4.33 \text{ kW} \quad \text{--- (2)}$$

Now, solving equations (1) and (2)

$$W_1 = 7.165 \text{ W}, W_2 = 2.835 \text{ W}$$

Polyphase System

16. A balanced star connected load is supplied from a symmetrical 3 phase 400 V system. The current in each phase is 30 A and lags by 30° behind the voltage. find

- (i) The impedance in each phase (ii) Total power drawn.

Solution:

For a star connected load

$$V_L = \text{supply voltage} = 400 \text{ V}$$

$$V_{ph} = V_L / \sqrt{3} = 400 / \sqrt{3} = 230.94 \text{ V}$$

$$I_{ph} = 30 \text{ A (lagging } V_{ph} \text{ by } 30^\circ \text{ i.e. load is inductive)}$$

$$Z_{ph} = V_{ph} / I_{ph} = 230.94 / 30 = 7.698 \Omega = 7.698 \angle 30^\circ \Omega$$

$$\text{ii) Total power} = 3 V_{ph} I_{ph} \cos \phi = 3 \times 230.94 \times 30 \times \cos 30^\circ = 18000 \text{ W}$$

17. A balanced 3 phase load of 3 kW at pf of 0.8 (lagging) is connected across a 3 phase supply. If the line current is 12.5 A, calculate resistance and reactance in each branch of same load is connected in delta? Draw the phasor diagram in each case.

Solution:

$$\text{In star: } \cos \phi = 0.8 \text{ (lag)}$$

$$I_L = 12.5 \text{ A}$$

$$I_{ph} = 12.5 \text{ A}$$

$$\text{Power (P)} = \sqrt{3} I_L V_L \cos \phi$$

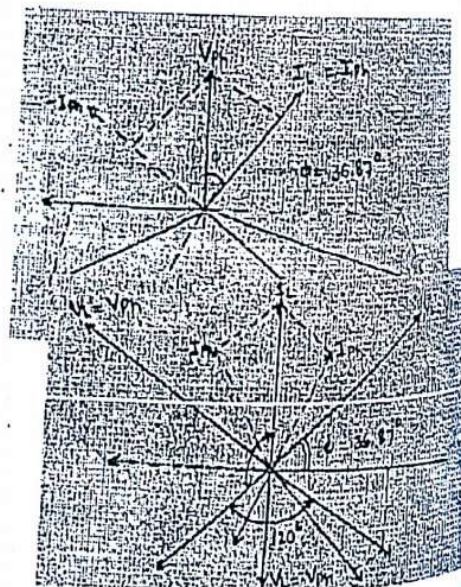
$$\Rightarrow 3000 = \sqrt{3} \times 12.5 \times V_L \times 0.8 \Rightarrow V_L = 173.2 \text{ V}$$

$$\& V_{ph} = 173.2 / \sqrt{3} = 100 \text{ V}$$

$$\therefore Z_{ph} = V_{ph} / I_{ph} = 100 / 12.5 = 8 \Omega$$

$$R_{ph} = Z_{ph} \cos \phi = 8 \times 0.8 = 6.4 \Omega$$

$$X_{ph} = Z_{ph} \sin \phi = 8 \times 0.6 = 4.8 \Omega$$



Polyphase System

$$\text{In delta: } V_L = V_{ph} = 173.2 \text{ V}$$

$$I_L = 12.5 \text{ A}, \cos \phi = 0.8, \sin \phi = 0.6, I_{ph} = I_L / \sqrt{3} = 7.22 \text{ A}$$

$$\therefore \text{Reactive power} = \sqrt{3} V_L I_L \sin \phi = 2250 \text{ VAR}$$

$$\text{Power loss} = \sqrt{3000^2 + 2250^2} = 1984.3 \text{ KW}$$

18. A balanced delta connected load having a Impedance of $Z_L = (300 + j210) \Omega$ in each phase is supplied from 400 V, 3 phase supply through a 3 phase line having an impedance of $Z_S = (4 + j8) \Omega$ in each phase. Find the total power supplied to the load as well as the current and voltage in each phase of the load. (PEC Internal 2013)

Solution:

$$Z_L = (300 + j210) \Omega = \sqrt{300^2 + 210^2} = 366.19 \Omega$$

$$V_L = 400 \text{ V}$$

$$Z_S = Z_{ph} = (4 + j8) \Omega = \sqrt{4^2 + 8^2} = 8.94 \Omega$$

$$I_L = \sqrt{3} I_{ph} \text{ or } \frac{400}{366.19} = \sqrt{3} I_{ph}$$

$$I_{ph} = 0.63 \text{ A}, I_L = V_L / Z_L = 1.09 \text{ A}$$

$$V_{ph} = I_{ph} * Z_{ph} = 0.63 * 8.94 = 5.63 \text{ V}$$

$$\text{Power (P)} = \sqrt{3} V_L * I_L \cos \phi = \sqrt{3} \times 400 \times 1.09 \times \frac{300}{366.19} = 618.67 \text{ W}$$



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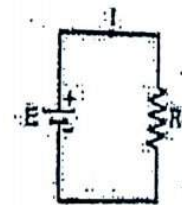
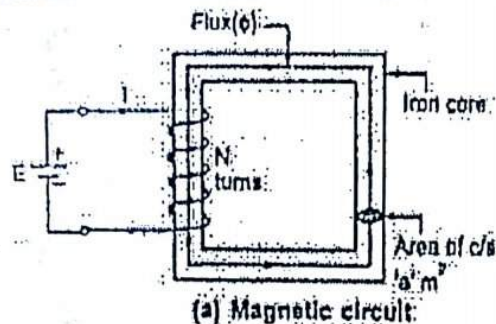
Chapter: 5 ELECTRIC MACHINES

REVIEW OF MAGNETIC CIRCUITS:

The magnetic circuit can be defined as, the closed path traced by the magnetic lines of force i.e. flux. Such a magnetic circuit is associated with different magnetic quantities as magneto-motive force (m.m.f.), flux reluctance, permeability etc.

Consider simple magnetic circuit shown in the figure (a). This circuit consists of an iron core with cross-sectional area of ' a ' m² with a mean length ' l ' m. (This is mean length of the magnetic path which flux is going to trace.) A coil of N turns is wound on one of the sides of the square core which is excited by a supply. This supply drives a current I through the coil. This current carrying coil produces the flux (ϕ) which completes its path through the core as shown in the fig (a).

This is analogous to simple electric circuit in which a supply i.e. e.m.f. of E volts drives a current I which completes its path through a closed conductor having resistance R . This analogous electrical circuit is shown in fig (b).



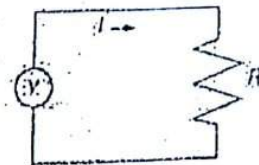
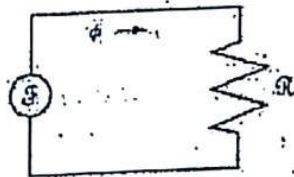
Tips:

Electrical	Magnetic	Magnetic Units
Voltage V	Magneto-motive Force $\mathcal{F} = NI$	Amp-turns
Current I	Magnetic flux ϕ	Webers Wb
Resistance R	Reluctance $\mathcal{R}$	Amp-turns/Wb
Conductivity σ	Permeability μ	Wb/A-t-m
Current density J	Magnetic flux density B	Wb/m ² = Tesla T
Electric field E	Magnetic field intensity H	Amp-turn/m

Figure: Analogous between Electrical and Magnetic Circuit quantities

- Analogy between Magnetic and Electric Circuit. [PU 2003, PU 2007, PU 2009, PU 2013]

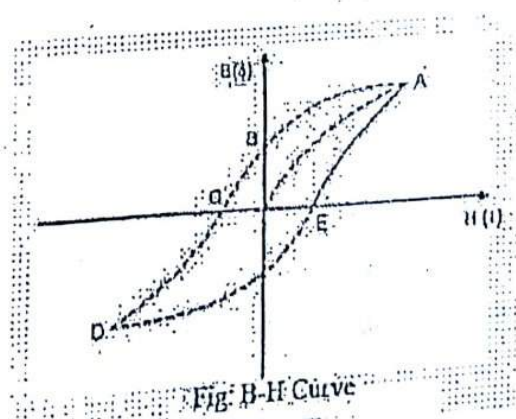
S.No.	Magnetic Circuit	Electric Circuit
1.	Path traced by the magnetic flux is defined as magnetic circuit.	Path traced by the current is called electric circuit.
2.	M.M.F. is the driving force in the magnetic circuit, the unit of which is ampere turns.	E.M.F. is the driving force in electric circuit, the unit of which is volts.
3.	There is flux (ϕ) in the magnetic circuit measured in webers.	There is current I in the electric circuit measured in amperes.
4.	The number of magnetic lines of force decides the flux.	The flow of electrons decides the current in the conductor.
5.	Resistance offered to the flow of flux is called reluctance. Unit is ampere turns/weber.	The property which obstructs the flow of current is called resistance. Unit is ohm.
6.	Kirchhoff's m.m.f law and flux law is applicable to the magnetic circuit.	Kirchhoff's current and voltage law is applicable to the electric circuits.
7.	The flux density 'B' is the line of forces per unit area i.e. $B = \frac{\phi}{A} \text{ Wb/m}^2$.	Current density is the current per unit area i.e. $J = \frac{I}{A} \text{ A/m}^2$.



Hysteresis and Eddy Current Loss: [PU 2002, PU 2003, PU 2006, PU 2007, PU 2008, PU 2009, PU 2010, PU 2011]

a) Hysteresis loss:

- In every magnetizing cycle, magnetic field 'H' is reversed during half cycle and magnetic flux density 'B' lags behind 'H' as shown in figure which is known as Hysteresis loop.
- During the complete cycle, the elementary magnet within the magnetic material try to line up first in one direction and then in reversed direction which causes the mechanical stress in magnetic materials and produces heat. This so generated heat



energy during the cycle of magnetization is given by the area of hysteresis loop and it is also known as hysteresis loop.

b) Eddy Current loss:

- When a conductor is either moved relative to magnetic field or present in changing magnetic field, the emf is induced on it (according to the Faraday's Law of Electromagnetic Induction). This induced emf set-up a current which circulate in a closed path of conductor within itself. This current is known as eddy current.
- Due to eddy current, the conducting material core becomes hot and the heat energy represents the loss of useful energy and known as eddy current loss.
- When the current in Primary coil changes, the magnetic flux linked with the Iron core of transformer changes and eddy current is produced in the core. Due to this, some of the electrical energy is lost in the form of heat energy.
- Eddy Current loss can be reduced either by making the conductor (core) of laminated thin sheets or by using the conducting material of high resistivity.

❖ Some Formulas:

- Magnetic Flux density (B) = flux/Area = Φ / A (units are weber/metre² or tesla)
Also, $B = \mu H$
- Permeability (μ) = Absolute permeability of air (vacuum) * Relative permeability
= $\mu_0 \mu_r = 4\pi * 10^{-7} * \mu_r$ (unit: weber per ampere-turn or Wb/AT)
- Magnetomotive force (mmf) = Number of turns * Current = NI (unit: Ampere-turn i.e. AT)
- Magnetic field intensity (H) = mmf/length = NI / l (unit: AT/m)
- Reluctance (S) = mmf/flux

$$\text{Also, } S = \frac{1}{\mu} \frac{L}{A} = \frac{1}{\mu_0 \mu_r} \frac{L}{A}$$

Where, L = length of magnetic path; A = Area of cross-section of magnetic path; $\mu_0 = 4\pi * 10^{-7}$ H/m = permeability of the free space; μ_r = relative permeability of the magnetic material and μ = permeability of magnetic material.

Numerical problems:

1. An electromagnet as shown in figure has cross-section area of 12 cm^2 , mean length of iron path is 50 cm and air gap is 0.4 cm . A coil of 800 turns is placed in one side of iron bar which is taking current of 1 A from the source. Find the flux density in the air gap given that relative permittivity $(\mu_r)=1300$. $[\mu_0 = 4\pi \times 10^{-7}]$

Solution:

Mean length of Iron path (L_1) = $50 \text{ cm} = 0.5 \text{ m}$

Mean length of air gap (L_2) = $0.4 \text{ cm} = 0.04 \text{ m}$

Cross-sectional area (A) = $12 \text{ cm}^2 = 12 \times 10^{-4} \text{ m}^2$

m.m.f. = 800 AT ($1 \text{ Ampere} \times 800 \text{ turns}$)

For iron part:

$$\text{Reluctance } (S_1) = \frac{1}{\mu} \frac{L}{A} = \frac{1}{\mu_0 \cdot \mu_r} \cdot \frac{0.5}{12 \times 10^{-4}} = 255056 \text{ AT/Wb}$$

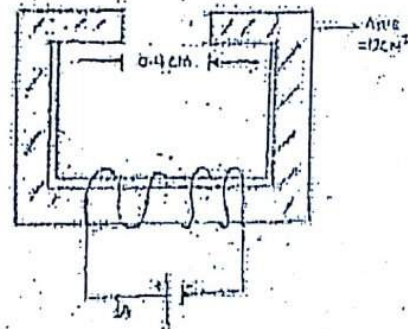
For air-gap:

$$\text{Reluctance } (S_2) = \frac{1}{\mu} \frac{L}{A} = \frac{1}{\mu_0 \cdot \mu_r} \cdot \frac{L}{A} = 2040.44 \text{ AT/Wb}$$

$$\text{Total reluctance } (S) = S_1 + S_2 = (255056 + 2040) \text{ AT/Wb} = 257096 \text{ AT/Wb}$$

$$\text{Flux } (\phi) = \frac{\text{m.m.f.}}{S} = \frac{800}{257096} = 3.11 \times 10^{-3} \text{ Wb}$$

$$\text{Flux density } (B) = \frac{\phi}{A} = \frac{3.11 \times 10^{-3}}{12 \times 10^{-4}} = 2.59 \text{ wb/m}^2.$$



2. An iron ring 15 cm in diameter, and 10 cm^2 in cross-section is wound with 200 turns of wire. For a flux density of 1 Wb/m^2 , and a permeability of 500 , Find the exciting current.

Solution:

Length (l) = $\pi(0.15) = 0.4712 \text{ m}$; $N = 200$; $B = 1 \text{ Wb/m}^2$; $\mu_r = 500$; current (I) = ?

$$\text{Mmf} = \text{AT} = I \times 200 = H l = \frac{B}{\mu_0 \cdot \mu_r} l = \frac{1}{4\pi \times 10^{-7} \cdot 500} \times 0.4712$$

$$\Rightarrow I = \frac{1}{4\pi \times 10^{-7} \cdot 500 \cdot 200} \times 0.4712 = 3.75 \text{ A}$$

Electric Machines

3. An iron ring 100 cm mean diameter, and 10 cm^2 cross-section has 1,000 turns of copper wire wound on it. If the permeability of the material is 1500, and it is required to produce a flux density of 1 Wb/m^2 in an air-gap of 2 mm width in the ring, find: (i) reluctance of the ring; (ii) flux required (iii) mmf required (iv) Current produced. Neglect leakage, and fringing.

Solution

- (i) Reluctance $(S) = \frac{1}{\mu_0 \cdot \mu_r \cdot A} \cdot L = \frac{\pi \cdot 1}{4\pi \cdot 10^{-7} \cdot 1500 \cdot 1 \cdot 10^{-3}} = 1.67 \cdot 10^6 \text{ AT/Wb}$
(ii) Flux $(\phi) = BA = 1 \text{ Wb/m}^2 \cdot 10 \cdot 10^{-4} \text{ m}^2 = 10^{-3} \text{ Wb}$
(iii) Mmf required $= \frac{B}{\mu_0} [L_g + (L_i/\mu_r)] = \frac{1}{4\pi \cdot 10^{-7}} [2 \cdot 10^{-3} + \frac{\pi}{1500}] = 3258 \text{ AT}$
(iv) Current $(I) = \text{mmf}/N = 3258/1000 = 3.258 \text{ A}$

4. An iron ring of circular-section 10 cm^2 , and mean circumference 30 cm has air-gap of 2 mm. If the ring is wound with 500 turns, find the exciting current to establish a flux of 0.4 mWb in the air-gap. The relative permeability of iron is assumed to be 2500.

[PU 2002]

Solution:

$$\text{Area } (A) = 10 \text{ cm}^2 = 10^{-3} \text{ m}^2$$

$$\text{Length of iron ring } (L_i) = 30 \text{ cm} = 0.3 \text{ m}$$

$$\text{Number of turns } (N) = 500$$

$$\text{Flux} = 0.4 \cdot 10^{-3} \text{ Wb}$$

$$\mu_r = 2500$$

$$\text{Current } (I) = ?$$

We know that,

$$\text{Mmf} = \frac{B}{\mu_0} [L_g + (L_i/\mu_r)] = \frac{\phi}{\mu_0 \cdot A} [L_g + (L_i/\mu_r)] = 674.81 \text{ AT}$$

$$\text{Also, mmf} = NI$$

$$\text{Or, } I = \text{mmf}/N$$

$$\therefore I = 1.349 \text{ A}$$

TRANSFORMER:

The transformer is a static piece of apparatus by means of which an electrical power is transformed from one alternating current circuit to another with the desired change in voltage and current, without any change in the frequency.

Transformer can be of two types:

[PU 2002]

- i) Step-up transformer is one which receives A.C. electrical energy at one end at certain voltage and delivers it at higher voltage at the other end.
- ii) Step-down transformer is one which receives A.C. electrical energy at one voltage at one end and delivers at the lower voltage.

The use of transformer in transmission system is shown below:



Fig. Use of transformers in transmission system.

Advantages:

1. Increase or decrease the alternating voltage.
2. It has high efficiency (90 – 99 %).
3. There is no loss (friction and windage) in transformer so that wear and tear is less.

Construction of Transformer:

[PU 2002, PU 2010]

1. Iron Core:

→ Iron core is made up of thin laminated silicon steel. Precut insulated sheets are placed on the top of each other. The sheets are overlapped to avoid air-gap. Cooling is provided by air convection, forced air, insulating liquids, or gas.

2. Windings:

→ The winding is made of copper or aluminum conductor insulated with paper or synthetic insulating material. The windings are manufactured in several layers and insulation is placed between winding. The primary and secondary windings are placed on top of each other but insulated by several layers of insulating sheets. The windings are dried in vacuum and impregnated to eliminate moisture.

Types of transformer:

The two basic types of transformer construction used for power and distribution applications are shown below in the figures. Note that the high voltage coils are wound with a greater number of turns of smaller cross-section conductor than the low-voltage coils. The core type, shown in figure (a), has primary and secondary coils wound on different legs, and the shell type, shown in figure (b), has both coils wound on the same leg. The wider spacing between primary and secondary in the core-type transformer gives it an advantage in high-voltage applications. The shell type, however, has advantage of less leakage flux.

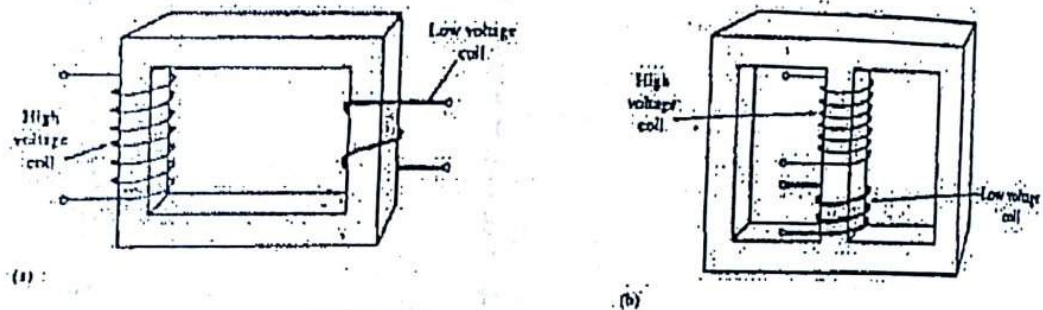


FIGURE
Transformer construction: (a) core type; (b) shell type.

S.No.	Core-type transformer	Shell-type transformer
1.	The windings surround a considerable part of the core.	The core surrounds a considerable portion of the windings.
2.	More suitable for high voltage transformers.	More economical for low voltage transformers.
3.	Mean length of coil turn is shorter.	Mean length of coil turn is longer.

Working Principle:

→ Transformer works on the principle of electromagnetic induction, which states that:

Induced emf is proportional to the rate of change of magnetic flux.

→ An alternating voltage V_1 applied to the primary winding of N_1 turns circulates an alternating current through it. This produces an alternating flux ϕ in the core. Due to this alternating flux, an emf is induced in primary as well as in secondary windings.

$$\rightarrow \text{Then, } V_1 = N_1 \frac{d\phi}{dt}, V_2 = N_2 \frac{d\phi}{dt}$$

$$\Rightarrow \frac{V_2}{V_1} = \frac{N_2}{N_1} = k, \text{ where } k = \text{transformation ratio}$$

$$\text{Also, since } (mmf)_1 = (mmf)_2$$

$$\Rightarrow I_1 N_1 = I_2 N_2$$

$$\Rightarrow \frac{N_1}{N_2} = \frac{I_2}{I_1}$$

$$\text{So, } \frac{V_1}{V_2} = \frac{N_2}{N_1} = \frac{I_1}{I_2}$$

EMF Equation of a Transformer:

[PU 2002, PU 2006, PU 2010]

- ⇒ When the primary winding is excited by an alternating voltage V_1 , it circulates alternating current, producing an alternating flux ϕ . The primary winding has N_1 number of turns. The alternating flux ϕ linking with the primary winding itself induces an e.m.f. in it denoted as E_1 . The flux links with secondary winding through the common magnetic core. It produces induced e.m.f. E_2 in the secondary winding. This is mutually induced e.m.f. Let us derive the equations for E_1 and E_2 .
- ⇒ As ϕ is sinusoidal, the induced e.m.f. in each turn of both the windings is also sinusoidal in nature. For sinusoidal quantity,

$$\text{Form factor} = \frac{\text{R.M.S. Value}}{\text{Average Value}} = 1.11$$

$$\text{Therefore, R.M.S. Value} = 1.11 \times \text{Average value}$$

$$\begin{aligned} \text{So, R.M.S. value of induced e.m.f. per turn} \\ &= 1.11 \times 4 f \phi_m \\ &= 4.44 f \phi_m \end{aligned}$$

- ⇒ There are N_1 number of primary turns hence the R.M.S. value of induced e.m.f. of primary denoted as E_1 is,

$$E_1 = N_1 \times 4.44 f \phi_m \text{ volts}$$

While as there are N_2 number of secondary turns the R.M.S. value of induced e.m.f. of secondary denoted as E_2 is,

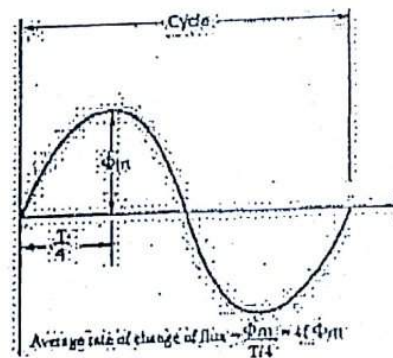
$$E_2 = N_2 \times 4.44 f \phi_m \text{ volts}$$

The expressions of E_1 and E_2 are called e.m.f equations of a transformer.

Thus, e.m.f. equations are,

$$E_1 = N_1 \times 4.44 f \phi_m \text{ volts}$$

$$E_2 = N_2 \times 4.44 f \phi_m \text{ volts}$$



Transformer on No-load condition:

[PU 2012]

→ A transformer is said to be no-load if its secondary side is open and primary side is considered to be sinusoidal alternating voltage V_1 . The alternating applied voltage will cause the flow of alternating current in primary winding which will create alternating flux. The primary input current (I_0) under no-load condition supply.

- i. Iron losses in core
- ii. Copper loss (In primary)

Thus, I_0 lags V_1 by angle $\phi_0 < 90^\circ$

No-load power Input: $W_0 = V_1 I_0 \cos \phi_0$

Where, $\cos \phi_0$ = primary power factor under no-load conditions.

The primary current I_0 has two components:

- (i) Active or working or Iron loss component [I_w]:

$$I_w = I_0 \cos \phi_0$$

- (ii) Magnetising components [I_m]:

$$I_m = I_0 \sin \phi_0$$

$$\text{Also, } I_0 = \sqrt{I_w^2 + I_m^2}$$

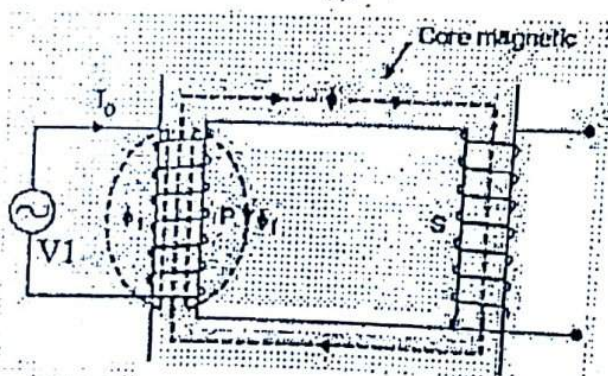


Fig. Transformer on No-Load condition

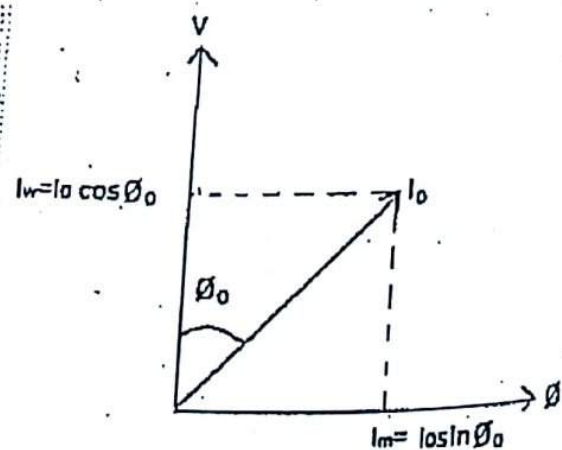


Fig: Phasor diagram of transformer under no load condition.

Transformer on loaded condition:

- When the transformer is loaded, the current I_2 flows through the secondary winding. The magnitude and phase of I_2 is determined by the load. If load is inductive, I_2 lags V_2 . If load is capacitive, I_2 leads V_2 while for resistive load I_2 is in phase with V_2 .
- There exists a secondary m.m.f. $N_2 I_2$ due to which secondary current sets up its own flux ϕ_2 . This flux opposes the main flux ϕ which is produced in the core due to magnetizing component of no load current. Hence the m.m.f. $N_2 I_2$ is called demagnetizing ampere-turns, which is shown in fig (a).
- The flux ϕ_2 momentarily reduces the main flux, due to which the primary induced e.m.f. E_1 also reduces. Hence, the vector difference $V_1 - E_1$ increases due to which primary draws more current from the supply. This additional current drawn by primary is due to the load hence called load component of primary current denoted as I'_2 as shown in fig (b).
- This current I'_2 is in anti-phase with I_2 . The current I'_2 sets up its own flux ϕ'_2 which opposes the flux ϕ_2 and helps the main flux ϕ . This flux ϕ'_2 neutralizes the flux ϕ_2 produced by I_2 . The m.m.f. i.e. ampere turns $N_1 I'_2$ balances the ampere turns $N_2 I_2$ hence the net flux in the core is again maintained at constant level.

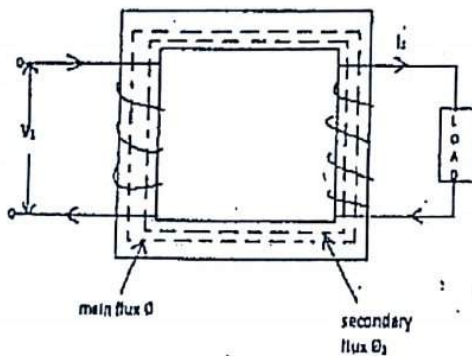


Fig. ϕ_2 opposes ϕ

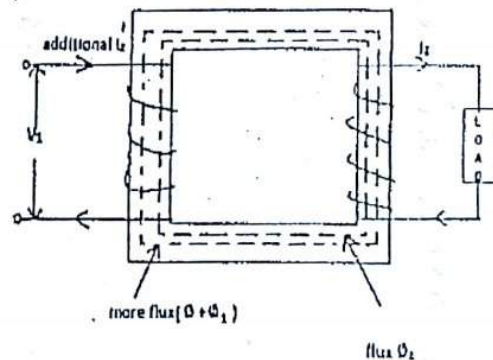


Fig. Primary draws more current

Equivalent circuit of a single phase Transformer:

[PU-2007-F, PU-2008]

→ The complete circuit model (commonly called as equivalent circuit) is drawn below. It comprises following circuit elements.

⇒ Magnetizing shunt branch – R_o and X_o in parallel:

- Resistance (R_o) is connected in parallel with the primary so that when it is connected to the input (supply) voltage, the power absorbed by it equals the core loss in the iron (i.e., hysteresis + eddy-current losses)
- Inductance (X_o) is connected in parallel with the primary so that when it is connected to the input voltage, it takes the magnetizing current I_m responsible for producing flux in the core.

⇒ Primary resistance R_1 and leakage reactance X_1 in series.

⇒ Ideal transformer (turn ratio $N_1/N_2 = \text{constant}$).

⇒ Secondary resistance R_2 and leakage reactance X_2 in series.

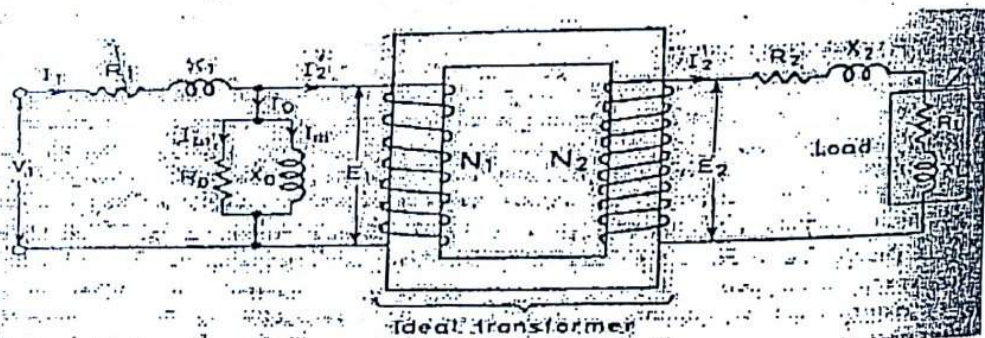


Fig: Equivalent circuit of a transformer

OR

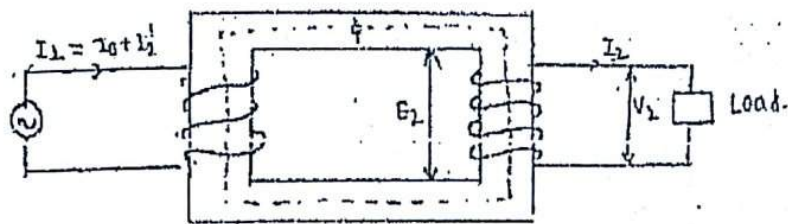


Fig: Ideal Transformer

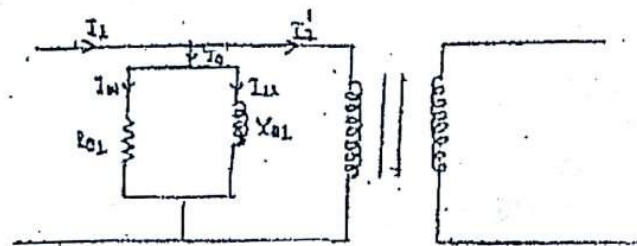


Fig: Considering Iron Loss Component

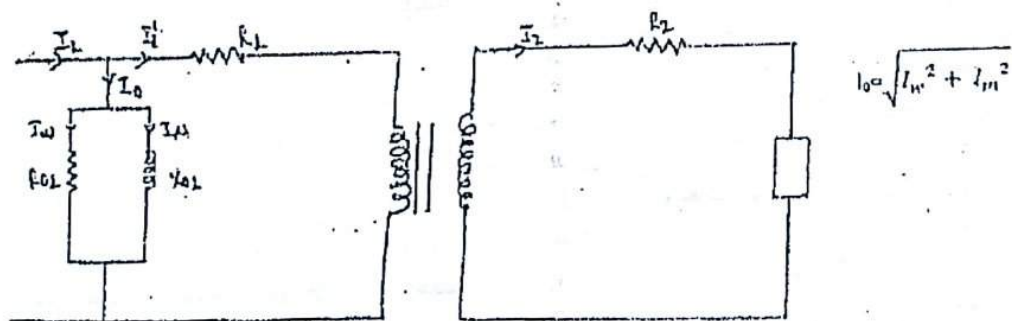


Fig: Considering Primury and Secondary Winding resistance

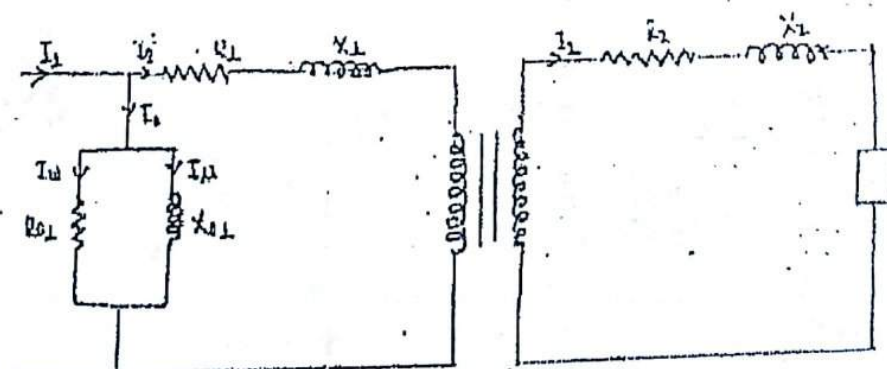


Fig: Considering Leakage flux in Primary and secondary Windings.

Phasor diagram of the transformer at 0.6 pf leading:

[PU-2007F]

→ Consider a transformer supplying the load as shown in the Fig. above.
The various transformer parameters are,

The various transformer parameters are,

R_1 = Primary winding resistance

X_1 = Primary leakage reactance

R_2 = Secondary winding resistance

X_2 = Secondary leakage reactance

Z_L = Load Impedance

I_1 = Primary current

$$I_2 = \text{Secondary current} = I_L = \text{Load current}$$

Now, $I_1 = I_0 + I_2'$.

Where, I_0 = No load current

I_2' = Load component of current decided by the load
 $= K I_1$ where K is transformer

$= K I_2$ where K is transformer component

→ The primary voltage V_1 has now three components,
E: the induced e.m.f. E_1

1. $-E_1$, the induced e.m.f. which opposes V_1
2. $I_1 R_1$, the drop across the resistance, in phase with I_1
3. $I_1 X_1$, the drop across the reactance, leading I_1 by 90°

$$\tilde{V}_1 = -\tilde{E}_1 + \sqrt{1-K_1} + \sqrt{1-X_1}$$

$$V_1 = -E_1 + \sqrt{Z_1}$$

...ရှမ်းပြည်နယ်

→ The secondary induced e.m.f. has also three components,

1. V_2 , the terminal voltage across the load
2. $I_2 R_2$, the drop across the resistance, in phase with I_2
3. $I_2 X_2$, the drop across the reactance, leading I_2 by 90°

$$E_2 = V_2 + I_2 R_2 + I_2 X_2$$

$$\bar{V}_1 = \bar{E}_1 - \bar{I}_1 (R_1 + jX_1)$$

$$V_1 = \bar{E}_1 - \sqrt{2}$$

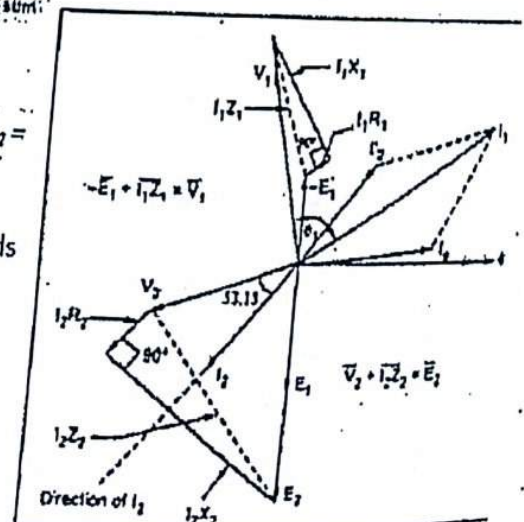
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Leading Power Factor Load, $pf = 0.6$ i.e. $\cos\phi_2 = 0.6 \Rightarrow \phi_2 = 53.13^\circ$

As load power factor is leading, the current I_2 leads V_2 by angle Φ_1 .

Steps to draw the phasor diagram are,

1. Consider flux Φ as reference
2. E_1 lags Φ by 90° . Reverse E_1 to get $-E_1$.
3. E_1 and E_2 are in phase.
4. Assume V_2 in a particular direction
5. I_2 leads V_2 by angle 53.13° (i.e. $\text{pf}=0.6$).



Electric Machines

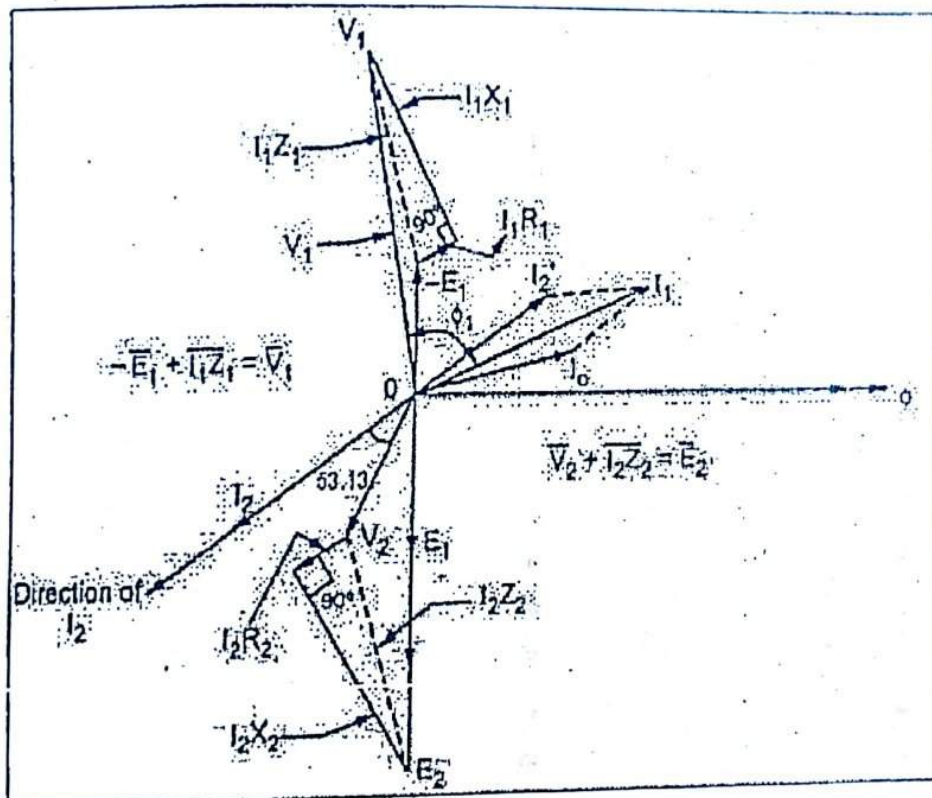
6. Add $I_2 Z_2$ and V_2 to get E_2 .
7. Reverse I_2 to get I_2' .
8. Add I_0 and I_2' to get I_1 .
9. Add $I_1 Z_1$ and to $-E_1$ to get V_1 .

Angle between V_1 and I_1 is Φ_1 and $\cos\Phi_1$ is primary power factor. Remember that $I_1 X_1$ leads I_1 direction by 90° and $I_2 X_2$ leads I_2 by 90° as current through Inductance lags voltage across Inductance by 90° .

The complete phasor diagram is shown in the Fig. aside.

Phasor diagram of the transformer at 0.6 pf lagging:

[PU 2006 S]



⇒ As load power factor is lagging $\cos\Phi_2$, the current I_2 lags V_2 by angle Φ_2 . So only changes in drawing the phasor diagram is to draw I_2 lagging V_2 by Φ_2 in step 5 discussed earlier. Accordingly direction of $I_2 R_2$, $I_2 X_2$, I_2' , I_1 , $I_1 R_1$ and $I_1 X_1$ will change. Remember that whatever may be the power factor of load, $I_2 X_2$ leads I_2 by 90° and $I_1 X_1$ leads I_1 by 90° .

Lagging Power Factor Load, $pf = 0.6$ i.e. $\cos\Phi_2 = 0.6 \Rightarrow \Phi_2 = 53.13^\circ$

LOSSES & EFFICIENCY IN A TRANSFORMER

→ There are mainly two types of losses in a transformer:

(i) Core loss or Iron loss = Hysteresis loss + eddy current loss

⇒ It is the power loss due to heating of iron core and the main causes for this heating are eddy current and hysteresis loss. This power loss is equal to the no-load power loss and remains constant at any load.

(ii) Copper loss = $I_1^2 R_1 + I_2^2 R_2$

⇒ It is the power loss due to heating of primary and secondary coil. The main cause of copper loss is the heat generation due to resistance of coils.

→ Transformer being a static device, the friction and windage losses are absent. So, the efficiency of a transformer is much higher than that of rotating machine's efficiency.

$$\text{Efficiency } (\eta) = \frac{\text{Output Power}}{\text{Input Power}} = \frac{\text{Output Power}}{\text{Output Power} + \text{losses}} = \frac{\text{Input Power} - \text{Losses}}{\text{Input Power}}$$

Now, Primary Input power = $V_1 I_1 \cos \phi_1$

Losses: Cu loss (W_c) = $I_1^2 R_{eq}$

Iron loss (W_I) = $W_h + W_e$

$$\eta = \frac{\text{Input Power} - \text{Losses}}{\text{Input Power}}$$

$$= \frac{V_1 I_1 \cos \phi_1 - I_1^2 R_{eq} - W_I}{V_1 I_1 \cos \phi_1}$$

$$= 1 - \frac{I_1^2 R_{eq}}{V_1 I_1 \cos \phi_1} - \frac{W_I}{V_1 I_1 \cos \phi_1}$$

$$= 1 - \frac{I_1 R_{eq}}{V_1 \cos \phi_1} - \frac{W_I}{V_1 I_1 \cos \phi_1}$$

Differentiating on both sides with respect to I_1

$$\frac{d\eta}{dI_1} = 0 - \frac{R_{eq}}{V_1 \cos \phi_1} + \frac{W_I}{V_1 I_1^2 \cos \phi_1}$$

For maximum η , $\frac{d\eta}{dI_1} = 0$

$$\text{Then, } \frac{R_{eq}}{V_1 \cos \phi_1} = \frac{W_I}{V_1 I_1^2 \cos \phi_1}$$

$$W_I = I_1^2 R_{eq}$$

W_I = Cu loss

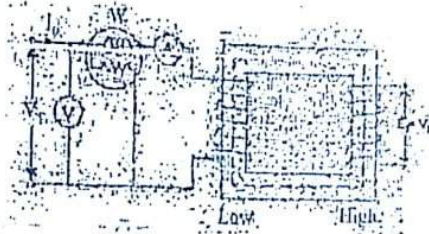
∴ At maximum efficiency, Iron Loss = Copper Loss.

Tests on Transformer:

[PU 2011]

a) *Open Circuit or No-load Test:*

⇒ Open circuit test is used to estimate the iron loss of the transformer and no load component. The transformer is connected in such a way that one of the two winding (high voltage winding) is kept open while the other is connected to supply of rated voltage. An ammeter, voltmeter and wattmeter are connected across low voltage side to measure current, voltage and power respectively as shown in figure.



⇒ Under no load condition, the normal flux ϕ will set-up and there is negligible copper loss. Therefore, wattmeter reading represents the iron loss and ammeter reading represents no load primary current.

⇒ The no-load vector diagram is shown above in Transformer on No-load condition. If 'W' be the wattmeter reading, then

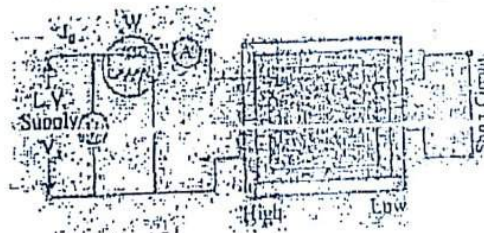
$$W = V_1 I_0 \cos \phi_0 ; \cos \phi_0 = W / (V_1 I_0)$$

$$I_m = I_0 \sin \phi_0 ; I_w = I_0 \cos \phi_0$$

$$X_0 = V_1 / I_m \text{ and } R_0 = V_1 / I_w$$

b) *Short Circuit or Impedance Test:*

⇒ In this test low voltage side of the transformer is short circuited with thick wire in high voltage side. The measurements are taken as shown in figure.



⇒ A small voltage usually (5-10) % of rated voltage is applied to high voltage side. The voltage is adjusted so that full load current will flow in low voltage side.

⇒ Since the applied voltage is only few percentage of rated voltage, the flux produced in the core is very low and iron loss can be neglected. In other words, the wattmeter reading is supposed to be equal to full copper loss in transformer.

⇒ If V_{sc} is the voltage required to circulate full load current in secondary then the equivalent impedance, $Z_{eq} = V_{sc} / I_{sc}$; $X_{eq} = \sqrt{Z_{eq}^2 - R_{eq}^2}$

$$W_{sc} = (I_{sc})^2 \cdot R_{eq} ; R_{eq} = W_{sc} / (I_{sc})^2$$

* Why transformer rating in KVA?

→ As seen, Cu loss of a transformer depends on current and iron loss on voltage. Hence, total transformer loss depends on volt-ampere (VA) and not on phase angle between voltage and current i.e. It is independent of load power factor. That is why rating of transformers is in KVA and not in KW.

* Ideal Transformer:

→ An ideal transformer is that which has:

- ⇒ No magnetic leakage.
- ⇒ No winding resistance (i.e. pure inductive coil).
- ⇒ No core losses (i.e. NO hysteresis and eddy current loss).
- ⇒ Efficiency of 100%.

Key point: No transformer is ideal in practice but the value of E_2 is almost equal to V_1 for properly designed transformer.

Points to remember:

1. Ideal transformer: $I_1 N_1 = I_2 N_2$; $V_1 I_1 = V_2 I_2$
 $\frac{E_1}{E_2} = \frac{V_1}{V_2} = \frac{I_2}{I_1} = \frac{N_1}{N_2} = \frac{1}{k}$; k = transformation ratio
2. $E_1 = N_1 \times 4.44 f \phi_m$ volts
 $E_2 = N_2 \times 4.44 f \phi_m$ volts
3. Transformer on no-load:
 - ⇒ $I_w = I_0 \cos \phi_0$
 - ⇒ $I_m = I_0 \sin \phi_0$
 - ⇒ $I_0 = \sqrt{I_w^2 + I_m^2}$
 - ⇒ Power at no load, $w_0 = V_1 I_0 \cos \phi_0$
4. Equivalent resistance referred to primary, $R_{eq} = R_1 + \left(\frac{N_1}{N_2}\right)^2 R_2$
Equivalent reactance referred to primary, $X_{eq} = X_1 + \left(\frac{N_1}{N_2}\right)^2 X_2$
5. Equivalent resistance referred to secondary, $R'_{eq} = R_2 + \left(\frac{N_2}{N_1}\right)^2 R_1$
Equivalent reactance referred to secondary, $X'_{eq} = X_2 + \left(\frac{N_2}{N_1}\right)^2 X_1$
6. % voltage regulation = $\frac{V_{2nl} - V_{2fl}}{V_{2fl}} \times 100$
Where, V_{2nl} and V_{2fl} are the effective values across secondary terminals under no load and full load conditions respectively.
Also, % voltage regulation = $\frac{I_2 (R_2 \cos \phi \pm X_2 \sin \phi)}{V_2} \times 100$.
Where, '+' → lagging p.f.
 '-' → leading p.f.

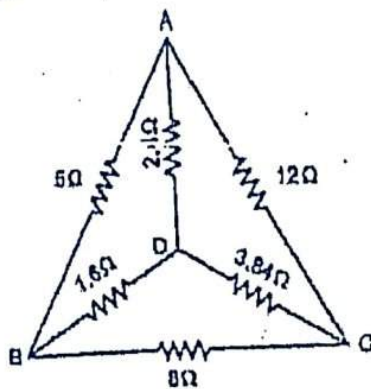


Fig.1

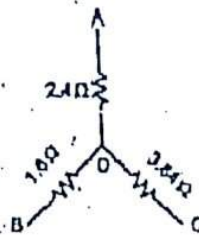


Fig.2

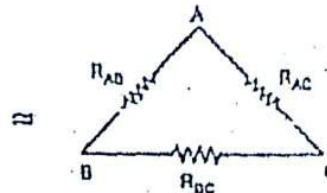
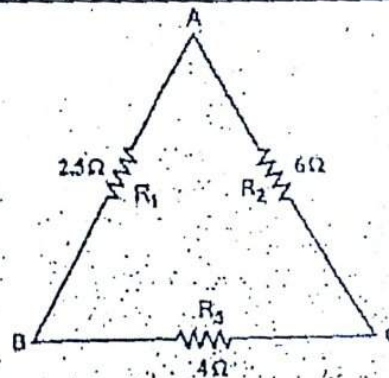


Fig.3



Solution:-

By converting the star ADCB into equivalent delta we get the fig. 2

$$\therefore R_{AB} = 2.4 + 1.6 + \frac{2.4 \times 1.6}{5.84} = 5\Omega$$

$$\therefore R_{AC} = 2.4 + 3.84 + \frac{2.4 \times 3.84}{1.6} = 12\Omega$$

$$\therefore R_{BC} = 1.6 + 3.84 + \frac{1.6 \times 3.84}{2.4} = 8\Omega$$

From fig.3

$$\text{Resistance between A \& B} = (6 + 4) \parallel 2.5 = 2\Omega$$

$$\text{Resistance between B \& C} = (2.5 + 6) \parallel 4 = 2.72\Omega$$

$$\text{Resistance between C \& A} = (4 + 2.5) \parallel 6 = 3.12\Omega$$

25. Use Mesh analysis to find mesh current in the given figure.

[PU 2010 S]

$$\frac{V_1}{V_2} = \frac{N_1}{N_2} \Rightarrow N_2 = 80 \text{ turns}$$

$$\text{Again, } V/\text{turns} = V_1/N_1 \text{ or } V_2/N_2 = 5.5 \text{ V}$$

2. A three-phase 11000/440, 330 KVA connected transformer has 55 turns on secondary side. Find the number of primary turn and secondary phase current.

→ **Solution:**

Since, $V_1 > V_2$ so, it is step-down transformer

$$V_1 = 11000 \text{ V} \quad V_2 = 440 \text{ V} \quad Q = 330 \text{ KVA}$$

$$N_1 = ? \quad N_2 = 55 \text{ turns}$$

We have,

$$\frac{V_1}{V_2} = \frac{N_1}{N_2}$$

On solving: $N_1 = 1375 \text{ turns}$

In Secondary side:

$$I_2 = \frac{\text{KVA}}{\text{KV}} = \frac{330 \cdot 10^3}{440} = 750 \text{ A}$$

3. A single phase transformer has 1110/555 V and 50 Hz supply which is having a maximum value of flux density 1 Wb/m^2 in the iron core. If the net cross sectional area of core is 100 cm^2 , Calculate the primary and secondary turns of the transformer.

→ **Solution:**

$$V_1 = 1110 \text{ V}; \quad V_2 = 555 \text{ V}; \quad f = 50 \text{ Hz}$$

$$B_{\text{max}} = 1 \text{ Wb/m}^2; \quad A = 100 \text{ cm}^2 = 100 \cdot 10^{-4} \text{ m}^2; \quad \phi_m = B_{\text{max}} \cdot A = 100 \cdot 10^{-4} \text{ Wb}$$

$$N_1 = ? \quad N_2 = ?$$

We have,

$$N_2 = \frac{V_2 \cdot N_1}{V_1} = \frac{V_2 \cdot N_1}{N_1 \times 4.44 f \phi_m} = \frac{V_2}{4.44 f \phi_m} = \frac{555}{4.44 \cdot 50 \cdot 100 \cdot 10^{-4}} = 250 \text{ turns}$$

$$N_1 = \frac{V_1 \cdot N_2}{V_2} = \frac{V_1 \cdot N_2}{N_2 \times 4.44 f \phi_m} = \frac{V_1}{4.44 f \phi_m} = \frac{1110}{4.44 \cdot 50 \cdot 100 \cdot 10^{-4}} = 500 \text{ turns}$$

4. A 2200/200V transformer draws a no load primary current of 0.6 A and absorbs 400 watts. Find the magnetizing and iron loss currents.

→ Since, $V_1 > V_2$ so, it is step-down transformer which is at no load condition.

$$V_1 = 2200 \text{ V} \quad V_2 = 200 \text{ V}$$

$$I_0 = 0.6 \text{ A} \quad W_0 = 400 \text{ Watts}$$

$$\text{Magnetizing current loss } (I_m) = ?$$

$$\text{Iron loss current } (I_w) = ?$$

$$\text{Power at no-load, } W_0 = V_1 I_0 \cos \phi_0$$

$$\text{On solving, we get, } \phi_0 = 72.36^\circ$$

Electric Machines

- $\Rightarrow$ Magnetizing current loss (I_m) = $I_0 \sin \phi_0 \Rightarrow 0.57$ A
 $\Rightarrow$ Iron loss current (I_w) = $I_0 \cos \phi_0 \Rightarrow 0.18$ A

5. The maximum flux density in the core of a 250/3000 V 50 Hz single phase transformer is 1.2 Wb/m^2 . If the e.m.f. per turn is 8V, determine primary and secondary turns and area of the core.

$\rightarrow$ In step-up transformer,

$$B = 1.2 \text{ Wb/m}^2 \quad E_1 = 250 \text{ V} \quad E_2 = 3000 \text{ V} \quad f = 50 \text{ Hz}$$

$$E/\text{turn} = 8 \text{ V} \quad N_1 = ? \quad N_2 = ? \quad \text{Area (A)} = ?$$

We know that,

$$E/\text{turns} = E_1/N_1 \text{ or } E_2/N_2$$

So, $E/\text{turns} = E_1/N_1$; upon solving, $N_1 \sim 31$

Also, $E/\text{turns} = E_2/N_2$; upon solving, $N_2 = 375$

Again, $E/\text{turns} = 4.44 f \Phi_m = 4.44 f B.A$ ($\Phi_m = B.A$); upon solving, $A = 0.03 \text{ m}^2$.

6. A 25 KVA 2200/220 V, 50 Hz single phase transformer has the following resistance and leakage reactances,

$$R_1 = 0.8 \Omega \quad R_2 = 0.009 \Omega$$

$$X_1 = 3.2 \Omega \quad X_2 = 0.03 \Omega$$

Calculate R_{eq} and X_{eq} referred to primary and secondary side.

$\rightarrow$ Solution:

We know that, $V_1/V_2 = N_1/N_2 = 10 = \text{transformation ratio} \dots \dots (i)$

a) Referred to primary side:

$$R_{eq} = R_1 + \left(\frac{N_1}{N_2}\right)^2 R_2$$

$$= 0.8 \Omega + (10)^2 \cdot 0.009 \Omega$$

$$= 1.7 \Omega$$

$$X_{eq} = X_1 + \left(\frac{N_1}{N_2}\right)^2 X_2$$

$$= 3.2 \Omega + (10)^2 \cdot 0.03$$

$$= 6.2 \Omega$$

b) Referred to secondary side:

$$R'_{eq} = R_2 + \left(\frac{N_2}{N_1}\right)^2 R_1$$

$$= 0.009 \Omega + (0.1)^2 \times 0.8 \Omega$$

$$= 0.017 \Omega$$

$$X'_{eq} = X_2 + \left(\frac{N_2}{N_1}\right)^2 X_1$$

$$= 0.03 \Omega + (0.1)^2 \times 3.2 \Omega$$

$$= 0.062 \Omega$$

7. Single phase, 3 KVA, 230/115 V, 50 Hz transformer has the following constants.

Resistance: Primary 0.3Ω , Secondary 0.09Ω

Reactance: Primary 0.4Ω , Secondary 0.1Ω

Resistance of equivalent exciting circuit referred to the Primary, $R_0 = 600 \Omega$.

Reactance of equivalent exciting circuit referred to Primary, $X_0 = 200 \Omega$.

What would be the reading of the instrument when the transformer is connected for?

(i) O.C. Test

(ii) S.C. Test

In Both test, Supply is given to H.V. side.

→ Solution:

Primary Resistance (R_1) = 0.3Ω

Primary Reactance (X_1) = 0.4Ω

Secondary Resistance (R_2) = 0.09Ω

Secondary Reactance (X_2) = 0.1Ω

$R_0 = 600 \Omega$

$X_0 = 200 \Omega$

We know that, $V_1/V_2 = N_1/N_2 = 2/1 =$ transformation ratio

(i) O.C. Test :

Magnetizing components of no-load current,

$$I_m = \frac{V_1}{X_0} = \frac{230}{200} = 1.15 \text{ A}$$

Wattful Components of no-load current,

$$I_w = \frac{V_1}{R_0} = \frac{230}{600} = 0.383 \text{ A}$$

$$\therefore \text{No-load Current, } I_0 = \sqrt{I_w^2 + I_m^2} = 1.21 \text{ A}$$

$$\text{Input on no-load to H-V Winding} = V_1 I_w = 230 \times 0.383 = 88.09 \text{ W}$$

Hence, the reading of the instruments are:

Voltmeter reading = 230V

Ammeter reading = 1.21 A

Wattmeter reading = 88.09 W

(ii) S.C. Test :

Referred to HV side:

$$R_{eq} = R_1 + R_2' = R_1 + R_2 \left(\frac{N_1}{N_2} \right)^2 = 0.3 + 0.09 \left(\frac{230}{115} \right)^2 = 0.66 \Omega$$

$$X_{eq} = X_1 + X_2' = X_1 + X_2 \left(\frac{N_1}{N_2} \right)^2 = 0.4 + 0.1 \left(\frac{230}{115} \right)^2 = 0.8 \Omega$$

$$Z_{eq} = \sqrt{R_{eq}^2 + X_{eq}^2} = \sqrt{0.66^2 + 0.8^2} = 1.03 \Omega$$

$$\text{Full-load current in the H.V. winding, } I_L = \frac{3 \times 10^3}{230} = 13.04 \text{ A}$$

Voltage to be applied to H.V. winding on short circuit L.V. winding to pass full-load current of 13.04 A through H.V. winding:

$$V_{sc} = I_L Z_{eq} = 13.04 \times 1.03 = 13.43 \text{ V}$$

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Power Supplied to the H.V. winding on short-circuit,
 $P = I_L^2 R_{eq} = (13.04)^2 \times 0.66 = 112.22 \text{ W}$

Hence, the reading of instruments are:

Voltmeter reading = 13.52 V

Ammeter reading = 13.04 A

Wattmeter reading = 112.22 W

8. The equivalent circuit for a 200/400 V step-up transformer has the following parameters referred to the low-voltage side.

Equivalent resistance = 0.15Ω ;

Equivalent reactance = 0.37Ω

Core-loss component resistance = 600Ω ;

Magnetizing reactance = 300Ω .

When the transformer is supplying a load at 10 A at a power factor of 0.8 lag. Calculate:

- The primary current
- Secondary terminal voltage

→ Solution:

We are given the following:

$$R_{eq} = 0.15 \Omega \quad X_{eq} = 0.37 \Omega \quad R_0 = 600 \Omega \quad X_0 = 300 \Omega$$

Using the approximate equivalent circuit (as in qn. No. 14). We can have,

$$I_m = \frac{V_1}{X_0} = \frac{200}{300} = \frac{2}{3} \text{ A}$$

$$I_w = \frac{V_1}{R_0} = \frac{200}{600} = \frac{1}{3} \text{ A}$$

$$I_0 = \sqrt{I_w^2 + I_m^2} = 0.745 \text{ A}$$

As seen from figure aside

$$\tan \theta = \frac{I_w}{I_m} = \frac{1}{2} \quad \theta = 26.6^\circ$$

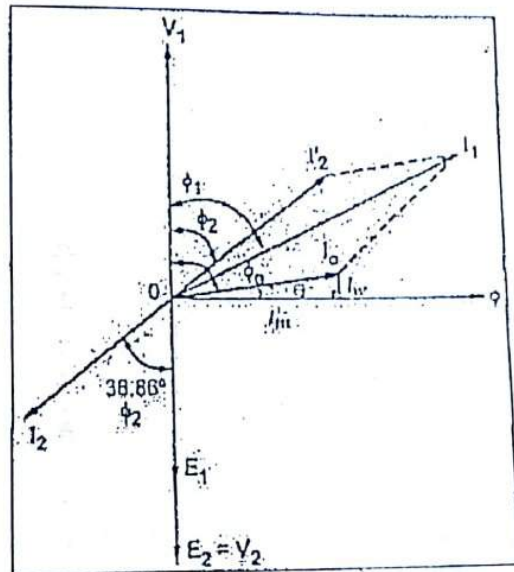
$$\therefore \phi_0 = 90^\circ - 26.6^\circ = 63.4^\circ$$

$$I_2' = 63.4^\circ - 36.86^\circ = 26.54^\circ \quad k = \frac{400}{200} = 2$$

$$I_2' = KI_2 = 2 \times 10 = 20 \text{ A}$$

$$a. \quad I_1 = \sqrt{(0.745^2 + 20^2 + 2 \times 0.745 \times 20 \times \cos 26.54^\circ)} \\ = 20.67 \text{ A}$$

$$b. \quad R_2 = R_{eq} (N_2/N_1)^2 = R_{eq} (K)^2 = 0.15(2)^2 = 0.6 \Omega \text{ [referred to H.V. Side]}$$



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$$X_2 = X_{eq} (N_2/N_1)^2 = X_{eq} (K)^2 = 0.37 (2)^2 = 1.48 \Omega$$

Approximate voltage drop

$$= I_2 (R_2 \cos \phi + X_2 \sin \phi)$$

$$= 10 (0.6 \times 0.8 + 1.48 \times 0.6)$$

$$= 13.7 \text{ V}$$

$$\therefore \text{Secondary terminal voltage} = 400 - 13.7 = 386.3 \text{ V}$$

9. A 4 KVA, 400/200 V, 50 Hz 1 ϕ transformer has the following test data:

O.C. test: 200V 1A 64W (on LV side)

S.C. test: 15V 10A 80W (on HV side)

Determine:

(i) Circuit referred to L.V side

(ii) Secondary load voltage on full load at 0.8 power factor lagging.

→ Solution:

We know that, $V_1/V_2 = N_1/N_2 = 2/1 = \text{transformation ratio}$

(i)

O.C. test on LV side:

$$V_2 = 200 \text{ V}; I_0 = 1 \text{ A}; W = 64 \text{ W}$$

$$\therefore \cos \phi_0 = W/(V_2 I_0) = 0.32, \text{ and } \sin \phi_0 = \sqrt{1 - \cos^2 \phi} = 0.94$$

Magnetizing component of no-load current:

$$I_m = I_0 \sin \phi_0 = 0.94 \text{ A}$$

Wattful component of no-load current:

$$I_w = I_0 \cos \phi_0 = 0.32 \text{ A}$$

Magnetising reactance:

$$X_{m2} = V_2/I_m = 212.76 \Omega$$

Resistance representing the core loss:

$$R_{w2} = V_2/I_w = 625 \Omega$$

S.C. test on HV side:

$$V_{sc} = 15 \text{ V}; I_{sc} = 10 \text{ A}; W_{sc} = 80 \text{ W}$$

$$R_{eq} = W_{sc}/(I_{sc})^2 = 0.8 \Omega$$

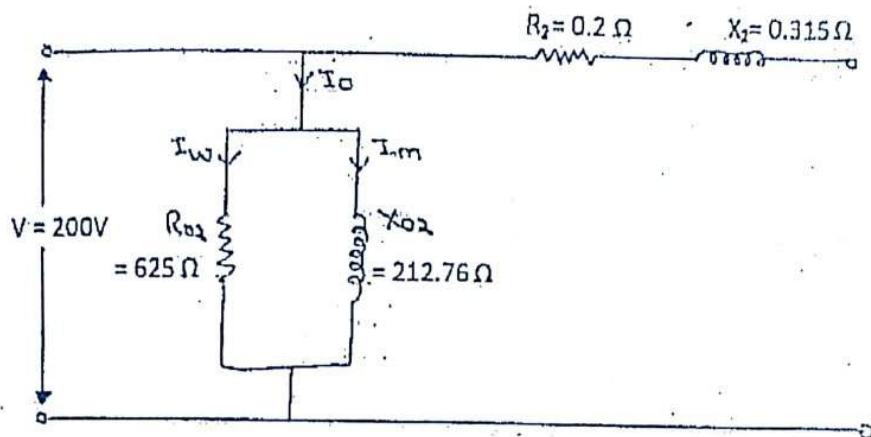
$$Z_{eq} = V_{sc}/I_{sc} = 1.5 \Omega; X_{eq} = \sqrt{Z_{eq}^2 - R_{eq}^2} = 1.26 \Omega$$

∴ Required Parameters of equivalent circuit (Resistance and reactance) referred to L.V. side are:

$$R_2 = R_{eq} (N_2/N_1)^2 = 0.8 (200/400)^2 = 0.2 \Omega$$

$$X_2 = X_{eq} (N_2/N_1)^2 = 1.26 (200/400)^2 = 0.315 \Omega$$

$$Z_2 = R_2 + jX_2 = 0.2 + j0.315 \text{ [Rectangular form]} = 0.374 \angle 57.76^\circ \text{ [Polar Form]}$$

Equivalent circuit:

(ii)

Secondary load voltage (V_2) = ?Secondary Full-load current, $I_2 = \frac{4 \times 10^3}{200} = 20 \text{ A}$ Given: $\cos \phi_2 = 0.8$ (lagging).so, $I_2 = 20 \angle -36.86^\circ \text{ A}$ ($\phi_2 = 36.86^\circ$)

Secondary Load voltage:

$$V_2 = 200 - I_2 Z_2$$

$$= 200 - (20 \angle -36.86^\circ \times 0.374 \angle 57.76^\circ)$$

$$= 200 - (7.48 \angle 20.9^\circ)$$

$$\therefore V_2 \sim 193 \text{ V}$$

10. Obtain the approximate equivalent circuit of a given 200/2000V single phase 30 KVA transformer having the following test results: [Similar to PU: 2007 F]

O.C. test;	200V	6.2A	360W (on LV side)
S.C. test;	75V	18A	600W (on HV side)

→ Solution:O.C. test on LV side:

$$V_1 = 200\text{V}; I_0 = 6.2 \text{ A}; W = 360 \text{ W}$$

$$\therefore \cos \phi_0 = W / (V_1 I_0) = 0.29, \text{ and } \sin \phi_0 = \sqrt{1 - \cos^2 \phi_0} = 0.95$$

Magnetizing component of no-load current:

$$I_m = I_0 \sin \phi_0 = 5.89 \text{ A}$$

Wattful component of no-load current:

$$I_w = I_0 \cos \phi_0 = 1.79 \text{ A}$$

Magnetising reactance:

$$X_0 = V_1 / I_m = 33.95 \Omega$$

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Resistance representing the core loss:

$$R_o = V_1 / I_w = 111.73 \Omega$$

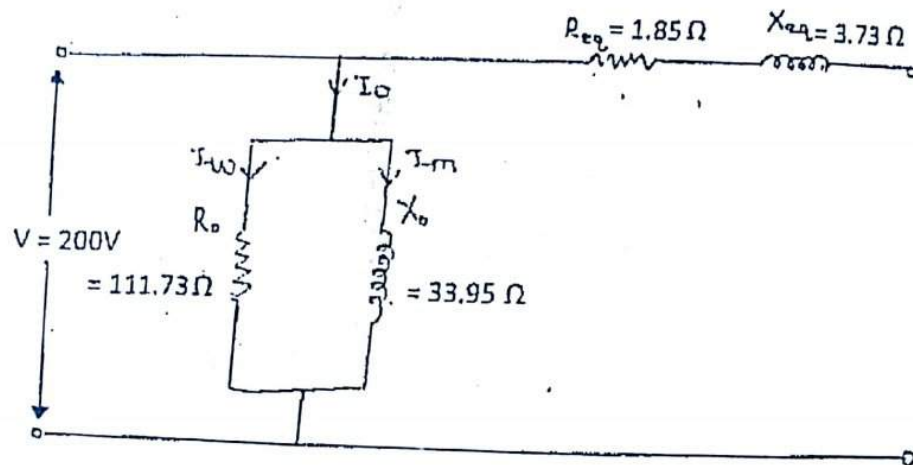
S.C. test on HV side:

$$V_{sc} = 75 \text{ V}; I_{sc} = 18 \text{ A}; W_{sc} = 600 \text{ W}$$

$$R_{eq} = W_{sc} / (I_{sc})^2 = 1.85 \Omega$$

$$Z_{eq} = V_{sc} / I_{sc} = 4.166 \Omega; X_{eq} = \sqrt{Z_{eq}^2 - R_{eq}^2} = 3.73 \Omega$$

Equivalent Circuit:



11. Open Circuit and short circuit test on a 4 KVA, 200/400 V, 50 Hz, single-phase transformer gave the following results:

OC test:	200V	1 Amp	100W (Carried on LV side)
SC test:	15V	10 Amp	85W (Carried on HV side)

Draw the equivalent circuit referred to L.V. side.

Solution:

We know that, $V_1/V_2 = N_1/N_2 = 1/2 = \text{transformation ratio}$

O.C. test on LV side:

$$V_1 = 200 \text{ V}; I_o = 1 \text{ A}; W = 100 \text{ W}$$

$$\therefore \cos \phi_o = W / (V_1 I_o) = 0.5, \text{ and } \sin \phi_o = \sqrt{1 - \cos^2 \phi_o} = 0.866$$

$$I_m = I_o \sin \phi_o = 0.866 \text{ A}, \text{ and } I_w = I_o \cos \phi_o = 0.5 \text{ A}$$

$$X_o = V_1 / I_m = 231 \Omega \text{ and } R_o = V_1 / I_w = 400 \Omega$$

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S.C. test on HV side:

$$V_{sc}=15 \text{ V}; I_{sc}=10 \text{ A}; W_{sc}=85 \text{ W}$$

$$R_{eq} = W_{sc}/(I_{sc})^2 = 0.85 \Omega$$

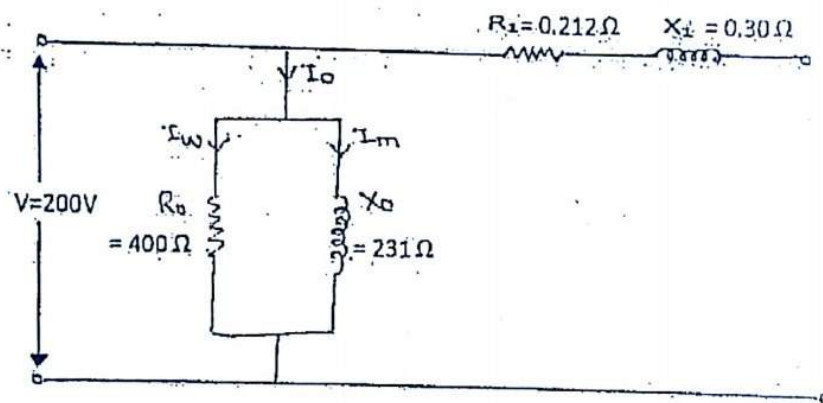
$$Z_{eq} = V_{sc}/I_{sc} = 1.5 \Omega; X_{eq} = \sqrt{(Z_{eq}^2 - R_{eq}^2)} = 1.23 \Omega$$

∴ Resistance and reactance referred to L.V. sides are:

$$R_1 = R_{eq}(N_1/N_2)^2 = 0.85(1/2)^2 = 0.212 \Omega$$

$$X_1 = X_{eq}(N_1/N_2)^2 = 1.23(1/2)^2 = 0.30 \Omega$$

∴ Equivalent circuit referred to L.V. side is:



12. A transformer has a primary winding of 800 turns and secondary windings of 200 turns. When the load current on the secondary is 80 A at 0.8 p.f. lagging, the primary current is 25 A at 0.707 p.f. lagging. Determine the no load current of the transformer. Draw the vector diagram.

[PU

2003]

→ Solution:

$$\phi_2 = \cos^{-1}(0.8) = 36.86^\circ \text{ (lag)} \quad \phi_1 = \cos^{-1}(0.707) = 45^\circ \text{ (lag)}$$

$$\text{Primary Current } (I_1) = 25 \angle -45^\circ \text{ A}$$

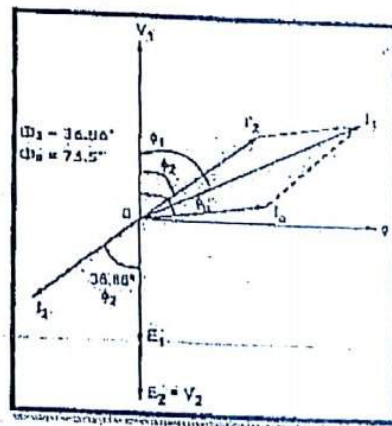
$$\text{Secondary Current } (I_2) = 80 \angle -36.86^\circ \text{ A}$$

$$\text{Additional Current in Primary side } (I_2') = I_2 \frac{N_2}{N_1} =$$

$$80 \cdot \frac{200}{800} \Rightarrow 20 \angle -36.86^\circ \text{ A}$$

$$\text{No load current } (I_0) = ?$$

We know that,



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$$I_1 = I_0 + I_2'$$

$$\text{Or, } I_0 = I_1 - I_2'$$
$$= (25 \angle -45^\circ - 20 \angle -36.86^\circ) \text{ Amp}$$

$$\therefore I_0 = 5.92 \angle -73.5^\circ \text{ Amp}$$

Fig: Phasor diagram

13. A 50 KVA, 4400/220 V transformer has $R_1=3.45\Omega$, $R_2=0.009\Omega$. The values of reactance are $X_1=5.2\Omega$ and $X_2=0.015\Omega$. For the transformer calculate: [PU 2006]

- equivalent resistance as referred to both primary and secondary
- equivalent impedance as referred to both primary and secondary
- Total copper loss, first using individual resistances of the two windings and secondly using equivalent resistances as referred to each side.

→ Solution:

We know that, $V_1/V_2 = N_1/N_2 = 20/1 = \text{transformation ratio}$

- equivalent resistance as referred to both primary and secondary

Referred to primary side:

$$R_{eq} = R_1 + \left(\frac{N_1}{N_2}\right)^2 R_2$$
$$= 3.45 \Omega + (20)^2 \cdot 0.009 \Omega$$
$$= 7.05 \Omega$$

Referred to secondary side:

$$R'_{eq} = R_2 + \left(\frac{N_2}{N_1}\right)^2 R_1$$
$$= 0.009 \Omega + (0.05)^2 \times 3.45 \Omega$$
$$= 0.017 \Omega$$

- equivalent impedance as referred to both primary and secondary

Referred to primary side:

$$Z_{eq} = Z_1 + \left(\frac{N_1}{N_2}\right)^2 Z_2$$
$$= (3.45 \Omega + 5.2j) + (20)^2 (0.009 \Omega + 0.015j)$$
$$= 7.05 \Omega + 11.2j$$

Referred to secondary side:

$$Z'_{eq} = Z_2 + \left(\frac{N_2}{N_1}\right)^2 Z_1$$
$$= (0.009 \Omega + 0.015j) + (0.05)^2 \times (3.45 \Omega + 5.2j)$$

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$$= 0.017 \Omega + 0.028 \Omega$$

c. Total copper loss, first using individual resistances of the two windings and secondly using equivalent resistances as referred to each side.

⇒ We know that, $KVA = E_1 I_1 / 1000 = E_2 I_2 / 1000$; Solving: $I_1 = 11.36 \text{ A}$; similarly, $I_2 = 227.27 \text{ A}$

First part:

$$W_c = I_1^2 R_1 + I_2^2 R_2 = 445.22 + 464.86 = 910 \text{ W (approx.)}$$

Second part:

$$W_c = I_1^2 R_{eq} \text{ (Primary side)} = 910 \text{ W}$$

$$\text{Similarly, } I_2^2 R'_{eq} \text{ (Secondary side)} = 878 \text{ W}$$

14. A 220/440 V, 50 Hz transformer gave the following data on tests. Obtain the parameters of equivalent circuit.

[PU2007]

O.C. test:	220V	0.6A	66W (on LV side)
S.C. test:	17V	10A	80W (on HV side)

Solution:

O.C. test on LV side:

$$V_1 = 220 \text{ V}; I_0 = 0.6 \text{ A}; W = 66 \text{ W}$$

$$\therefore \cos \phi_0 = W / (V_1 I_0) = 0.5, \text{ and } \sin \phi_0 = \sqrt{1 - \cos^2 \phi_0} = 0.866$$

$$I_m = I_0 \sin \phi_0 = 0.51 \text{ A}, \text{ and } I_w = I_0 \cos \phi_0 = 0.3 \text{ A}$$

$$X_0 = V_1 / I_m = 431.37 \Omega \text{ and } R_0 = V_1 / I_w = 733.34 \Omega$$

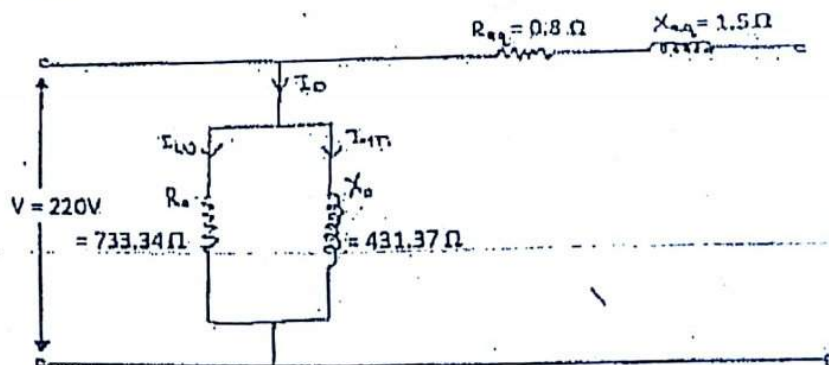
S.C. test on HV side:

$$V_{sc} = 17 \text{ V}; I_{sc} = 10 \text{ A}; W_{sc} = 80 \text{ W}$$

$$R_{eq} = W_{sc} / (I_{sc})^2 = 0.8 \Omega$$

$$Z_{eq} = V_{sc} / I_{sc} = 1.7 \Omega; X_{eq} = \sqrt{Z_{eq}^2 - R_{eq}^2} = 1.5 \Omega$$

Equivalent circuit (optional):



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15. The following data refer to a 50KVA, 2200/110 V, 50 Hz, single phase transformer.
- | | | |
|-----------------------------------|-------|------|
| O.C. test with H.V. open: 380W | 9.5A | 110V |
| S.C. test with L.H. shorted: 820W | 22.7A | 100V |
- Determine the equivalent circuit parameters referred to LV side.

[PU 2008]

Solution:

We know that, $V_1/V_2 = N_1/N_2 = 20/1 = \text{transformation ratio}$

O.C. test on LV side:

$$V_2 = 110V; I_0 = 9.5 A; W = 380 W$$

$$\therefore \cos \phi_0 = W/(V_2 I_0) = 0.3636, \text{ and } \sin \phi_0 = \sqrt{1 - \cos^2 \phi_0} = 0.931$$

$$I_m = I_0 \sin \phi_0 = 8.84 A, \text{ and } I_w = I_0 \cos \phi_0 = 3.45 A$$

$$X_{m2} = V_2/I_m = 12.44 \Omega \text{ and } R_{w2} = V_2/I_w = 31.88 \Omega$$

S.C. test on HV side:

$$V_{sc} = 100 V; I_{sc} = 22.7 A; W_{sc} = 820 W$$

$$R_{eq} = W_{sc}/(I_{sc})^2 = 1.59 \Omega$$

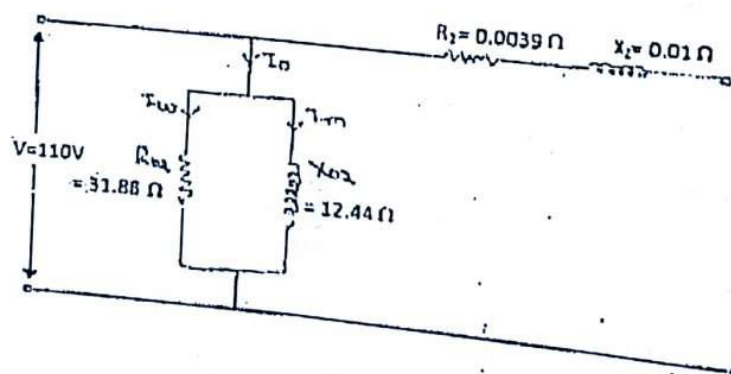
$$Z_{eq} = V_{sc}/I_{sc} = 4.40 \Omega; X_{eq} = \sqrt{Z_{eq}^2 - R_{eq}^2} = 4.10 \Omega$$

$\therefore$ Required Parameters of equivalent circuit (Resistance and reactance) referred to L.V. sides are:

$$R_2 = R_{eq}(N_2/N_1)^2 = 1.59(1/20)^2 = 0.0039 \Omega$$

$$X_2 = X_{eq}(N_2/N_1)^2 = 4.10(1/20)^2 = 0.01 \Omega$$

Equivalent circuit(Optional):



16. A 12 KVA transformer having Primary Voltage of 2000 V at 50 Hz has $N_1=180$, $N_2= 30$.

Find:

- (i) The full load primary and secondary current.
- (ii) The no-load secondary induced EMF.
- (iii) The maximum flux in the core.

[PU: 2008 S]

→ Solution:

$$\frac{V_2}{V_1} = \frac{N_2}{N_1} \Rightarrow V_2 = 333.34 \text{ V}$$

$$\therefore \text{Secondary EMF } (V_2) = 333.34 \text{ V}$$

$$\text{Primary Current } (I_1) = \frac{\text{Power}}{\text{Voltage } (V_1)} = \frac{12000}{2000} \Rightarrow 6 \text{ Amp.}$$

$$\text{Secondary Current } (I_2) = \frac{\text{Power}}{\text{Voltage } (V_2)} = \frac{12000}{333.34} \Rightarrow 36 \text{ Amp.}$$

If ϕ_m is the maximum core-Flux in Wb,

$$E_1 = 4.44 N_1 f \phi_m$$

On Solving: $\phi_m = 0.05 \text{ Wb or } 50\text{-mWb.}$

17. A 10 KVA, 200/400 V, 50-Hz, 1-phase transformer gives the following test results;

O.C. test (HV side open): 200V, 103A, 120W.

S.C. test (LV side shorted): 22V, 30A, 200W.

Find the parameters of equivalent circuit as referred to LV side.

[PU 2009 S]

→ Solution:

O.C. test on LV side:

We know that, $V_1/V_2 = N_1/N_2 = 200/400 = \text{transformation ratio}$

$$V_1=200\text{V}; I_0=103 \text{ A}; W=120 \text{ W}$$

$$\therefore \cos\phi_0 = W/(V_1 I_0) = 0.0058, \text{ and } \sin\phi_0 = \sqrt{1 - \cos^2\phi_0} = 0.999$$

$$I_m = I_0 \sin\phi_0 = 102.99 \text{ A, and } I_w = I_0 \cos\phi_0 = 0.5974 \text{ A}$$

$$X_{o1} = V_1/I_m = 1.94 \Omega \text{ and } R_{o1} = V_1/I_w = 334.78 \Omega$$

S.C. test on HV side:

$$V_{sc}=22 \text{ V}; I_{sc}= 30 \text{ A}; W_{sc}=200 \text{ W}$$

$$R_{eq} = W_{sc}/(I_{sc})^2 = 0.22 \Omega$$

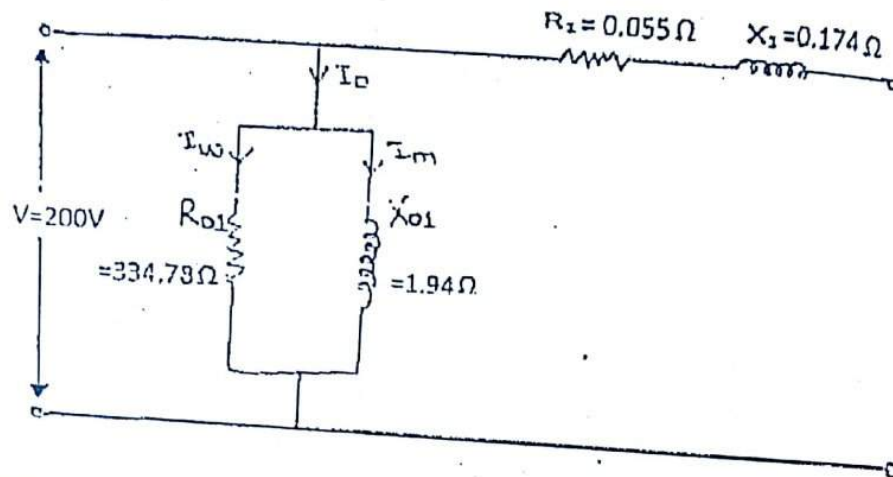
$$Z_{eq} = V_{sc}/I_{sc} = 0.73 \Omega; X_{eq} = \text{sqrt. } (Z_{eq}^2 - R_{eq}^2) = 0.696 \Omega$$

∴ Required-Parameters-of-equivalent-circuit (Resistance-and reactance)-referred-to-L.V. sides are:

$$R_1 = R_{r\eta} (N_1/N_2)^2 = 0.22 (200/400)^2 = 0.055 \Omega$$

$$X_1 = X_{e\eta} (N_1/N_2)^2 = 0.696 (200/400)^2 = 0.174 \Omega$$

Equivalent circuit: (optional)



18. A single phase, 50 KVA, 2200/220 V, 50 Hz, transformer has HV winding resistance of 0.65 ohm and leakage reactance of 0.85 ohm. The corresponding values of LV winding are 0.006 ohm and 0.008 ohm respectively. The shunt branch parameters G_m and b_m determined at rated voltage and frequency, which viewed from LV side are 0.0035 and 0.0215. Draw the equivalent circuit referred to LV side and HV side. [PU 2011]

Solution:

We know that, $V_1/V_2 = N_1/N_2 = 10/1 =$ transformation ratio

Given Data: In a single phase, 50KVA, 2200/220V; 50 Hz transformer

Case: I (HV)

Winding resistance (R_1) = 0.65 Ω

Leakage reactance (X_1) = 0.85 Ω

Impedance (Z_1) = 0.65 + j0.85 Ω

Case II (LV)

Winding resistance (R_2) = 0.006 Ω

Leakage reactance (X_2) = 0.008 Ω

Impedance (Z_2) = 0.006 + j0.008 Ω

As viewed from LV Side:

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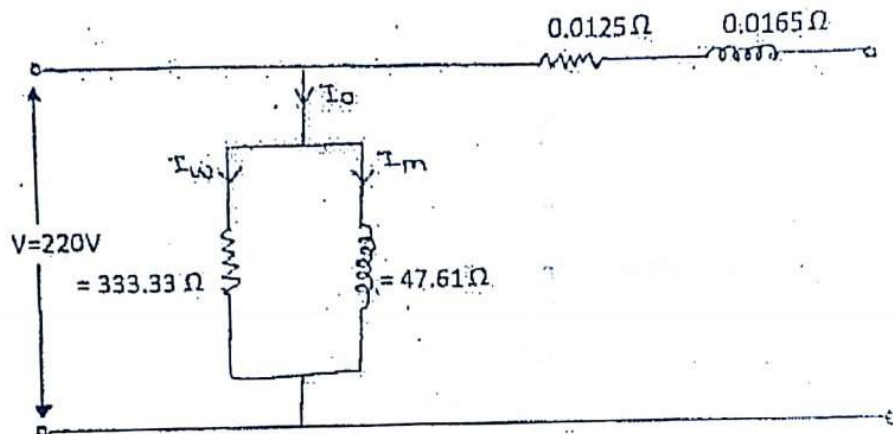
$$g_m = 0.0035 = 333.33 \Omega \text{ [conversion: siemens} \rightarrow \text{ohm]}$$

$$b_m = 0.0215 = 47.61 \Omega$$

Equivalent circuit referred to L.V. side:

$$Z_{LV} = Z_2 + \left(\frac{N_2}{N_1}\right)^2 Z_1$$

$$= (0.006 + j0.008 \Omega) + (0.1)^2 (0.65 + j0.85 \Omega) = 0.0125 + j0.0165 \Omega$$



Equivalent circuit referred to H.V. side:

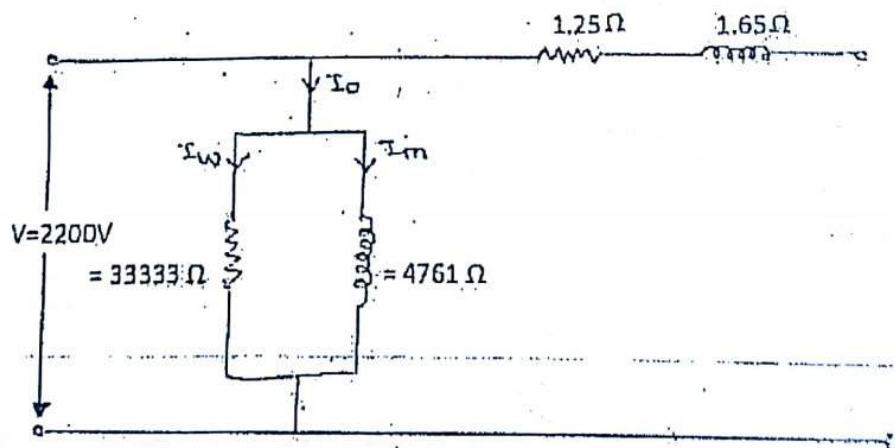
$$Z_{HV} = Z_1 + \left(\frac{N_1}{N_2}\right)^2 Z_2$$

$$= (0.065 + j0.85 \Omega) + (10)^2 (0.006 + j0.008 \Omega) = 1.25 + j1.65 \Omega$$

Shunt branch as seen from HV side:

$$g_m = 333.33 * \left(\frac{N_1}{N_2}\right)^2 = 33333 \Omega$$

$$b_m = 47.61 * \left(\frac{N_1}{N_2}\right)^2 = 4761 \Omega$$



10. A 50KVA, 500/250V transformer has a primary winding resistance of 0.5 ohm and leakage reactance of 0.7 ohm. The secondary winding resistance is 0.8 ohm and leakage reactance of 0.9 ohm. Calculate the equivalent resistance, reactance and impedance of transformer referred to:
a) Primary b) secondary

[PU 2013F]

Solution:

Given data: $R_1 = 0.5 \Omega$, $X_1 = 0.7 \Omega$; $R_2 = 0.8 \Omega$, $X_2 = 0.9 \Omega$

We know that, $V_1/V_2 = N_1/N_2 = 2/1 =$ transformation ratio

$$Z_1 = 0.5 + j0.7 \Omega; Z_2 = 0.8 + j0.9 \Omega$$

→ Referred to Primary:

- i. Equivalent resistance: $R_{eq} = R_1 + \left(\frac{N_1}{N_2}\right)^2 R_2$
 $= 0.5 + (2)^2 \cdot 0.8$
 $= 3.7 \Omega$
- ii. Equivalent reactance: $X_{eq} = X_1 + \left(\frac{N_1}{N_2}\right)^2 X_2$
 $= 0.7 + (2)^2 \cdot 0.9$
 $= 4.3 \Omega$
- iii. Equivalent impedance: $Z_{eq} = Z_1 + \left(\frac{N_1}{N_2}\right)^2 Z_2$
 $= (0.5 + j0.7 \Omega) + (2)^2 \cdot (0.8 + j0.9 \Omega)$
 $= 3.7 + j4.3 \Omega$

→ Referred to Secondary:

- i. Equivalent resistance: $R'_{eq} = R_2 + \left(\frac{N_2}{N_1}\right)^2 R_1$
 $= 0.8 + (0.5)^2 \cdot 0.5$
 $= 0.925 \Omega$
- ii. Equivalent reactance: $X'_{eq} = X_2 + \left(\frac{N_2}{N_1}\right)^2 X_1$
 $= 0.9 + (0.5)^2 \cdot 0.7$
 $= 1.075 \Omega$
- iii. Equivalent impedance: $Z'_{eq} = Z_2 + \left(\frac{N_2}{N_1}\right)^2 Z_1$
 $= (0.8 + j0.9 \Omega) + (0.5)^2 \cdot (0.5 + j0.7 \Omega)$
 $= 0.925 + j1.075 \Omega$

Electric Machines

20. A 25 KVA, 400/200 V, 1 ϕ transformer has high voltage winding resistance and reactance are 0.2 Ω and 0.5 Ω respectively. The values of low voltage are 0.6 Ω and 0.8 Ω respectively. Calculate the equivalent resistance, reactance and impedance referred to; [PU 2013 S]

i. LV side

ii. HV side

→ Solution:

Given:

$$R_1 = 0.2 \Omega, X_1 = 0.5 \Omega; R_2 = 0.6 \Omega, X_2 = 0.8 \Omega$$

We know that, $V_1/V_2 = N_1/N_2 = 2/1 =$ transformation ratio

$$Z_1 = 0.2 + j0.5 \Omega; Z_2 = 0.6 + j0.8 \Omega$$

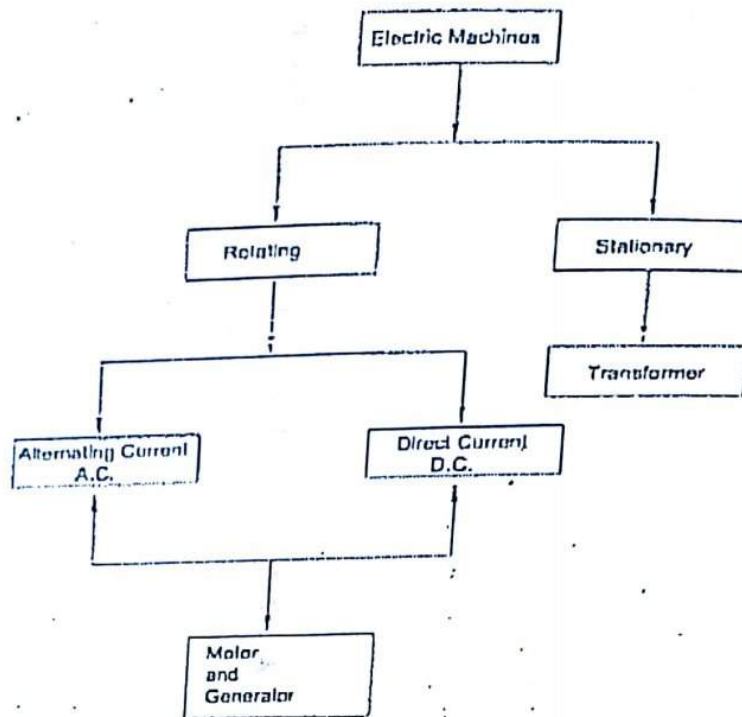
(i) Referred to LV side (Secondary):

- Equivalent resistance: $R'_{eq} = R_2 + \left(\frac{N_2}{N_1}\right)^2 \cdot R_1$
 $= 0.6 + (0.5)^2 \cdot 0.2$
 $= 0.65 \Omega$
- Equivalent reactance: $X'_{eq} = X_2 + \left(\frac{N_2}{N_1}\right)^2 \cdot X_1$
 $= 0.8 + (0.5)^2 \cdot 0.5$
 $= 0.925 \Omega$
- Equivalent Impedance: $Z'_{eq} = Z_2 + \left(\frac{N_2}{N_1}\right)^2 \cdot Z_1$
 $= (0.6 + j0.8 \Omega) + (0.5)^2 \cdot (0.2 + j0.5 \Omega)$
 $= 0.65 + j 0.925 \Omega$

(ii) Referred to HV side (Primary):

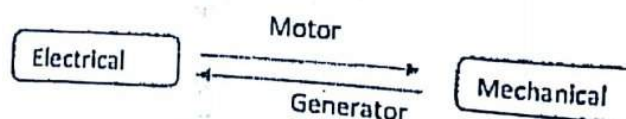
- Equivalent resistance: $R_{eq} = R_1 + \left(\frac{N_1}{N_2}\right)^2 \cdot R_2$
 $= 0.2 + (2)^2 \cdot 0.6$
 $= 2.6 \Omega$
- Equivalent reactance: $X_{eq} = X_1 + \left(\frac{N_1}{N_2}\right)^2 \cdot X_2$
 $= 0.5 + (2)^2 \cdot 0.8$
 $= 3.7 \Omega$
- Equivalent Impedance: $Z_{eq} = Z_1 + \left(\frac{N_1}{N_2}\right)^2 \cdot Z_2$
 $= (0.2 + j0.5 \Omega) + (2)^2 \cdot (0.6 + j0.8 \Omega)$
 $= 2.6 + j 3.7 \Omega$

Electrical Machines:



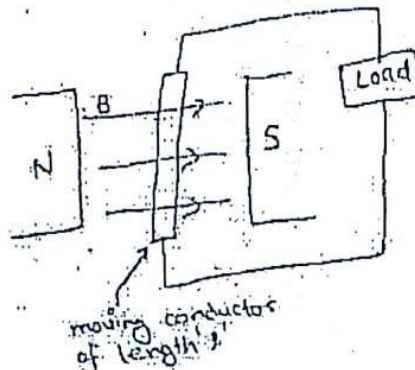
→ Electrical Machines are the rotating devices (machines) and works as electromechanical energy conversion. Thus, they are even termed as electromechanical energy conversion devices. This involves the interchange of energy between an electrical system and a mechanical system through the medium of magnetic field or electric field. The conversion process is essentially reversible except for a small amount, which is lost as heat energy. When the conversion takes place from electrical to mechanical form, the device is called motor. The word motor is derived from Latin which means *mover*. When the mechanical energy is converted to electrical energy, the device is called a generator, which means *producer* in Latin. When the electrical system is characterized by A.C., the devices are referred as A.C. motors and A.C. generators. Similarly, when the electrical system is characterized by direct current, the device are called D.C. motors and D.C. generators.

Principle of energy conversion:

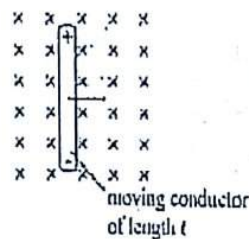


Electric Machines

- The law of motor action is given by ampere's law which states that, if a conductor of length ' l ' carrying current is placed in a magnetic field of flux density ' B ', it will experience a force ' F ' which is given by: $F = BIl$.



- Similarly, The generator action is given by Faraday's law which states that, if a conductor of length ' l ' is moved with a velocity ' v ' in a magnetic field of a constant flux density ' B ' then, induced emf ' E ' is given by: $E = Blv$



DC Machine:

- DC machines [PU 2011] are rotating electrical device/machine which convert mechanical energy to electrical energy or vice-versa. When mechanical energy is converted into electrical energy then the machine is known as generator and when electrical energy is converted into mechanical energy, the machine is known as motor. Generally motors and generators are very similar to each other in construction and essential parts.

DC machine (a generator or a motor)

Construction:

a. Yoke:

- ⇒ It is the outermost frame of the machine. It serves as double purpose in the DC machine:

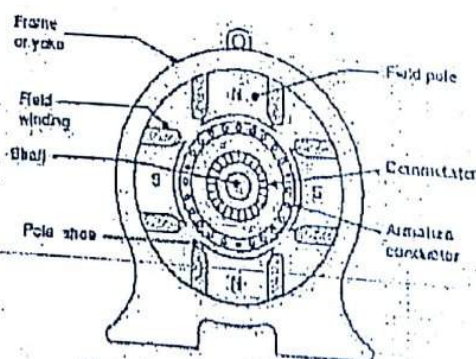


Fig. Cross-sectional view of a 4-pole dc machine.

Electric Machines

- o It provides mechanical support for the field pole and acts as protecting cover for the whole machine.
- o It carries the magnetic flux produced by the field poles.
- b. Field pole:
 - ⇒ Field poles are the iron core projected from the yoke. Upper part of pole connected to yoke is pole-core and lower part is pole-shoe. Combination of pole-shoe surface forms a circular gap at the center of the machine.
- c. Field Winding:
 - ⇒ It is the copper wire or strip wound on the field pole. The windings are insulated from the pole core and each turns of winding are also insulated from each other to protect from turn to turn short circuit. The enamel insulated copper wire is used for that purpose when DC current is passed through these coils. They will magnetize the pole core and produce the magnetic field in the central space of the machine.
- d. Armature:
 - ⇒ It is the rotating part of the machine. The various parts of an Armature are shaft, armature core, commutator and armature winding.
- e. Commutator:
 - ⇒ It is another cylinder fitted on the shaft little away from the armature core. It is made of up copper segments insulated from each-other and from the shaft.
- f. Armature winding
 - ⇒ It is the enamel insulated copper wire wound in the slots of the armature core.
- g. Brushes:
 - ⇒ It is made from carbon graphite block. The function of the brush is to collect current from the rotating armature coil to the stationary load. The carbon brushes are housed in a brush holder usually box-type. The carbon brushes are fixed and touching over the surface of the commutator segments.

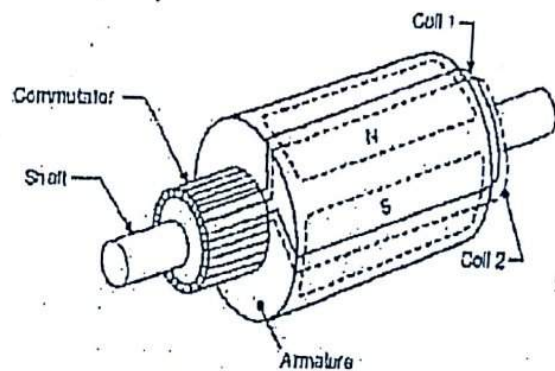


Fig. Longitudinal view of armature and commutator.

*Lap winding VS Wave-winding

S.No.	Lap winding	Wave winding
1.	Number of parallel paths is equal to the number of poles.	Only two parallel paths are there, irrespective of the number of poles.
2.	Full current is equally distributed between the parallel paths.	Full current is equally distributed between the only two parallel paths.

Electric Machines

3.	Number of brush-sets needed equals the number of paths, and hence, the number of poles.	Only two brush-sets are needed.
4.	It is employed for obtaining low voltage, but high current.	It is employed for obtaining high voltage, but low current.

Emf equation of DC machines:[PU 2003, PU 2011]

- The magnetic field produced by the stator poles induces a electromagnetic force (emf) in the rotor (or armature) coils when the machine rotates or is rotated.
- Let, ϕ = flux/pole
 - Z = Total number of conductors on armature
 - P = no of poles
 - A = no of parallel paths
 - N = speed of armature
 - E = EMF Induced in a armature conductor

EMF generated by armature = EMF generated in one of the parallel path

Also,

EMF generated = flux cut per second

----- Flux cut by one conductor in making one revolution = ϕP

Flux cut by one conductor per second = $(\phi P N)/60$

EMF generated in one conductor = $(\phi P N)/60$

Since, ' Z ' no. of conductors are arranged in ' A ' no's of parallel paths.

Therefore, no. of conductors per path = Z/A

Total EMF generated = $(\phi P N)/60 \times Z/A$

If we consider angular velocity as ' ω ' rad/sec. Then, $N = \omega \times (60/2\pi)$

$\therefore E = (Z\phi\omega P) / 2\pi A$

Case I: For a wave-wound armature:

No. of parallel paths (A) = 2

$\therefore E = (\phi Z N P) / 120$

Case II: For a lap-wound armature:

No. of parallel paths (A) = P

$\therefore E = (\phi Z N P) / 60 A$
 $= (\phi Z N P) / 60 P$

Electric Machines

$$= (\phi Z N) / 60$$

Torque equation of DC Machine: [PU 2003]

→ Let, τ_a be the torque developed in the armature rotating at 'N' r.p.m.
∴ Power developed = $\tau_a * (2\pi N)/60$

Also we have,

Electrical power in armature = $E_a * I_a$

Where, E_a = EMF induced in conductor

I_a = Current in armature

For any machine:

Electrical power = mechanical power developed

$$\Rightarrow E_a I_a = \tau_a * (2\pi N)/60$$

$$\text{Also, } E_a = (\phi Z N P) / 60 A$$

$$\Rightarrow I_a * (\phi Z N P) / 60 A = \tau_a * (2\pi N)/60$$

$$\Rightarrow \tau_a = ((\phi Z P) / 2\pi A) * I_a$$

LOSSES IN ROTATING MACHINES/DC Machines

[PU 2013 S]

→ Mainly following losses take place in a rotating machine:

⇒ Copper loss: These are the I^2R loss and takes place in the armature conductors ($I_a^2 R_a$), series field winding ($I^2 R_{se}$) and shunt field winding $\frac{V^2}{R_{sh}}$ or $I_{sh}^2 R_{sh}$.

⇒ Rotational losses: In a DC machine, the core loss is the true rotational loss. Except in the pole faces the flux in the stator remains constant in the steady state operation. The flux in the armature core is constantly alternating as the machine rotates and at zero speed, the core loss is zero. The rotational loss is then consists of the sum of friction windage and core losses. In a generator or a constant speed motor. These are the considered part of the constant losses.

⇒ Brush drop loss: Brush contact loss occurs due to the voltage drop across the brush and is equal to the $2V_d I_a$ where V_d is the voltage drop across one brush.

⇒ Stray load loss: We calculate armature copper loss on the basis of DC resistance whereas the current in the armature winding is alternating. An incremental loss should be included to the account for skin effect, eddy current in the conductors and core loss due to leakage flux in the iron surrounding the armature conductors. The hysteresis and eddy current losses in the pole faces are also somewhat increased at high armature currents. There is no way by which to measure these stray load losses. They are often considered negligible in small DC machines.

Electric Machines

Commutation:

- The armature conductors of D.C. generator produces an alternating induced current, i.e., the current in conductors under N-pole is exactly opposite to that in conductors under the influence of S-pole. As the conductors exit the influence of a pole of one polarity and enter the pole of reverse polarity, the direction of current is exactly reversed. The phenomenon of reversal of directional induced current in armature conductors actually takes place along the so called magnetic neutral axis or brush axis. Consequently, as the brush has an area of contact the particular coil undergoing reversal of the direction of current passing through gets short circuited. The process by which the direction of current in armature while crossing the magnetic neutral axis is short-circuited is called commutation and the brief period during which short circuiting of coils remains is known as commutation period. If current reversal takes more time sparking between commutator and brush occurs.

For sparkless commutation:

- ⇒ Interpoles can be used.
- ⇒ By increasing brush contact resistance.
- ⇒ Shifting the brush areas.

Armature Reaction:[PU 2002, PU 2010]

- Armature reaction is the effect, in a DC motor or generator, of current in the armature creating a magnetic field that distorts the main field.
- The armature winding of a DC machine, when carrying current produces a magnetic field. As a result the main magnetic field across the air-gap is weakened and distorted; and heavy sparking on the commutator occurs, and EMF generated is reduced.
- This effect can be minimized by applying the following techniques:
- ⇒ By providing extra ampere-turns to the magnetic field. This reduces demagnetizing effect.
 - ⇒ By using compensating winding.
 - ⇒ By increasing the length of air-gap at pole tips.
 - ⇒ By shifting brushes to magnetic neutral plane.
 - ⇒ By using commutating poles.

Types of Excitation:

The field of a DC machine is excited by either of the following two methods or combination of those:

(a) Voltage excitation:

- ⇒ The field winding has a large number of turns of thin wire and is excited from a voltage source (Self or Separate). The field resistance is therefore high and the field carries a small current. Such a field carries a small current. Such a field winding is known as *Shunt field winding*.

Electric Machines

(b) Current excitation:

⇒ The field is excited here by a few turns of thick wire (low resistance) by means of the armature current. This excitation therefore varies load on the machine.

Types of DC Generator: [PU 2002, PU 2009]

→ Depending upon the way of excitation, DC generator are classified into two types.

a) Separately excited generator

b) Self-excited generator

⇒ Shunt wound generator

⇒ Series wound generator

⇒ Compound wound generator

○ Short shunt compound

○ Long shunt compound

a) Separately excited generator:

= In this type of generator, field winding are excited from a separate DC source as shown in figure.



b) Self-excited generator:

= A self-excited generator is the one in which excitation is produced to the field windings by its own armature voltage or by its own current. Depending on the connection, they are further divided as:

i) Shunt wound generator:

= In this type of generator, field coils are in parallel with armature coil. Such a generator passes full voltage generated. Usually field coil consist of many turns of fine gauge copper wire.

ii) Series wound generator:

= In this type of generator, field coils and the armature coils are joined in series. Evidently, the current flowing through the coil is the same as flowing through the load. Since, the field coils carry full load current they consist of relatively fewer turns of thick wire strip.

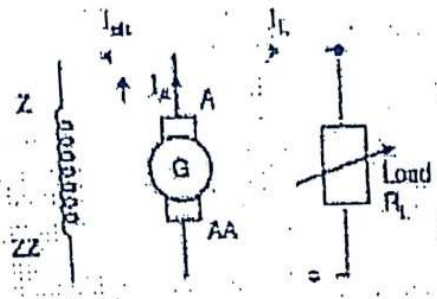


Fig: Shunt-wound

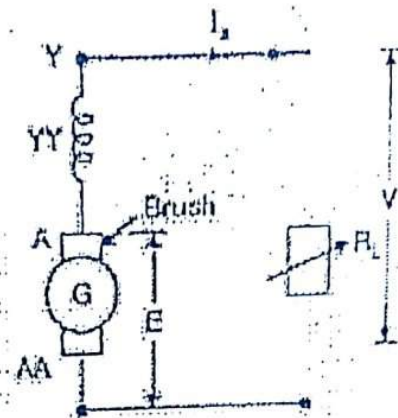


Fig: Series-wound

- iii) **Compound wound generator:**
 = This type of generator is the combination of few series and a few shunt windings i.e. Total excitation is supplied partly by series coil and partly by shunt coils. However, the major part of the excitation current is supplied by the shunt field if the shunt coils are connected across the armature only, then the generator is called *short shunt generator* and when series are also included then it is called *long shunt generator*.

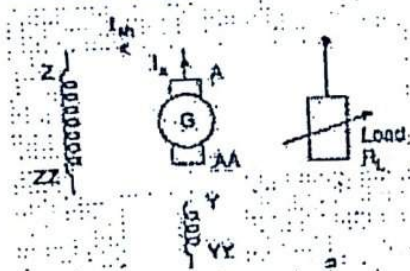


Fig: Short Shunt

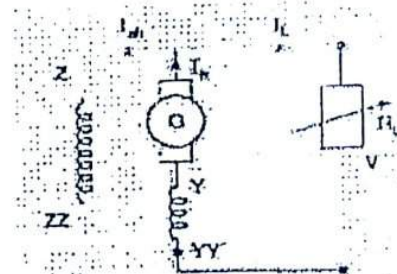


Fig: Long shunt

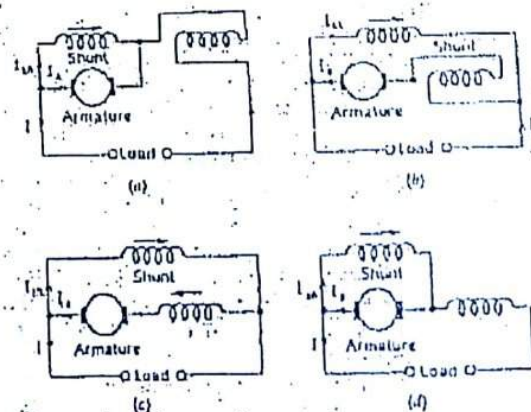
Generators can further be of two types each, namely, cumulative short (or long) - shunt compound generator or differential compound short (or long)-shunt generator.

Fig: (a) Cumulative short-shunt compound generator

(b) Cumulative long-shunt compound generator

(c) Differential compound long-shunt generator

(d) Differential compound short-shunt generator.

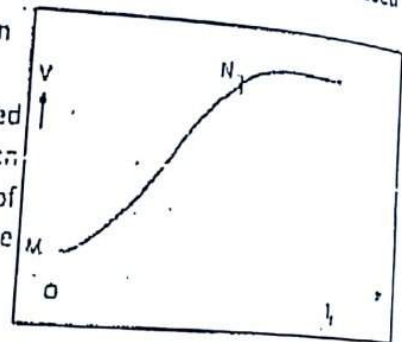


Characteristics of DC Generators:

- The working behavior of D.C. generator (as well as motors) under different conditions are studied, and the graphs or curves exhibiting in inter-relations between various conditions/quantities, are called characteristics or characteristics curves.
- Typical characteristics of D.C. generators, which are generally studied, are discussed below:

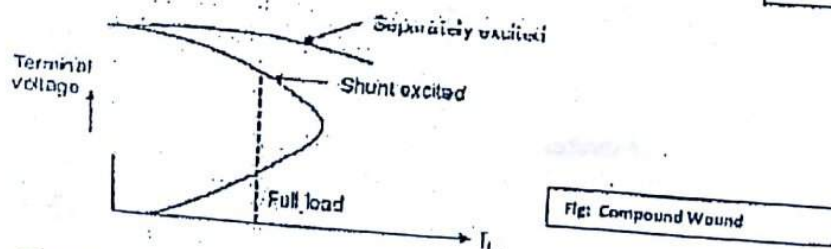
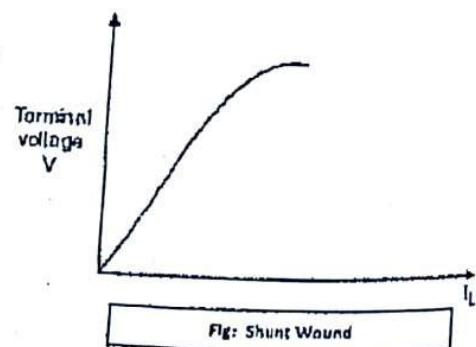
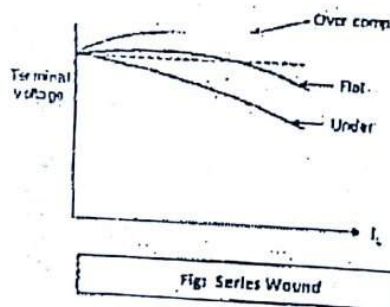
i) Open circuit or No load or Magnetization Characteristics

- ⇒ It gives the relation between the EMF generated in the armature and the field current at a given speed. Basically, this is the magnetization curve of the electromagnets. Its shape is practically same for all types of generator.



ii) External or load characteristics

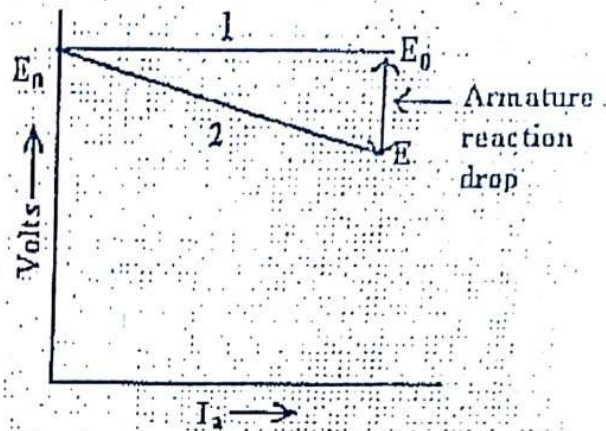
- ⇒ This gives the relation between the terminal voltage and load current. The voltage drop across the armature circuit resistance $I_a R_a$ is subtracted from the generated voltage E_a to get the terminal voltage, $V_t = E_a - I_a R_a$.
- ⇒ In a series wound generator, load and field current are same so that increase in load current increases magnetic flux thereby increasing generated e.m.f.
- ⇒ In Shunt wound generator, field current is constant and hence generated emf is also constant to a large extent. But the increase in load current increases voltage drop across armature resistance and thus load voltage decreases.
- ⇒ In compound wound generator, the curve is the mixed series and shunt wound generator. Therefore, the load voltage is nearly constant.



Electric Machines

iii) Internal Characteristics:

- ⇒ It gives the relationship between emf generated in armature and armature current. This generated emf decreases as the load is increased because of demagnetization effect of armature reaction.



D.C. Motors:

- A D.C. Motor is similar in construction to a DC generator. In fact, same DC machine can be used as generator as well as motor. In case of DC motor its field and armature windings are connected to DC source. Field winding produces the necessary magnetic field; while the current flowing through the armature conductor produces mechanical force or torque which rotates the armature.

Back EMF in a DC motor:

[PU 2002]

- The rotating conductors of armatures between poles of magnets in a D.C. motor cut the magnetic flux thereby developing an induced EMF which opposes the applied voltage. The induced EMF set up in the coil of D.C. motor opposing the current flowing through the conductors when the armature rotates is called Back EMF. The value of back emf depends upon the speed of the rotation of the armature conductor. The magnitude of back emf is given by:

$$E_b = V - I_a R_a$$

Similarly, current in the armature conductor is given by,

$$I_a = (V - E_b) / R_a$$

- At no load, the armature peaks speed and back emf starts increasing, the motor thus goes on peaking up speed until the back emf equals the applied voltage. Under such conditions, the speed of motor becomes constant and takes no current.

Importance of Back EMF:

- ⇒ It acts as energy conversion agent to convert electrical energy into mechanical energy.
⇒ From the equation it is clear that without back EMF, armature current will be extremely high which results in damage of motor.

Types of DC motors:

[PU 2010]

- Similar to generators, DC motors are also classified according to the method of excitation. Thus, they may be classified as:

- a) **Series-wound motor:** It possess field winding of a few turns of heavy conductors connected in series with the armature. These motors are used where high starting torque is required.

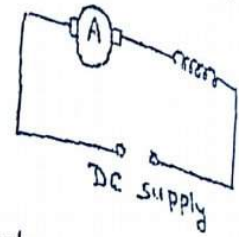


Fig: series-wound motor

Example / Applications [PU 2011]: Cranes, trains, trolley cars.

- b) **Shunt-wound motor (constant flux motor):** It possess field winding of large number of turns and high resistance which is connected in parallel with the armature.

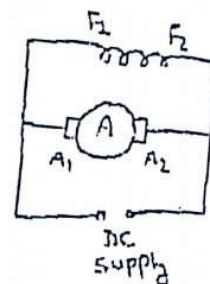


Fig: shunt-wound motor

These type of motors are used in pumps, lathe, drill, printing press etc. [PU 2011]

- c) **Compound-wound motor:**

→ According to the effect of field winding, characteristics of DC compound generator lies between the characteristics of DC shunt motor and DC series motor. If both field winding acts in same direction then the motor has cumulative characteristics and the DC motor is known as cumulative compound motor and if both field winding acts in opposite direction then the motor is known as differential compound motor.

- Cumulative compound:** series field assist shunt field.
- Differential compound:** Series field opposes shunt field.

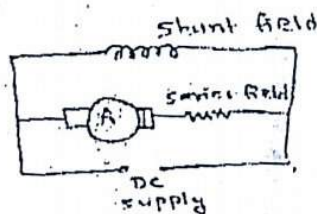


Fig: cumulative

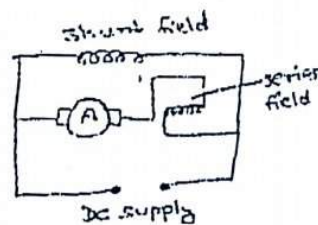


Fig: differential compound

Characteristics of DC Motor: [PU 2002]

- The performance of a DC motor is judged from its performance of characteristics curves. The following are the important characteristics of DC motors.
- Torque and Armature current characteristics:

Electric Machines

⇒ It is a curve for torque (τ_a) vs. armature current (I_a) of the DC motors.

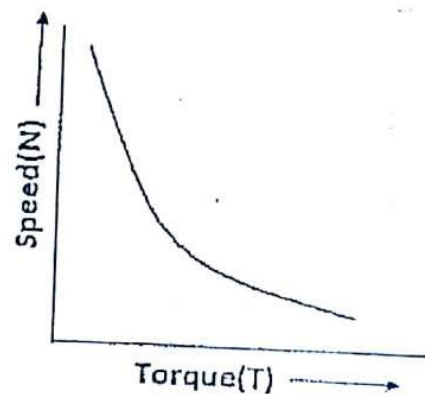
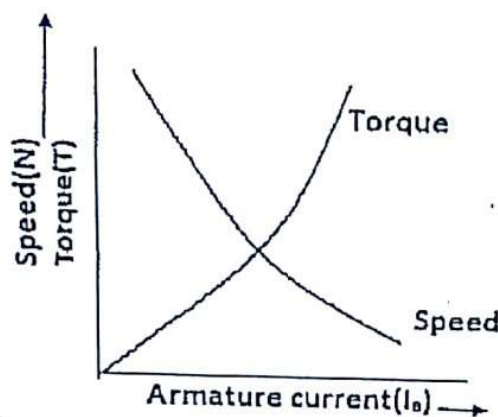
2. Speed-torque Characteristics:

⇒ It is a curve between speed ' N ' and armature torque ' τ_a ' of the motor. It is a mechanical characteristics of the DC motor.

3. Speed-armature current characteristics:

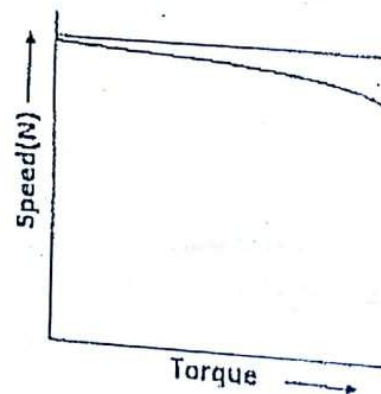
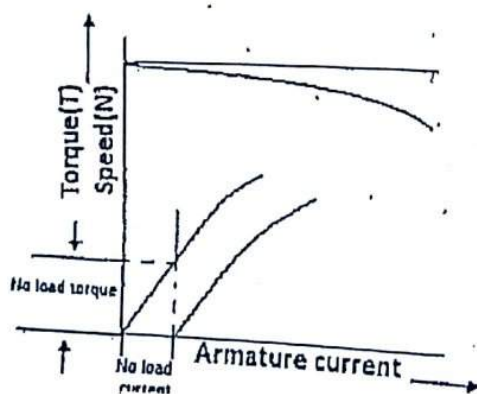
⇒ It is a curve between speed ' N ' and armature current ' I_a ' of the motor. This characteristics is very important in the selection of DC motor for particular application.

Characteristics (series wound motor): [PU 2006, PU 2008]



⇒ With increase in load ' I_a ', the speed ' N ' decreases. Consequently, at no-load, the speed of the motor is very high. Hence, series-wound motor should never be used without load.

Characteristics (shunt wound motor) : [PU 2006, PU 2008]

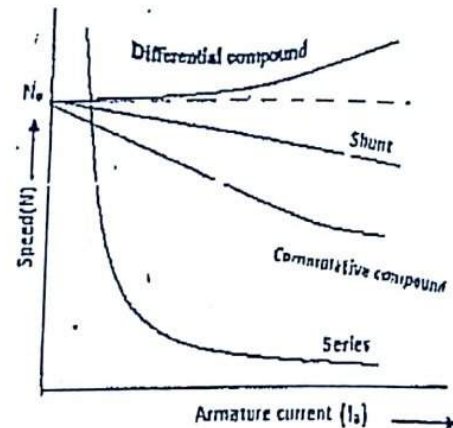
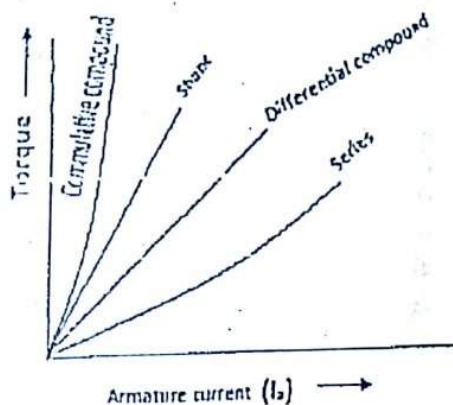


⇒ Since ϕ is constant, torque of the DC shunt motor is proportional to armature current.

Electric Machines

- ⇒ Speed of the DC shunt motor decreases gradually as the torque increases. Since shunt field is connected directly to supply, flux remains constant. Therefore, speed of the motor decreases slightly with increasing armature current.

Characteristics (Compound wound motor)



Starting of a DC motor:

- At the time of starting, the back emf is zero. Consequently the current flowing through the armature is very large i.e.

$$I_a = \frac{V - E_b}{R_a}$$

Where $E_b = 0$ (In case of starting)

$$I = \frac{V}{R_a}$$

- ⇒ If such current is allowed to flow, it would cause intolerable heavy sparking at the brushes which may destroy the commutator and the brush gear. Also certain development of large torque may cause mechanical shock to the shaft reducing its life. In order to save the motor from developing high initial current during the starting period or rheostat resistance called the stator is inserted in series with the armature of DC motor. Thus a motor stator provides a suitable resistance in the beginning thereby avoiding the damage to the motor. However once the motor has gained the speed the starting resistance must be fully removed from the circuit since the back emf developed in the motor now limits the current to a desired value.

Speed Control of DC motor:

[PU 2010 S, PU 2007 F, PU 2009 S, PU 2012 S]

- For DC motor,

$$E = \frac{\phi Z N \Gamma}{60 A} \dots \dots \dots (i)$$

$$E = V - I_a R_a \dots \dots \dots (ii)$$

From (i) and (ii)

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$$\frac{\phi ZNP}{60 A} = V - I_a R_a$$

$$\text{Or, } N = \frac{(V - I_a R_a) \cdot 60 A}{\phi ZP}$$

$$\text{Or, } N = \frac{(V - I_a R_a) \cdot k}{\phi}$$

$$\text{Where, } k = \frac{60 A}{ZP}$$

$$\text{Since } V \gg I_a R_a \quad N \cong \frac{V}{\phi} K$$

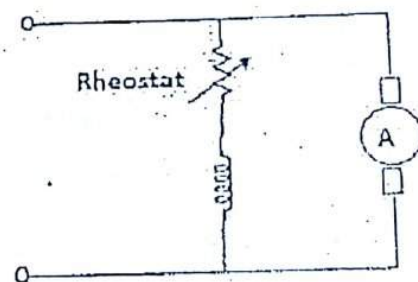
⇒ This equation gives us two methods of affecting of speed changes i.e. the variation of field excitation I_f and that of terminal voltage V .

- Speed Control of DC machine: [PU 2007, PU 2012]

→ The first method causes a change in flux and is known as Field Control Method. Second method is known as Armature Control Method.

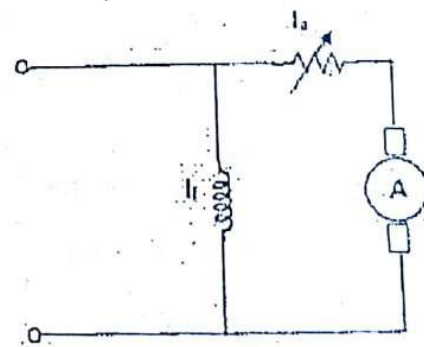
→ Field Control:

⇒ This method is the simplest way to control speed. Thus, the flux is adjusted varying the field current. This is done simply by rheostat in the field. As the field rheostat is increased, the air gap flux decreases resulting in higher speed. By this method, speed lowers than the rated speed cannot be obtained because the field cannot be made any stronger, it can only be weakened.



→ Armature Control:

⇒ In this method, the voltage applied to the armature is varied. The simplest way to achieve this is to connect an external resistor in the armature circuit. The only requirement is that, the external resistor must be of suitable value and size to be able to carry armature current. By this method, only speed below the rated value can be obtained. As the value of external resistor is increased, the speed decreases.



General Steps to Solve Problems on Speed Control:

- 1) Identify the method of speed control i.e. in which winding of the motor, the external resistance is to be inserted.
- 2) Use the torque equation $\tau \propto \phi I_a$ to determine the new armature current according to the condition of torque given. Load condition indicates the condition of torque.
- 3) Use the speed equation $N \propto E_b / \phi$ to find the unknown back emf or field current.

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- 4) From the term calculated above, & using voltage current relationship of the motor, the value of extra resistor to be added can be determined.

The above steps may vary little bit, according to the nature of problem but are always the base of any speed control problem.

Points to remember:

- 1) Principle of operation of dc generator is based upon faradays law of electromagnetic induction.
- 2) The direction of induced emf in generator is determined by Fleming's right hand rule.
- 3) The direction of rotation for motor is given by Fleming's left hand rule.
- 4) For satisfactory performance of operation, dc machines consists of armature, main field poles, yoke, commutator, interpoles, brushes, brush holders, bearing and proper enclosure.
- 5) The armature is laminated to reduce the eddy current losses. The thickness of the lamination being 0.4 to 0.5 mm.
- 6) The commutators converts the alternating emf in the armature to direct emf at the brushes.
- 7) Emf of dc machine, $E = \frac{\phi Z N P}{60 A}$
 $P = \text{no. of poles}$
 $\phi = \text{Flux per pole}$
 $N = \text{speed in rpm}$
 $Z = \text{total no. of armature conductor}$
 $A = \text{the number of parallel paths in the armature winding}$
 - For a generator: $E = V + I_a R_a$
 - For a motor: $E = V - I_a R_a$
- 8) Types of d.c machines separately excited machines, shunt machines, series machines & compound machines.
- 9) The terminal voltage falls as the load generator increases.
- 10) Fall in terminal voltage is much greater in case of shunt generators as compared to separately excited generators.
- 11) The terminal voltage increases as the load increases in case of series generator.

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- 12) The terminal voltage is practically constant on increasing the load in case of level compound generator.
- 13) In over compounded generator, the terminal voltage increases with an increase in the load on the generator.
- 14) The external characteristics is highly dropping in case of differentially compounded generator.
- 15) Torque: For Shunt motor... $\tau_a \propto I_a$; For series motor... $\tau_a \propto I_a^2$
- 16) Armature copper losses = $I_a^2 R_a$ (R_a = Armature resistance)
- 17) Constant losses = Iron losses + mechanical losses

Numerical Problems:

1. An 8 pole lap wound dc generator has 960 conductors, a flux of 40mWb and is driven at 400 rpm. Find the Induced EMF.

→ Solution:

We know that,

$$E = \frac{\phi Z N P}{60 A}$$
$$= \frac{\phi Z N}{60} \quad [\text{being } A=P]$$
$$= \frac{40 \times 10^{-3} \times 960 \times 400}{60}$$

$$\therefore E = 4.26 \text{ V}$$

2. An 8 pole wave wound dc generator has 900 armature conductors and a flux pole of 0.04 Wb. At what speed it must be driven to generate 500 V?

→ Solution:

We know that,

$$E = \frac{\phi Z N P}{60 A}$$

Since, $A=2$,

We get,

$$E = \frac{\phi Z N P}{120}$$

On Solving: $N = 208 \text{ rps}$

3. A 220V DC shunt motor has an armature resistance of 0.5Ω and is excited to produce constant flux. At full load, the motor runs at 1000 r.p.m. taking an armature current of 40 A. If a resistance of 0.8Ω is inserted in series with the armature, determine speed at the full load.

Electric Machines

Solution:

$$V=220 \quad R_a=0.5 \, \Omega \quad N_{fl}=1000 \text{ r.p.m.}$$

$$E_{b1}=V-I_{a1}R_a=220-40 \times 0.5=200V$$

$$I_{a1}=40A$$

$$\tau \propto \phi I_a \propto I_a \dots \phi \text{ is constant.}$$

$$\tau_1/\tau_2 = I_{a1}/I_{a2}=1$$

Now,

$$E_{b2}=V-I_{a2}(R_a+R_f)=220-40(0.5+0.8)=168V$$

$$\text{And, } N \propto \frac{E_b}{\phi} \propto E_b \dots \phi \text{ is constant.}$$

$$\text{So, } N_1/N_2=E_{b1}/E_{b2}$$

On solving: we get, $N_2=840 \text{ rpm.}$

4. An eight pole, DC shunt generator has 778 wave-connected conductors on its armature while running at 500 rpm. It supplies power to a load of 12.5 ohm at 250 V. The armature and the shunt field resistances are 0.24 ohm and 250 ohm respectively. Determine the armature current, the emf induced and the flux per pole.

→ Solution:

$$\text{The load current, } I_L = \frac{V}{R_L} = \frac{250}{12.5} = 20A$$

$$\text{The shunt field current, } I_{sh} = \frac{V}{R_{sh}} = \frac{250}{250} = 1A$$

$$\therefore \text{Armature current, } I_a = 20 + 1 = 21A$$

$$\text{The EMF induced, } E = V + I_a R_a = 250 + (21 \times 0.24) = 255.04V$$

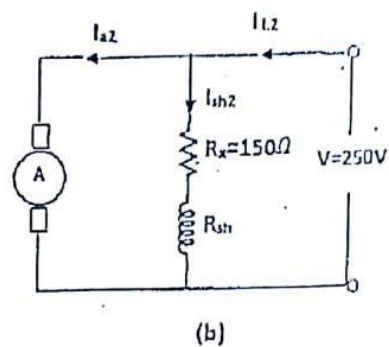
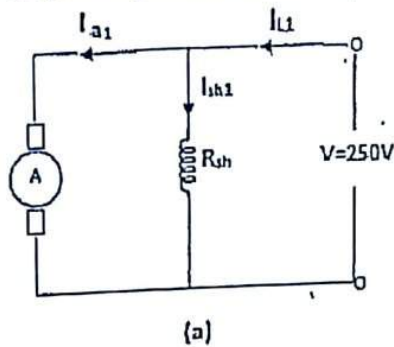
$$\text{Since, } E = \frac{\phi ZNP}{60A} \Rightarrow \phi = \frac{E \cdot 60A}{ZNP} = \frac{255.04 \cdot 60 \cdot 2}{778 \cdot 500 \cdot 2} = 9.83 \text{ mWb}$$

5. A 250V d.c. shunt motor has a shunt field resistance of 200 Ω and an armature resistance of 0.3 Ω . For a given load, motor runs at 1500 rpm drawing a current of 22 A from supply. If a resistance of 150 Ω is added in series with the field winding, find the new armature current and the speed. Assume load torque constant and magnetization curve be linear.

Solution:

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The following figure shows the two conditions:



This method is *flux control method*.

$$V = 250V \quad R_a = 0.3 \Omega \quad R_{sh} = 200 \Omega$$

In 1st case:

$$I_{L1} = 22 A$$

$$I_{sh1} = V/R_{sh} = 250/200 = 1.25 A$$

$$I_{a1} = I_{L1} + I_{sh1} = 22 + 1.25 A = 20.75 A$$

Now,

$$\tau \propto \phi I_a \propto I_{sh} I_a \text{ (as: } \phi \propto I_{sh} \text{)}$$

$$\tau_1 / \tau_2 = (I_{sh1} / I_{sh2}) * (I_{a1} / I_{a2})$$

As load torque is constant, $\tau_1 = \tau_2$

$$I_{sh1} I_{a1} = I_{sh2} I_{a2} \dots \dots (i)$$

$$\text{Now, } I_{sh2} = V / (R_{sh} + R_x) = 250 / (200 + 150) = 0.7142 A$$

Substituting this value in (i),

$$1.25 * 20.75 = I_{a2} * 0.7142$$

$$\Rightarrow I_{a2} = 36.316 A$$

$$\text{Hence, } E_{b1} = V - I_{a1} R_a = 250 - 20.75 * 0.3 = 243.775 V$$

$$\& E_{b2} = V - I_{a2} R_a = 250 - 36.316 * 0.3 = 239.10 V$$

Using speed equation,

$$N \propto \frac{E_b}{\phi} \propto \frac{E_b}{I_{sh}}$$

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$$\frac{N_1}{N_2} = \frac{E_{b1}}{E_{b2}} \cdot \frac{I_{sh2}}{I_{sh1}}$$

Substituting the values, we get, $N_2 = 2575.03$ r.p.m

This shows as flux decreases, the speed increases.

6. A 220 V shunt motor with an armature resistance of 0.5Ω is excited to give constant main field. At full load the motor runs at 500 rev. per minute and takes an armature current of 30 A. If a resistance of 1Ω is placed in the armature circuit, find the speed at:

(a) Full-load torque (b) double full-load torque

→ Solution:

Since flux remains constant, the speed formula becomes: $\frac{N_2}{N_1} = \frac{E_{b2}}{E_{b1}}$

(a) Full load torque:

With no-additional resistance in the armature circuit.

$$N_1 = 500 \text{ r.p.m.}; I_{a1} = 30 \text{ A}; E_{b1} = 220 - (30 \times 0.5) = 205 \text{ V}$$

Now,

$\tau \propto \Phi$ (since $\tau \propto \Phi$)

$$\therefore \frac{\tau_2}{\tau_1} = \frac{I_{a2}}{I_{a1}} \quad \text{since } \tau_2 = \tau_1; I_{a2} = I_{a1} = 30 \text{ A}$$

When additional resistance of 1Ω is introduced in the armature circuit,

$$E_{b2} = 220 - 30(1 + 0.5) = 175 \text{ V} \quad N_2 = ?$$

$$\frac{N_2}{500} = \frac{175}{205} \approx 427 \text{ r.p.m.}$$

(b) Double full-load torque:

$$\frac{\tau_2}{\tau_1} = \frac{I_{a2}}{I_{a1}} \quad \text{or} \quad \frac{2\tau_1}{\tau_1} = \frac{I_{a2}}{30}$$

On solving, we get: $I_{a2} = 60 \text{ A}$

$$E_{b2} = 220 - 60(1 + 0.5) = 130 \text{ V}$$

$$\frac{N_2}{500} = \frac{130}{205} \approx 317 \text{ r.p.m.}$$

7. A 250-V shunt motor with armature resistance of 0.5Ω runs at 600 rpm on full load and takes an armature current of 20 A. If resistance of 1Ω is placed in the armature circuit, find the speed at: [PU 2006]

a) Full load torque
b) Half load torque

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→ **Solution:**

Since, the flux remains constant, the speed formula becomes: $\frac{N_2}{N_1} = \frac{E_{b2}}{E_{b1}}$

(i) In the first case, full-load torque is developed.

$$N_1 = 600 \text{ r.p.m.} \quad E_{b1} = V - I_{a1}R_{a1} = 250 - (20 \times 0.5) = 240 \text{ V}$$

Now, $\tau \propto \phi I_a \propto I_a$ [since ϕ is constant]

$$\frac{\tau_2}{\tau_1} = \frac{I_{a2}}{I_{a1}} \quad \text{since } \tau_2 = \tau_1; I_{a2} = I_{a1} = 20 \text{ A}$$

$$E_{b2} = V - I_{a2}R_{a2} = 250 - (20 \times 1.5) = 220 \text{ V}$$

$$N_2 = \frac{N_1 \cdot E_{b2}}{E_{b1}} = \frac{600 \times 220}{240} = 550 \text{ r.p.m.}$$

(ii) In this case, the torque developed is half the full-load torque.

$$\frac{\tau_2}{\tau_1} = \frac{I_{a2}}{I_{a1}} \quad \text{or} \quad \frac{\tau_1}{\tau_1} = \frac{I_{a2}}{I_{a1}}$$

On solving, we get: $I_{a2} = 10 \text{ A}$

$$E_{b2} = 250 - (10 \times 1.5) = 235 \text{ V}$$

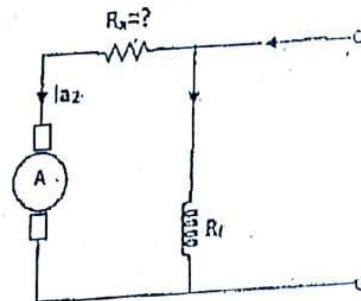
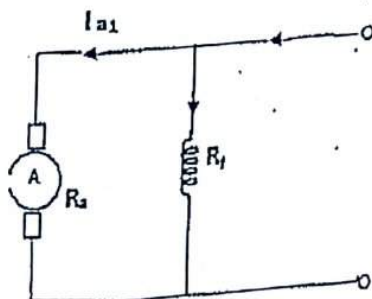
$$N_2 = \frac{N_1 \cdot E_{b2}}{E_{b1}} = \frac{600 \times 235}{240} = 587.5 \text{ r.p.m.}$$

8. A 220 V dc shunt motor draws a current of 40 A at full load and runs with a speed of 1400 RPM. Calculate the value of resistance required to be inserted in the armature circuit so that the speed drops to 1200 RPM at constant load. Given that $R_a = 0.02 \Omega$ and $R_f = 100 \Omega$. [PU 2010]

Solution:

Given: $I_{a1} = 40 \text{ A}$ $N_1 = 1400 \text{ RPM}$ $R_a = ?$

$N_2 = 1200 \text{ RPM}$ $R_f = 0.02 \Omega$



This method is *armature control method*.

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Here, shunt resistance is not changed. So, flux remains constant, which in turn gives relation: $I_{a1} = I_{a2}$

We know that,

$$E_{b1} = V - I_{a1}R_a = 220 - 40(0.02) = 219.2$$

$$\& E_{b2} = V - I_{a2}(R_a + R_x) \text{ [where, } R_x = \text{unknown resistance}]$$

$$= 220 - 40(0.02 + R_x)$$

We have the relation,

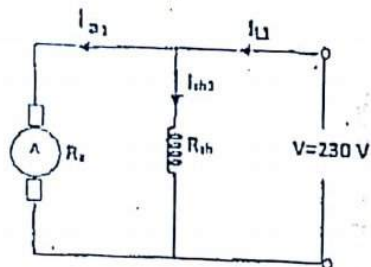
$$\frac{N_1}{N_2} = \frac{E_{b1}}{E_{b2}}$$

Substituting the values, we get: $R_x = 0.78 \Omega$.

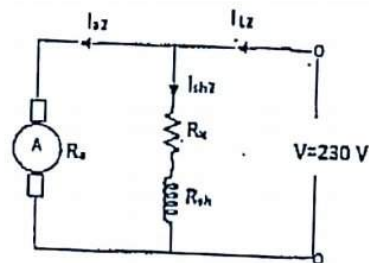
9. A 230V dc shunt motor has an armature resistance of 0.1 ohm and a shunt field resistance of 275 ohm. It runs at speed of 1000 rpm when drawing an armature current of 75A. Calculate the additional resistance to be inserted in the field circuit to raise the motor speed to 1200 rpm at an armature current of 125A. Assume linear magnetization characteristics.

[PU 2011]

Solution:



Case (I)



Case (II)

$$E_{b1} = V - I_{a1}R_a =$$

$$230 - 75(0.1) = 222.5V$$

$$E_{b2} = V - I_{a2}R_a = 230 - 125(0.1) = 217.5V$$

$$I_{sh1} = V/R_{sh1} = 230/275 = 0.83A$$

$$I_{sh2} = V/R_{sh2} = 230/(275 + R_x)$$

Now,

$$\frac{N_1}{N_2} = \frac{E_{b1}}{E_{b2}} \cdot \frac{I_{sh2}}{I_{sh1}}$$

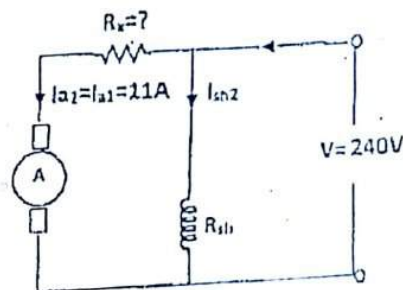
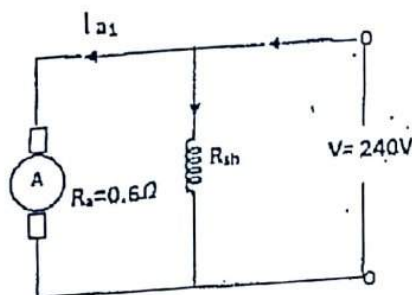
Substituting the values, we get: $R_x = 65.17\Omega$.

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10. A 240 V shunt motor runs at 1450 rpm at full load with the armature current 11A. The total resistance of armature and brush is 0.6 ohm. If the speed to be reduced to 1000 rpm with the same armature current, Calculate the value of resistance to be connected in series with the armature. [PU 2009 F, Similar to PU 2009 S, PU 2013 F]

Solution:

Given: $V=240V$ $I_{a1}=I_{a2}=11A$ $N_1=1450 \text{ rpm}$ $R_x=?$ $N_2=1000 \text{ rpm}$
 $R_a=0.6\Omega$



The method is *armature control method*.

We know that,

$$E_{b1} = V - I_{a1}R_a = 240 - 11(0.6) = 233.4$$

$$\& E_{b2} = V - I_{a2}(R_a + R_x) \text{ [where, } R_x = \text{unknown resistance]} \\ = 240 - 11(0.6 + R_x)$$

We have the relation,

$$\frac{N_1}{N_2} = \frac{E_{b1}}{E_{b2}}$$

Substituting the values, we get: $R_x = 6.58 \Omega$.

11. A 220 V dc shunt motor runs with 1000 rpm with an armature current of 40 A. The resistance of armature is 0.5 Ω . Calculate the value of resistance to be connected in series so that the speed drops to 600 rpm. [PU 2013 S]

→ Solution:

Given: $V=220V$ $I_{a1}=I_{a2}=40A$ $N_1=1000 \text{ rpm}$ $R_x=?$
 $N_2=600 \text{ rpm}$ $R_a=0.5 \Omega$

The method is *armature control method*.

We know that,

Electric Machines

$$E_{b1} = V - I_{a1} R_s = 220 - 40(0.5) = 200 \text{ V}$$

$$\& E_{b2} = V - I_{a2}(R_s + R_r) \text{ [where, } R_r = \text{unknown resistance]} \\ = 220 - 40(0.5 + R_r)$$

We have the relation,

$$\frac{N_1}{N_2} = \frac{E_{b1}}{E_{b2}}$$

Substituting the values, we get: $R_r = 2 \Omega$.

[Note: For figure see Q.No. 10]

A.C. MACHINE:

Single phase Induction Motor:

Construction:

The construction of a 1 ϕ induction motor is similar to that of 3 ϕ induction motor except that the stator is provided with the 1 ϕ between two poles. Thus, it has stator with slots and squirrel cage rotor with a small air gap in between.

Working principle:

Initially the rotor is stationary between the poles of the stator. As the alternating current is supplied to the stationary windings a sinusoidal pulsating magnetic field varying with time is produced which in turn produces pulsating current in the rotor. Since, this pulsating current in the

rotor is incapable of producing a rotating torque in the stationary rotor a single phase induction motor is not self-starting. In order to obtain a rotating field the stator is provided with two windings, the main winding and starting winding on the 4 poles instead of 2 poles. The phase difference of 90° between two windings is obtained by splitting the phase. The combined effect of the field setup by the 2 windings is more or less rotating field thereby providing a starting torque. After the starting, the rotation of a field is carried by the current setup in the main winding and starting winding is generally switched out of the stator circuit. Such a motor is also called as split phase induction motor.

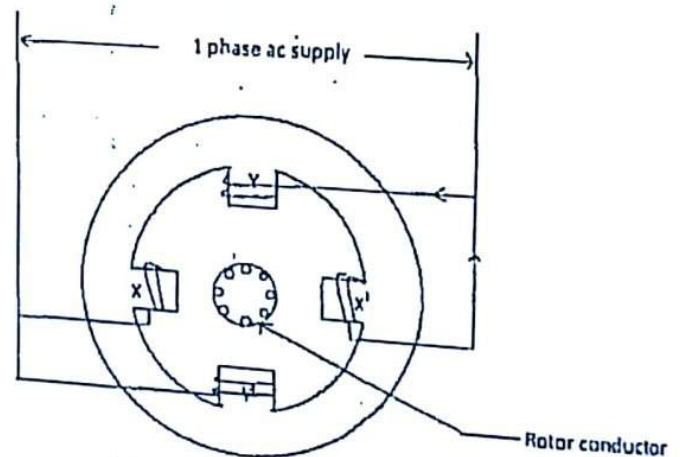


Fig. 4 pole 1 ϕ induction motor

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Three phase Induction Motor: [PU 2002, PU 2006, PU 2008, PU 2009, PU 2011, PU 2012, PU 2013 F, PU 2013 S]

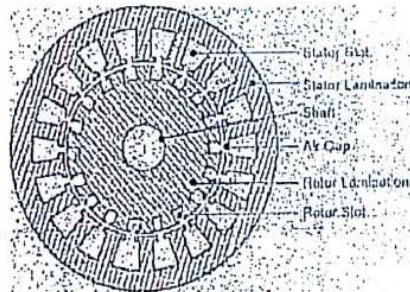
- They are most widely used A.C. motors. In this motor, only the field winding (stator) is connected to the voltage source, a rotating magnetic field is produced in it which induces emf in rotor winding by electromagnetic induction. The emf in the rotor in turn produces rotor current, the rotating field due to stator and rotor reacts to produce necessary torque and rotor conductor to rotate itself.
- Rotating magnetic field: $N_s = (120 f)/p$; where, N_s = Synchronous Speed, f =frequency, p =no. of poles
- Since, there is no electrical connection between rotor and the supply and the current in the rotor are set-up entirely by the effect of electromagnetic induction from the stator. Hence, the motor is known as *Induction Motor*.

Construction:

The 3 ϕ Induction motor consists of two main parts: Stator and Rotor.

(i) Stator:

- The stator is built up of laminations. It is hollow cylindrical in shape with slots cut parallel to its axis in its inner surface to carry windings. It is wound for two, four, six or eight poles depending upon speed required. The speed decreases as the number of poles increases. The stator is mounted on a frame and connections are taken out from the coils to a terminal box fitted to the side of a frame. Link plates are used to enable the connections to be changed into star or delta.

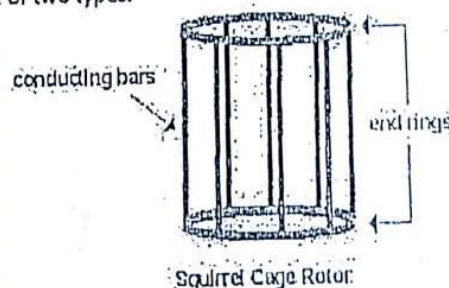


(ii) Rotor:

- It is the rotating part of the motor and are of two types:

⇒ Squirrel cage type:

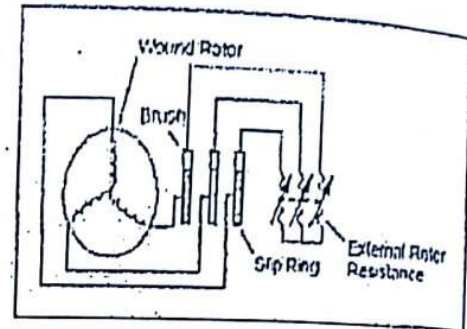
This type of rotor consists of copper bars or aluminum bars filling the slot of the core. These conductors are winding to the two end rings of the same material. Most of the 3 ϕ induction motor are of squirrel cage type as it has rugged construction. It is not possible to add any external rotor resistance.



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⇒ Wound Rotor (Slip ring rotor) type:

This type of rotor has 3 ϕ winding for same no of poles as of the stator. In this type winding is connected in rotor and other ends are brought to slip rings. In this type of rotor we can add external resistance.



Operating Principle:

- When three phase winding of stator is connected to three phase supply then 3 ϕ current in stator winding produces a rotating magnetic field which rotates round the stator at synchronous speed, $N_s = \frac{120 f}{P}$
- This rotating flux passes through air gap and cuts the rotor conductor. Due to relative velocity between rotating flux and the stationary rotor, e.m.f. is induced in rotor conductor. The frequency of induced e.m.f. is same as the supply frequency when the rotor is stationary.
- Since, rotor conductors circuit are closed, induced e.m.f. produce rotor current which follows through the rotor conductor. Due to this, rotor conductor experience a force and rotor starts rotating in the same direction as rotating magnetic field.
- Since, the direction of rotation of rotor as well as the rotation of magnetic field are same, the relative velocity between magnetic field and rotor decreases. A decrease in relative velocity decrease the induced e.m.f. in the current and rotor current but the rotor continues to rotate. When equilibrium condition is reached, the rotor runs at synchronous speed but in actual practice it is very near to synchronous speed and its value is always below the synchronous speed.
- If the rotor speed becomes synchronous speed, the relative velocity between magnetic field and rotor conductor becomes zero and hence e.m.f. induced is zero and torque produced is zero. Therefore, rotor of synchronous motor never runs at synchronous speed.

Application of Induction Motors:

1. Used in different types of Fans;
2. Driving pumps in power stations.
3. Fertilizers complexes, Irrigation project.
4. Driving Compressors in steel mills, paper mills and for crushers and coal conveyors.

Merits and Demerits of Induction Motors:

Merits:

Electric Machines

1. It is rugged, unbreakable in construction.
2. Simple design low maintenance cost.
3. Starting operation is simple.
4. High overload capacity.

Demerits:

1. Speed decreases with increase of load.
2. Speed cannot be varied without losing some of its efficiency.
3. Low power factor.

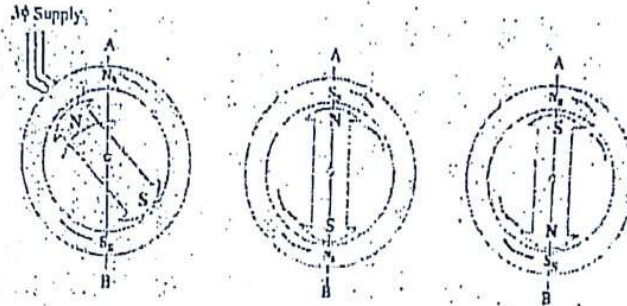
Synchronous Motor:

- It is a 3- ϕ electrical motor that runs only at synchronous speed. Therefore, a synchronous motor runs either at synchronous speed or not at all. While running, it maintains constant speed.
- Similar to the DC motor, rotating part of the motor requires DC supply to generate magnetic field.
- Similar to Induction Motor, its stationary part (Stator) is connected to three-phase supply.

Operating Principle:

- When the three-phase winding of a synchronous motor is fed with a balanced three-phase supply, a rotating magnetic field is produced. The rotating magnetic field has constant magnitude and revolves around the stator with synchronous speed, $N_s = \frac{120 f}{P}$. It is called synchronous speed because it synchronizes to the power frequency.
- Suppose at any instant N_s happens to face S. Due to mutual attraction N_s tries to pull S in clockwise direction. However, rotor has high inertia and takes little time to move. Meanwhile, the revolving pole N_s is replaced by S_s which helps in repelling the S in anticlockwise direction. Thus, the high inertia rotor is made to move first in one direction and then in the other direction in quick succession. The net result is that it just stays where it is. *That's why synchronous motor is not self-starting.*
- However, it can be made self-starting with the help of squirrel cage winding also known as damper winding or pole face winding. It consists of parallel copper bars at both ends. Now, when three phase supply is fed to stator winding of a synchronous motor while the rotor pole field circuit is left open. A rotating magnetic field is produced. The rotating field induces an EMF in the damper winding as it passes over. Hence, synchronous motor starts up as an induction motor. When it reaches near its synchronous speed, its rotor field circuit is closed. Hence, the rotor poles are magnetized and get immediately locked with the stator poles and start rotating at synchronous speed in same direction. The slip between the rotating stator poles and rotor poles becomes

zero i.e. the stator and rotor runs at same speed. In fact, torque in a synchronous motor never develop any other speed other than the synchronous speed.



Uses of Synchronous Motors:

1. Synchronous motor is used for constant speed power application in sizes above 20 H.P. and more often in sizes large than 100 H.P.
2. Typical applications are Fan, DC generator, rolling mills, Flour-mill line shaft, air and gas-compressors, rubber mills, mixers, crushers, etc.
3. Useful for power factor control and pulp grinders and refiners.
4. Used in paper making industry.

गो. परिवार आगम

पुष्प ६०-३५४४
महानगर ०२११८
पुष्प ६०-३५४४

कावाकुवा हाइमिङ्ग इन्स्टिट्यूट

लेन पा - ७, तालकोट

फोन - ०६१-२२२६२४

कार, बाइक, ट्रक, रेट्रो, तथा विभिन्न किसिमका सवारी साधन सिक्का लागि
गो. परिवार आगम, तालकोट, लाइसेन्स, निगम, सहयोगको लागि हामीलाई सम्पर्क गर्नुहोस्।
सोपानिक र सवारी साधनको लागि हामीलाई सम्पर्क गर्नुहोस्।